Toward a converse of Turyn's theorem: obstructions to Turyn pairs at composite orders

Jeffery Kline

July 2026 — consolidated working draft

Abstract

Turyn (1972) constructed Hadamard matrices of order $4t$ from pairs of symmetric circulants R, S of odd order t with $R^2 + S^2 = qI$, $q = 2t - 1$, whenever q is a prime power, and conjectured that such pairs exist *only* then. This paper assembles a program of exact obstructions toward that converse. The organizing result is a selection theorem: nonexistence at q is equivalent to emptiness of a finite marginal system at some divisor of t, reducing the conjecture to per-family firing criteria. Written criteria close every self-conjugacy-blind multi-prime composite $q \leq 2000$, including apparently first determinations at $q = 1469$, 1937, 1325 (and, under GRH, $q = 5185$ and 62305); a closed-table caveat applies and is stated in the text. On the lane $q = 6\ell - 1$ with $\ell \equiv 3 \pmod 4$ prime, the criteria fire deterministically: a Turyn pair at composite q forces the multiplier image modulo ℓ down to $\{\pm 1\}$. Consequences include an unconditional semiprime nonexistence family and a conditional density program whose remaining analytic inputs are stated as explicit, falsifiable hypotheses. Every principal claim carries a verification-status tier recorded in Appendix B.

Contents

1 The converse problem

1.1 Turyn pairs and Turyn's theorem

Let t be odd, let $G = \langle x \rangle$ be cyclic of order t, and put $q = 2t - 1$, so that $q \equiv 1 \pmod 4$ (as t is odd). Identify circulant matrices of order t with elements of the group ring $\mathbb{Z}[G]$.

Definition 1.1. A *Turyn pair* of order t is a pair $r = \sum_g r_g g,\ s = \sum_g s_g g$ in $\mathbb{Z}[G]$ such that

- $r_1 = 0$ and $r_g \in \{\pm 1\}$ for $g \neq 1$;
- $s_g \in \{\pm 1\}$ for all g;
- $r_{g^{-1}} = r_g$ and $s_{g^{-1}} = s_g$ (both symmetric); and
- $r^2 + s^2 = q \cdot 1$ in $\mathbb{Z}[G]$.

From a Turyn pair one obtains the Williamson quadruple [4] $A = R + I$, $B = R - I$, $C = D = S$, which satisfies $A^2 + B^2 + C^2 + D^2 = 4tI$ and hence yields a Hadamard matrix of order $4t$.

Theorem 1.2 (Turyn 1972 [1])**.** *If $q = 2t - 1$ is a prime power ($q \equiv 1 \pmod 4$), a Turyn pair of order t exists.*

Turyn built the pair by ordering the quadratic-character matrix on $\mathrm{PG}(1, q)$ along a Singer cycle (Section A); Whiteman [6] re-proved it cyclotomically, and Yamamoto–Yamada [7] identified the character values as relative Gauss sums for $\mathbb{F}_{q^2}/\mathbb{F}_q$.

1.2 The conjecture

The closing remark of [1] states that the methods of [3] show, "in certain cases," that no analogue of the pair exists when q is not a prime power, and that "it seems likely this is true in general." We take this as the destination.

Conjecture 1.3 (Turyn's converse)**.** *If a Turyn pair of order t exists, then $q = 2t - 1$ is a prime power.*

Conjecture 1.3 is open. It has the flavour of, but is not implied by, the computational classifications discussed below.

1.3 The same object in other clothing

The Turyn-pair equation is a hub; the following identifications are standard and are used to import necessary conditions and to position the results (Section 6).

- *Negacyclic conference matrices.* By alternating-sign conjugation a Turyn pair of order t is monomially equivalent to a negacyclic $W(2t, 2t-1)$; see Seberry [12], Lemmas 1.3–1.4. The Delsarte–Goethals–Seidel–Eades computation [11, 12] found that the only negacyclic $W(v, v-1)$ with $v < 1000$ have $v = q+1$ with q an odd prime power—the converse, verified by machine below $v = 1000$.

- *Almost-perfect autocorrelation sequences.* The pair corresponds to a binary sequence of period $4t$ whose out-of-phase periodic autocorrelations all vanish except at the half-period shift $2t$, a value of $4 - 4t$; these are Wolfmann's almost-perfect sequences [13], identified with cyclic divisible difference sets by Bradley–Pott [14].

- *Two-circulant-core Hadamard matrices.* The pair is exactly the input to the two-circulant-core construction and its cocyclic analysis [15].

- *Complex/Butson conference matrices.* With $\tau = r + is$, the matrix $P + iI$ (where P is the conference matrix built from R, S) is a complex Hadamard matrix of order $q+1$; see Section A.

1.4 Main results and reading guide

The contribution is a theorem roadmap, not a proof of the conjecture. Principal results, with verification tiers defined in Appendix B:

(1) **Selection theorem** (Theorem 4.66, Tier 0): nonexistence at q is equivalent to emptiness of some divisor node's marginal system; feasibility is monotone on the divisor lattice and the orbit model at every node is computed by the multiplier group $M(q)$ alone. This converts the converse into per-family firing theorems.

(2) **Closed-form firing criteria** (Tier 0): the transitive two-orbit representation-pair criterion, the $h = 3$ support/augmentation hierarchy, the full-torus row-sum bound — sharp exactly at Turyn's $q = 25$ — with its exact sub-threshold form, the full-unit prime-power nested criterion, and lattice-reduced period joins.

(3) **Finite closure** (Tier 0 per member): every self-conjugacy-blind multi-prime composite $q \leq 2000$ is closed by written quantified criteria, including apparently first determinations $q = 1469, 1937, 1325$; under GRH (Tier 1) also $q = 5185$ and $q = 62305$.

(4) **The lane $q = 6\ell - 1$, $\ell \equiv 3 \pmod 4$ prime**: a Turyn pair at composite q forces the multiplier image in $(\mathbb{Z}/\ell)^\times$ down to $\{\pm 1\}$ (Proposition 4.78, Tier 0); the semiprime case gives an unconditional nonexistence family (Theorem 4.71, Tier 0) whose infinitude is parity-blocked; a density program under the explicit hypothesis EQ_3 (Tier 2) gives the small-factor obstruction count (Proposition 4.82), with the passage to an infinite family isolated in one open structure count (Remark 4.83) — open also under GRH.

(5) **Construction side** (Tier 0): a uniform character-lifting normal form and fillability dichotomy, in Appendix A.

Sections 2–3 build the character and multiplier machinery; Section 4 is the obstruction hierarchy and contains results (1)–(4); Sections 6–8 record the closed-order ledger, related work, and what remains.

2 The character identity

All obstructions flow from one identity. Put

$$\tau = r + is \in \mathbb{Z}[i][G], \qquad \tau^* = r - is.$$

Because r, s are symmetric, coefficient conjugation and the group-ring adjoint agree on τ, and the pair equation is

$$\tau\tau^* = q. \tag{1}$$

For a character χ of G of order d, the values $\chi(r), \chi(s)$ are real, so

$$\chi(\tau)\,\overline{\chi(\tau)} = \chi(r)^2 + \chi(s)^2 = q, \tag{2}$$

an equation in $\mathbb{Z}[\zeta_d, i] = \mathbb{Z}[\zeta_{4d}]$. Thus each character value $\chi(\tau)$ is a cyclotomic integer of absolute value $\sqrt{q}$ at every archimedean place: the Turyn-pair equation is a family of *norm-q conditions*, one per character, coupled by integrality of the coefficients. Every result below analyzes these conditions—at the trivial character (§4.1), over inert towers (§4.2), under common decomposition subgroups of the full Galois action (§3), by orbit-reduced full-group and marginal decisions (§§4.3–4.4), or after descent to a prime quotient (§§4.7–4.9).

3 Multiplier theorems

3.1 The prime-power case: exact signs

Before the composite obstructions we record the rigidity that the prime-power case already carries; it both frames the method and pins the symmetry the converse must contend with. For $u \in (\mathbb{Z}/t)^\times$ let $r^{(u)}$ denote decimation $x \mapsto x^u$. Say u is a *multiplier* of (r, s) if $r^{(u)} = \epsilon r$ and $s^{(u)} = \delta s$ for signs $\epsilon, \delta \in \{\pm 1\}$. Symmetry makes -1 a trivial multiplier with signs $(1, 1)$.

Theorem 3.1 (Frobenius multiplier, exact signs)**.** *Let (r, s) be a Turyn pair of order t with $q = 2t - 1 = p^e$, p prime. Then p is a multiplier, and*

$$p \equiv 1 \pmod 4 \;\Rightarrow\; r^{(p)} = r,\; s^{(p)} = s; \qquad p \equiv 3 \pmod 4 \;\Rightarrow\; r^{(p)} = -r,\; s^{(p)} = s.$$

In the second case e is even, $\sum_g r_g = 0$, and $\sum_g s_g = \pm p^{e/2}$.

When $e=1$ this only restates the symmetry $p\equiv -1 \pmod t$; the first nontrivial content is at $e\ge 2$. Let $\sigma_p\in \mathrm{Gal}(\mathbb{Q}(\zeta_{4t})/\mathbb{Q})$ send $\zeta_{4t}\mapsto \zeta_{4t}^p$ (well-defined since $p\mid q$ is coprime to $4t$), and define the Frobenius twist $\tau_p=\sum_g \sigma_p(\tau_g)g^p = r^{(p)}+i^p s^{(p)}$, so that $\chi(\tau_p)=\sigma_p(\chi(\tau))$ for every character χ.

Lemma 3.2. *For every character χ of G of order d, the quotient $u_\chi=\sigma_p(\chi(\tau))/\chi(\tau)$ is a root of unity in $\mathbb{Q}(\zeta_{4d})$.*

Proof. Let $\alpha=\chi(\tau)\in\mathbb{Z}[\zeta_{4d}]$. Every Galois conjugate of α has absolute value $\sqrt{q}$: a conjugate sends χ to some power χ^m and i to $\pm i$, so its squared absolute value is $\chi^m(r)^2+\chi^m(s)^2=q$ by (2); in particular $\alpha\neq 0$. The identity $\alpha\overline{\alpha}=q=p^e$ shows (α) is a product of primes above p. As $p\nmid 4d$, p is unramified in $\mathbb{Q}(\zeta_{4d})$ and σ_p is its Frobenius, so σ_p fixes each prime above p; hence $\sigma_p((\alpha))=(\alpha)$ and u_χ is a unit. All its conjugates have absolute value 1, so Kronecker's theorem makes u_χ a root of unity; as d is odd, these are μ_{4d}. □

Lemma 3.3. *There exist $a\in\mathbb{Z}/4$ and $c\in\mathbb{Z}/t$ with $u_\chi=i^a\chi(x^c)$ for every character χ.*

Proof. For $m\in(\mathbb{Z}/t)^\times$ let γ_m fix i and send $\zeta_t\mapsto\zeta_t^m$; since the cyclotomic Galois group is abelian and τ has $\mathbb{Z}[i]$ coefficients, $\gamma_m(u_\chi)=u_{\chi^m}$. The characters of exact order d form one orbit under these γ_m (as $(\mathbb{Z}/t)^\times\to(\mathbb{Z}/d)^\times$ is onto), so for each $d\mid t$ there are $a_d\in\mathbb{Z}/4$ and $c_d\in\mathbb{Z}/d$ with $u_\psi=i^{a_d}\psi(x^{c_d})$ for every ψ of exact order d. We show these data cohere. Set $\beta=\tau_p\tau^*\in\mathbb{Z}[i][G]$, so $\chi(\beta)=\sigma_p(\chi(\tau))\overline{\chi(\tau)}=q\,u_\chi$. Mapping $\mathbb{Z}[i][G]\to\mathbb{Z}[i][C_e]$ and inverting the Fourier transform on C_e gives, for the coefficient of y^j,

$$\pi_e(\beta)_j=\frac{q}{e}\sum_{d\mid e} i^{a_d}R_d(c_d-j),\qquad R_d(n)=\sum_{m\in(\mathbb{Z}/d)^\times}\zeta_d^{mn}$$

the Ramanujan sum. Since $\pi_e(\beta)_j\in\mathbb{Z}[i]$ and $(q,e)=1$,

$$e\;\Big|\;\sum_{d\mid e} i^{a_d}R_d(c_d-j)\qquad \text{in } \mathbb{Z}[i] \tag{3}$$

for all j and all $e\mid t$. Induct on the number of prime factors of e (with multiplicity); $e=1$ is empty. If $e=P$ is prime, choose $j\not\equiv c_P \pmod P$; then $R_1\equiv 1$, $R_P(c_P-j)=-1$, so (3) gives $P\mid i^{a_1}-i^{a_P}$, impossible unless $a_P=a_1$ (a nonzero multiple of P has norm $\ge P^2>4$, while a difference of fourth roots of unity has norm ≤ 4). For general e, by induction $a_d=a_1=:a$ for proper $d\mid e$ and the c_d are compatible; choose c_* with $c_*\equiv c_d \pmod d$ for all proper $d\mid e$. Using $\sum_{d\mid e}R_d(n)=e$ or 0 according as $e\mid n$ or not, (3) reduces to

$$e\mid i^{a_e}R_e(c_e-j)-i^aR_e(c_*-j)\qquad\text{for all } j. \tag{4}$$

The formula $R_e(n)=\mu(e/(e,n))\varphi(e)/\varphi(e/(e,n))$ shows, for odd e: the value $\varphi(e)$ occurs only when $e\mid n$, the value $-\varphi(e)$ never occurs, and all values exceed $-\varphi(e)/(P-1)$ (P the least prime factor). If a_e-a is odd the two terms in (4) lie in $\mathbb{Z}$-independent directions, forcing $e\mid R_e(c_*-j)$ for all j—impossible at $j=c_*$ where the value is $\varphi(e)\in(0,e)$. If $a_e-a\equiv 2 \pmod 4$ then $e\mid R_e(c_e-j)+R_e(c_*-j)$; the bounds confine the sum to $(-e,2e)$, so it is 0 or e, and summing over a period gives 0, whence it vanishes identically—but at $j=c_*$ this reads $R_e(c_e-c_*)=-\varphi(e)$, impossible for odd e. Thus $a_e=a$, and (4) gives $e\mid R_e(c_e-j)-R_e(c_*-j)$. If e has two distinct prime factors $P<Q$ the difference is bounded by $\varphi(e)P/(P-1)<e$, so the two Ramanujan functions agree and $c_*\equiv c_e \pmod e$; if $e=P^k$ the values are $\varphi(e),-P^{k-1},0$ and one checks $c_*\equiv c_e$ $(\mathrm{mod}\ P^{k-1})$, exactly the compatibility needed. The induction closes. □

Proof of Theorem 3.1. By Lemma 3.3, $\chi(\beta) = q\, i^a \chi(x^c)$ for all χ, so $\beta = \tau_p \tau^* = q\, i^a x^c$ in $\mathbb{Z}[i][G]$. Multiplying by τ and using (1) gives $\tau_p = i^a x^c \tau$. The identity coefficient of τ is is_1 (modulus 1), while every other has modulus $\sqrt{2}$; the same holds for τ_p and decimation fixes the identity, so $c = 0$. Comparing identity coefficients gives $i^p s_1 = i^{a+1} s_1$, so $a \equiv p-1 \pmod 4$ and $\tau_p = i^{p-1}\tau$. For $p \equiv 1$ this is $r^{(p)} + is^{(p)} = r + is$; for $p \equiv 3$ it is $r^{(p)} - is^{(p)} = -(r + is)$, i.e. $r^{(p)} = -r$, $s^{(p)} = s$. Finally, if $p \equiv 3 \pmod 4$, at the trivial character $(\sum r_g + i \sum s_g)(\sum r_g - i \sum s_g) = p^e$; p is inert in $\mathbb{Z}[i]$, so e is even and the left factor is $i^\kappa p^{e/2}$; as $\sum s_g$ is odd its imaginary part is nonzero, forcing $\sum r_g = 0$ and $\sum s_g = \pm p^{e/2}$. $\square$

Corollary 3.4. *If $q = p^e$ then every Turyn pair is determined by its values on the $\langle p \rangle$-orbits of G: constant along them for $p \equiv 1 \bmod 4$, sign-alternating ($r_{g^p} = -r_g$) for $p \equiv 3 \bmod 4$.*

Empirically the multiplier group is *exactly* $\langle p \rangle$ on all 328 pairs of the exhaustive sweep over odd $t \le 41$—a strengthening, since the theorem asserts only that p is a multiplier—consistent with the uniqueness up to equivalence observed by Lang–Schneider [8] through order 99. Theorem 3.1 is best viewed as a normal-form refinement of classical multiplier theory [5]: no group-development hypothesis is made (any solution of (1) already carries the Frobenius symmetry of the finite-field examples), the zero diagonal removes the usual translate ambiguity, and the signs are pinned exactly. For the group-developed objects behind the finite-field examples (Bose's affine relative difference set and its balanced-generalized-weighing-matrix avatars) see Jungnickel–Kharaghani [18].

3.2 Composite common-decomposition multipliers

At composite q no single Frobenius σ_p fixes the prime ideals above all other rational primes dividing q. The part of the Galois group that survives is the intersection of the relevant decomposition groups.

Definition 3.5. Let $q = 2t - 1 = \prod_{p|q} p^{e_p}$ be composite. For each $p \mid q$ write $\langle p \rangle$ for the subgroup generated by p in $(\mathbb{Z}/4t)^\times$, and put

$$M(q) = \bigcap_{p|q} \langle p \rangle \subseteq (\mathbb{Z}/4t)^\times.$$

For $\sigma \in M(q)$ write $\sigma = (\epsilon, u)$, where $\epsilon = 1$ or -1 according as σ is 1 or 3 modulo 4, and $u = \sigma \bmod t \in (\mathbb{Z}/t)^\times$.

Lemma 3.6 (Composite coherence)**.** *Let $\sigma = (\epsilon, u) \in M(q)$ and put*

$$\tau_\sigma = \sum_g \sigma(\tau_g) g^u.$$

Then there are $a \in \mathbb{Z}/4$ and $c \in \mathbb{Z}/t$ such that

$$\tau_\sigma \tau^* = q\, i^a x^c.$$

Proof. For a character χ of order d, put $\alpha = \chi(\tau) \in \mathbb{Z}[\zeta_{4d}]$. Since $\alpha\overline{\alpha} = q$, the ideal (α) is supported only at primes above rational primes dividing q. The image of σ in $(\mathbb{Z}/4d)^\times$ lies in $\langle p \rangle$ for every $p \mid q$, hence in the decomposition group of every prime above p in the abelian field $\mathbb{Q}(\zeta_{4d})$. Thus $\sigma((\alpha)) = (\alpha)$ and $u_\chi = \sigma(\alpha)/\alpha$ is a unit. All conjugates of u_χ have absolute value 1, so Kronecker's theorem makes u_χ a root of unity.

The roots u_χ satisfy the same two coherence inputs used in Lemma 3.3. First, decimation commutes with σ, so for every $m \in (\mathbb{Z}/t)^\times$,

$$\gamma_m(u_\chi) = u_{\chi^m}.$$

Second, with $\beta = \tau_\sigma \tau^*$, the coefficient of y^j in the image of β in $\mathbb{Z}[i][C_e]$ is

$$\pi_e(\beta)_j = \frac{q}{e} \sum_{d|e} i^{a_d} R_d(c_d - j),$$

once the exact-order-d character orbit is written as $u_\psi = i^{a_d}\psi(x^{c_d})$. Since $\pi_e(\beta)_j \in \mathbb{Z}[i]$ and $(q, e) = 1$, the integrality congruence (3) holds for every $e \mid t$ and every j. The Ramanujan-sum induction in the proof of Lemma 3.3 uses only this congruence, the compatibility of the character orbits, and the coefficient ring $\mathbb{Z}[i]$; it does not use that the automorphism came from one prime Frobenius. Hence the a_d and c_d cohere: there are $a \in \mathbb{Z}/4$ and $c \in \mathbb{Z}/t$ with $u_\chi = i^a\chi(x^c)$ for every χ. Fourier inversion then gives $\beta = q\, i^a x^c$, as claimed. □

Theorem 3.7 (Composite multiplier)**.** *Let (r, s) be a Turyn pair of order t and let $q = 2t - 1$ be composite. For every $\sigma = (\epsilon, u) \in M(q)$,*

$$r^{(u)} = \epsilon r, \qquad s^{(u)} = s.$$

Proof. Let $\tau = r + is$ and $\tau_\sigma = \sum_g \sigma(\tau_g) g^u = r^{(u)} + \epsilon i\, s^{(u)}$. By Lemma 3.6, $\tau_\sigma \tau^* = q\, i^a x^c$ for some $a \in \mathbb{Z}/4$ and $c \in \mathbb{Z}/t$. Multiplying by τ and using $\tau\tau^* = q$ gives $\tau_\sigma = i^a x^c \tau$. The identity coefficient of τ_σ has modulus 1, while a nonidentity coefficient of τ has modulus $\sqrt{2}$; hence $c = 0$. Comparing identity coefficients gives $i^a = \epsilon$, and so $\tau_\sigma = \epsilon\tau$. Unpacking the real and imaginary parts gives the two displayed identities. □

Remark 3.8. For $q = p^e$ the group $M(q)$ is just $\langle p \rangle$, so Theorem 3.7 recovers the multiplier part of Theorem 3.1; the prime-power theorem additionally records the trivial-character consequences in the sign-alternating case. For composite q the theorem should be read with the same caution as the prime-power multiplier: it is an unconditional normal-form constraint, while novelty relative to the divisible-difference-set multiplier literature is not audited here.

4 A hierarchy of obstructions at composite q

4.1 The trivial character

At $\chi = 1$, (2) reads $\left(\sum_g r_g\right)^2 + \left(\sum_g s_g\right)^2 = q$.

Proposition 4.1. *If some prime $\ell \equiv 3 \pmod 4$ divides q to an odd power, no Turyn pair of order t exists.*

This elementary sum-of-two-squares test closes 211 of the composite $q \le 2000$.

Lemma 4.2 (Signed branch after two squares)**.** *Suppose the subgroup generated by $M(q)$ and the trivial symmetry contains an element with $\epsilon = -1$. Then every prime divisor of q is congruent to $3 \bmod 4$. Consequently, after Proposition 4.1, the only possible sign-alternating composite branch has*

$$q = n^2, \qquad p \mid n \Rightarrow p \equiv 3 \pmod 4.$$

In particular, every composite q with some prime divisor $p \equiv 1 \pmod 4$ is automatically in the all-plus multiplier regime.

Proof. The trivial symmetry has $\epsilon = +1$, so an $\epsilon = -1$ element in the generated group already comes from $M(q)$. Let $\sigma = (\epsilon, u) \in M(q)$ have $\epsilon = -1$. For every prime $p \mid q$, the definition of $M(q)$ gives $\sigma \in \langle p \rangle \subset (\mathbb{Z}/4t)^\times$, so $\sigma \equiv p^a \pmod{4t}$ for some a. Reducing modulo 4 gives $p^a \equiv 3 \pmod 4$, hence $p \equiv 3 \pmod 4$ and a is odd. Thus all prime divisors of q are 3 mod 4. If a Turyn pair exists, the trivial-character equation makes q a sum of two squares, so Proposition 4.1 forces every exponent of every such prime in q to be even. This is exactly the displayed square branch. □

4.2 The self-conjugacy boundary

The classical refinement is the self-conjugacy (inert-tower) argument of Turyn [2, 3], in the present normal form. It is the class of argument Turyn indicated for composite q in [1].

Proposition 4.3. *Let $q = 2t - 1$ and suppose some prime $\ell \equiv 3 \pmod 4$ has $\ell^k \,\|\, q$ with either* (i) *k odd, or* (ii) *k even and $\mathrm{ord}_d(\ell)$ odd for every divisor $d \mid t$. Then no Turyn pair of order t exists.*

Proof. If k is odd, the trivial character gives a sum of two squares equal to q, impossible (Proposition 4.1). Assume (ii). Let χ have exact order $d \mid t$ and put $\alpha = \chi(\tau) \in K_d = \mathbb{Q}(\zeta_{4d})$. Since $\ell \mid q = 2t - 1$ is odd and coprime to t, $\ell \nmid 4d$ is unramified in K_d. Let $\lambda \mid \ell$ in $\mathbb{Q}(\zeta_d)$; its residue degree is $\mathrm{ord}_d(\ell)$, odd by hypothesis, so the residue field has order $\ell^{\mathrm{ord}_d(\ell)} \equiv 3 \pmod 4$ and lacks a primitive fourth root of unity; hence λ is inert in $K_d = \mathbb{Q}(\zeta_d)(i)$ and every prime $\Lambda \mid \ell$ of K_d satisfies $\Lambda = \overline{\Lambda}$. From $\alpha\overline{\alpha} = q$, $2\nu_\Lambda(\alpha) = \nu_\Lambda(q) = k$, so $\ell^{k/2} \mid \chi(\tau)$ for every χ. Fourier inversion $t\,\tau_g = \sum_\chi \chi(\tau)\chi(g^{-1})$ shows $\ell^{k/2}$ divides $t\,\tau_g$, hence (as $\mathbb{Z}[i]$ is integrally closed and $(\ell, t) = 1$) every τ_g; but $|\tau_g|^2 \in \{1, 2\}$ while $\ell^k \geq 9$, a contradiction. □

This closes $q = 45$ ($t = 23$, $\mathrm{ord}_{23}(3) = 11$ odd), the first composite case beyond two squares, and 12 further $q \leq 2000$.

Remark 4.4 (The blind spot). Both Propositions 4.1 and 4.3 require a prime divisor $\ell \equiv 3 \pmod 4$ in a suitable position—an odd power for the two-squares obstruction, an order-parity condition for self-conjugacy. A composite q survives both exactly when every $\ell \equiv 3 \pmod 4$ dividing it occurs to an even power and the self-conjugacy test does not fire. The simplest such orders have all prime divisors $\equiv 1 \pmod 4$—e.g. $q = 65 = 5 \cdot 13$ ($t = 33$), for which no inert prime appears in any $\mathbb{Q}(\zeta_{4d})$ layer—but the blind spot is larger: of the 111 surviving composite $q \leq 2000$, 80 are all-$\equiv 1 \pmod 4$ and 31 carry a $\equiv 3 \pmod 4$ prime to an even power (e.g. $q = 245 = 5 \cdot 7^2$). These 111 orders are the domain of Sections 4.3–4.9. (Exact quotient descent nonetheless closes $q = 65$ via $\mathbb{Z}[C_{33}] \to \mathbb{Z}[C_{11}]$; the T3 and quotient mechanisms developed next attack this same blind spot by different means.)

4.3 The composite-multiplier reduced decision

Theorem 3.7 also gives a full-group, field-free decision method when the multiplier signs are all positive. Let

$$V = \langle\, u : (1, u) \in M(q),\ -1 \,\rangle \subseteq (\mathbb{Z}/t)^\times.$$

If every element of the multiplier group generated by $M(q)$ and the trivial symmetry (-1) has $\epsilon = +1$, then any Turyn pair is constant on the V-orbits of C_t. The pair equation is equivalently

$$\mathrm{PAF}_r(\delta) + \mathrm{PAF}_s(\delta) = 0 \qquad (\delta \neq 0),$$

and the periodic autocorrelation functions are themselves V-invariant. Thus there is one equation for each nonzero V-orbit of shifts, and the r-side and s-side can be enumerated independently.

Proposition 4.5 (Orbit-reduced decision)**.** *Assume the multiplier signs above are all* $+1$*. Enumerate the possible signs of* r *and* s *on* V*-orbits, with* $r_1 = 0$*, store the vectors* $(\mathrm{PAF}_r(\delta))$ *on nonzero shift-orbit representatives, and meet in the middle against* $-(\mathrm{PAF}_s(\delta))$*. If this finite search has no match, no Turyn pair of order* t *exists. If it has a match, the resulting rows are a Turyn pair exactly when the full integer check* $r^2 + s^2 = q$ *passes.*

Proof. Theorem 3.7 and the symmetry $j \mapsto -j$ force every putative pair into the enumerated V-invariant search space. The equation $r^2 + s^2 = q$ has zero off-identity coefficients precisely when the displayed PAF equalities hold for all nonzero shifts; since PAF is V-invariant, the listed representatives are enough. The final statement is the same identity checked in the full group ring, with exact integer arithmetic. □

The direct sign-reversing kill mechanism does not fire in the current range: the sweep through $q \le 2000$ finds zero multi-prime composite cases surviving the classical filters for which an $\epsilon = -1$ multiplier is present. The reduced all-plus decision is nonetheless strong. Over the 111 self-conjugacy-blind composite $q \le 2000$, it closes 85 cases. Of these, 52 overlap the rigidity scan and 33 are new relative to quotient rigidity; the new list is recorded in Section 5. The closures are field-free and unconditional. The implementation keeps a hard enumeration cap and refuses sign-alternating inputs rather than applying an unsound constant-orbit reduction; Proposition 4.9 is the theorem-level replacement for that guard when an $\epsilon = -1$ branch must be handled.

4.4 Sub-torus marginals and cross-prime gluing

The full T3 decision can be too large when $M(q)$ is small, but a putative pair has smaller marginals at every divisor of t. Let $t' \mid t$, put $h = t/t'$, and let $\pi : \mathbb{Z}[C_t] \to \mathbb{Z}[C_{t'}]$ be induced by $x \mapsto y$. Set $A = \pi(r)$ and $B = \pi(s)$. Then

$$AA^* + BB^* = q \cdot 1 \qquad \text{in } \mathbb{Z}[C_{t'}],$$

where $*$ is inversion in the group ring. The coefficients have forced boxes: the origin coefficient of A is an even sum of $h-1$ signs and has absolute value at most $h-1$, while every other coefficient of A and every coefficient of B is an odd sum of h signs and has absolute value at most h.

Let V be the multiplier group used in Section 4.3, and let V' be its image in $(\mathbb{Z}/t')^\times$. By Theorem 3.7, A and B are constant on the V'-orbits of $C_{t'}$.

Definition 4.6 (Marginal orbit algebra)**.** Let $\Omega = \Omega(t', V')$ be the set of V'-orbits on $\mathbb{Z}/t'$. For $O \in \Omega$ put

$$E_O = \sum_{j \in O} y^j, \qquad \mathcal{A}(t', V') = \mathbb{Z}[C_{t'}]^{V'} = \bigoplus_{O \in \Omega} \mathbb{Z}E_O.$$

For $O, P, Q \in \Omega$, define $N_{O,P}^Q$ to be the number of pairs $(a, b) \in O \times P$ with $a - b = k$ for any fixed $k \in Q$. This number is independent of the choice of $k \in Q$, and

$$E_O E_P^* = \sum_{Q \in \Omega} N_{O,P}^Q E_Q.$$

The marginal equation is therefore a finite quadratic system in the orbit values. Writing

$$A = \sum_{O \in \Omega} a_O E_O, \qquad B = \sum_{O \in \Omega} b_O E_O,$$

we get

$$\sum_{O,P} N^Q_{O,P}(a_O a_P + b_O b_P) = \begin{cases} q, & Q = \{0\}, \\ 0, & Q \neq \{0\}, \end{cases}$$

for each orbit $Q \in \Omega$. Equivalently,

$$\mathrm{PAF}_A(\delta) + \mathrm{PAF}_B(\delta) = 0 \quad (\delta \neq 0), \qquad \sum_j A_j^2 + \sum_j B_j^2 = q,$$

with one PAF equation for each nonzero V'-orbit of shifts. The last equation is automatic at $h = 1$, but is a real counting constraint for proper marginals.

Lemma 4.7 (Marginal algebra lemma). *Assume the composite multiplier signs are all $+1$. Every Turyn pair of order t induces, for every divisor $t' \mid t$, orbit-value vectors $a = (a_O)_{O\in\Omega}$ and $b = (b_O)_{O\in\Omega}$ in $\mathcal{A}(t', V')$ satisfying the coefficient equations above, the trivial character equation*

$$\left(\sum_O |O| a_O\right)^2 + \left(\sum_O |O| b_O\right)^2 = q,$$

and the following boxes and parities: if $O = \{0\}$, then a_O is even and has $|a_O| \leq h-1$; if $O \neq \{0\}$, then a_O is odd and $|a_O| \leq h$; and every b_O is odd with $|b_O| \leq h$.

Proof. The map $\pi : \mathbb{Z}[C_t] \to \mathbb{Z}[C_{t'}]$ sends a Turyn pair to a marginal pair (A, B) satisfying $AA^* + BB^* = q \cdot 1$. Theorem 3.7 and the definition of V' put A, B in $\mathcal{A}(t', V')$, so expanding in the orbit-sum basis and using Definition 4.6 gives the coefficient equations. Evaluating at the trivial character gives the displayed two-square equation. The boxes and parities are exactly the fiber-sum boxes: the origin coefficient of A omits the zero diagonal and sums $h-1$ signs, while all other coefficients sum h signs. ☐

Definition 4.8 (Signed marginal module). Let $\widetilde{M}(q)$ be the subgroup generated by $M(q)$ and the trivial symmetry $(+1, -1)$, and let

$$E' = \{(\epsilon, u \bmod t') : (\epsilon, u) \in \widetilde{M}(q)\} \subseteq \{\pm 1\} \times (\mathbb{Z}/t')^\times.$$

Let H' be its decimation image. Define the signed marginal module

$$\mathcal{S}(t', E') = \{X = \sum_j X_j y^j \in \mathbb{Z}[C_{t'}] : X_{uj} = \epsilon X_j \text{ for all } (\epsilon, u) \in E'\}.$$

Thus a negative stabilizer of j forces $X_j = 0$. When all signs in E' are $+1$, this module is the invariant algebra $\mathcal{A}(t', V')$ above.

Proposition 4.9 (Signed marginal obstruction). *Let $q = 2t - 1$ be composite and let $t' \mid t$. Every Turyn pair induces marginals*

$$A = \pi(r) \in \mathcal{S}(t', E'), \qquad B = \pi(s) \in \mathcal{A}(t', H')$$

satisfying

$$AA^* + BB^* = q \cdot 1, \qquad A(1)^2 + B(1)^2 = q,$$

together with the same fiber boxes and parities as in Lemma 4.7. Consequently, if this finite signed marginal system is infeasible for some $t' \mid t$, then no Turyn pair of order t exists.

Proof. For $(\epsilon, u) \in \widetilde{M}(q)$, Theorem 3.7 and the symmetry $j \mapsto -j$ give $r_{uj} = \epsilon r_j$ and $s_{uj} = s_j$. Fiber summation commutes with decimation because u is prime to t and hence permutes the fibers over $C_{t'}$; therefore the marginals satisfy $A_{ua} = \epsilon A_a$ and $B_{ua} = B_a$. The group-ring equation, the trivial-character equation, and the boxes/parities are exactly the images of the Turyn pair equation and the fiber sums, as in Lemma 4.7. Thus any Turyn pair gives a point of the finite signed system, and emptiness of that system is an obstruction. ☐

Corollary 4.10 (Signed full-torus sign criterion)**.** *Assume the hypotheses of Proposition 4.9 and take the full torus $t' = t$. Then $h = 1$, so the boxes force*

$$A_0 = 0, \quad A_j \in \{\pm 1\} \ (j \neq 0), \qquad B_j \in \{\pm 1\} \quad (0 \leq j < t),$$

with A in the signed module $\mathcal{S}(t, E)$ and B invariant under the decimation image. If the resulting finite signed sign-join

$$AA^* + BB^* = q \cdot 1, \qquad A(1)^2 + B(1)^2 = q$$

is empty, then no Turyn pair of order t exists.

Proof. This is Proposition 4.9 with $h = 1$. The origin coefficient of A is forced to be 0, all other coefficients are signs, and the signed/invariant orbit relations are exactly those defining $\mathcal{S}(t, E)$ and $\mathcal{A}(t, H)$. ☐

Corollary 4.11 (Signed full-unit prime-torus obstruction)**.** *Let t be prime and $t > 5$. Suppose the image of $\widetilde{M}(q)$ in $(\mathbb{Z}/t)^\times$ is all of $(\mathbb{Z}/t)^\times$ and that some element of $\widetilde{M}(q)$ has sign $\epsilon = -1$. Then no Turyn pair of order t exists.*

Proof. If two elements of $\widetilde{M}(q)$ have the same decimation and opposite signs, then the multiplier equations force $r = -r$, impossible because $r_j = \pm 1$ for $j \neq 0$. Thus, for any putative pair, the sign descends to a nontrivial character of $(\mathbb{Z}/t)^\times$. Since $(\mathbb{Z}/t)^\times$ is cyclic, this character is the quadratic character η on $\mathbb{F}_t^\times$; it is even because the trivial symmetry $j \mapsto -j$ has sign $+1$. The multiplier equations force $s_j = s_*$ for all $j \neq 0$, and $r_j = r_1 \eta(j)$ for all $j \neq 0$, with $r_0 = 0$. For any nonzero shift δ, the standard quadratic-character correlation is

$$\sum_{j \in \mathbb{F}_t} \eta(j)\eta(j + \delta) = -1,$$

so $\mathrm{PAF}_r(\delta) = -1$. Writing s_0 for the origin value of s, the constant off-origin row gives

$$\mathrm{PAF}_s(\delta) = (t-2)s_*^2 + 2s_0 s_* = t - 2 + 2s_0 s_*.$$

The Turyn equation would require $0 = \mathrm{PAF}_r(\delta) + \mathrm{PAF}_s(\delta) = t - 3 + 2s_0 s_*$. Since $s_0 s_* \in \{\pm 1\}$ and $t > 5$, this is impossible. ☐

Proposition 4.12 (Sub-torus marginal obstruction)**.** *Assume the composite multiplier signs are all $+1$. If for some divisor $t' \mid t$ the V'-invariant marginal system above has no solution satisfying the stated boxes and parities, then no Turyn pair of order t exists.*

Proof. Every Turyn pair maps to a marginal pair (A, B) with the displayed group-ring identity, boxes, parities, and V'-invariance. If the finite marginal system is infeasible for one t', no such pair can have existed upstairs. ☐

Corollary 4.13 (Exact secondary marginal certificate)**.** *In the setting of Proposition 4.12, suppose the finite orbit-value system of Lemma 4.7 is exhaustively joined over the stated boxes and parities for a divisor $t' \mid t$. If the exact join is empty, then no Turyn pair exists at q, even if some other marginal divisor belongs to a quantified lane that passes locally.*

Proof. The exhaustive join is precisely the finite system whose infeasibility is forbidden by Proposition 4.12. Local feasibility at another divisor does not alter the necessity of the marginal equations at this divisor. □

Proposition 4.14 ($h = 3$ support-augmentation obstruction)**.** *Assume the hypotheses of Proposition 4.12 and let $t' = m$ be a divisor with fiber size $h = t/t' = 3$, so $q = 6m - 1$. Let $\mathcal{O}$ be the set of V'-orbits on C_m, and let $\mathcal{O}_0 = \{0\}$ be the zero orbit. For orbit-constant marginal values A_O, B_O, define*

$$N(A,B) = \sum_{\substack{O\in\mathcal{O}\\ O\neq\mathcal{O}_0}} |O|\,\mathbf{1}_{|A_O|=3} + \sum_{O\in\mathcal{O}} |O|\,\mathbf{1}_{|B_O|=3}.$$

Every marginal solution must satisfy

$$A_{\mathcal{O}_0} \in \{\pm 2\}, \qquad N(A,B) = \frac{m-1}{2}, \qquad A(1)^2 + B(1)^2 = q,$$

where

$$A(1) = \sum_{O\in\mathcal{O}} |O| A_O, \qquad B(1) = \sum_{O\in\mathcal{O}} |O| B_O.$$

Consequently, if the finite support-augmentation join over

$$A_{\mathcal{O}_0} \in \{\pm 2\}, \quad A_O \in \{\pm 1, \pm 3\}\ (O \neq \mathcal{O}_0), \quad B_O \in \{\pm 1, \pm 3\}$$

has no assignment satisfying the two displayed equations, then no Turyn pair exists at q.

Proof. The coefficient at the identity in $AA^* + BB^* = q \cdot 1$ is

$$q = \sum_{j\in C_m} A_j^2 + \sum_{j\in C_m} B_j^2.$$

For $h = 3$, the fiber boxes give $A_{\mathcal{O}_0} \in \{-2, 0, 2\}$, all other A-values are odd of absolute value at most 3, and all B-values are odd of absolute value at most 3. If all odd entries had magnitude 1, the non-origin A entries and all B entries would contribute $(m-1) + m = 2m - 1$. Each entry of magnitude 3 instead of magnitude 1 adds 8. Thus

$$6m - 1 = A_{\mathcal{O}_0}^2 + (2m-1) + 8N(A,B).$$

The case $A_{\mathcal{O}_0} = 0$ would force $N(A,B) = m/2$, impossible because m is odd. Hence $A_{\mathcal{O}_0} = \pm 2$ and $N(A,B) = (m-1)/2$. The trivial-character equation in Lemma 4.7 gives $A(1)^2 + B(1)^2 = q$. These are necessary conditions for the marginal system, so emptiness of this weaker finite join is an obstruction by Proposition 4.12. □

Corollary 4.15 ($h = 3$ support-PAF firing criterion)**.** *In the setting of Proposition 4.14, restrict the finite marginal system to rows satisfying*

$$A_{\mathcal{O}_0} \in \{\pm 2\}, \qquad N(A,B) = \frac{m-1}{2}, \qquad A(1)^2 + B(1)^2 = q.$$

If no such pair also satisfies the full orbit-algebra equation

$$AA^* + BB^* = q \cdot 1$$

in $\mathbb{Z}[C_m]^{V'}$, then no Turyn pair exists at q.

Proof. Proposition 4.14 shows that every marginal solution with $h = 3$ lies in this reduced finite row set. The displayed orbit-algebra equation is the remaining condition of Lemma 4.7. Hence an empty join on the reduced row set is an empty marginal system, and Proposition 4.12 applies. □

At a prime node the support stage collapses to a single congruence.

Corollary 4.16 (Prime support-sum closed form)**.** *In the setting of Proposition 4.14, suppose $t' = \ell$ is prime and let w be the order of the image of V' in $(\mathbb{Z}/\ell)^{\times}$; since that image acts freely on the nonzero residues, every nonzero orbit has size w. The support condition $A_{\mathcal{O}_0} = \pm 2$, $N(A, B) = (\ell - 1)/2$ is satisfiable if and only if*

$$\frac{\ell - 1}{2} \bmod w \in \{0, 1\}.$$

In particular, if $(\ell - 1)/2 \bmod w \notin \{0, 1\}$, the $h = 3$ marginal at ℓ fires and no Turyn pair exists at $q = 6\ell - 1$; and if V_ℓ is transitive ($w = \ell - 1$, $\ell \geq 5$), the condition always fails, so every transitive $h = 3$ node fires.

Proof. All nonzero orbits have size w and the B-origin has size 1, so $N(A, B) = wj + \beta$ with j the number of magnitude-3 nonzero orbits over both sides, $0 \leq j \leq 2(\ell - 1)/w$, and $\beta = \mathbf{1}_{|B_0|=3} \in \{0, 1\}$. Thus $N(A, B) = (\ell - 1)/2$ is solvable iff $(\ell - 1)/2 \equiv \beta \pmod{w}$ for some $\beta \in \{0, 1\}$; the range constraint never binds because $((\ell - 1)/2 - \beta)/w < 2(\ell - 1)/w$. In the transitive case one has $(\ell - 1)/2 \bmod (\ell - 1) = (\ell - 1)/2 \geq 2$ for $\ell \geq 5$. □

The recorded support-sum kills are its instances: $q = 425$ at $\ell = 71$ ($w = 10$, $35 \bmod 10 = 5$) and $q = 1625$ at $\ell = 271$ ($w = 18$, $135 \bmod 18 = 9$), while $q = 1445$ at $\ell = 241$ ($w = 20$, $120 \bmod 20 = 0$) passes the support stage and is closed by the PAF layer (Proposition 4.20); the transitive instances $q = 785$ at 131 and $q = 905$ at 151 fire with no computation. The closed form is machine-checked against the support DP on every prime $h = 3$ node: 19 nodes at $q \leq 2000$ and 71 at $q \leq 10000$, with no disagreement.

Note that the support congruence at prime ℓ fires automatically whenever $d = (\ell - 1)/w$ is odd and $w \geq 4$, since $(\ell - 1)/2 = (w/2)d \equiv w/2 \pmod{w}$. At composite t' the same counting argument gives a weaker congruence.

Corollary 4.17 (Composite support congruence)**.** *In the setting of Proposition 4.14, let g be the greatest common divisor of the nonzero orbit sizes of $\Omega(t', V')$ (computable from Lemma 4.64). Then $N(A, B) \equiv \mathbf{1}_{|B_0|=3} \pmod{g}$, so if $\frac{t'-1}{2} \bmod g \notin \{0, 1\}$, the $h = 3$ marginal at t' fires. This congruence is deliberately coarse: at mixed orbit sizes g is often 2 and the criterion is silent, while the exact subset-sum condition of Proposition 4.14 remains decisive.*

The augmentation stage also has a closed form at prime nodes, with the same representation-pair skeleton as Theorem 4.46: row sums are rigid modulo w, so feasibility is a residue condition on the Gaussian representations of q.

Proposition 4.18 ($h = 3$ augmentation representation criterion)**.** *Assume the setting of Proposition 4.14 with $t' = \ell$ prime, w the order of the image of V' in $(\mathbb{Z}/\ell)^{\times}$, $d = (\ell - 1)/w$, and suppose the support congruence passes: $\beta = \frac{\ell-1}{2} \bmod w \in \{0, 1\}$. Put $J = \left(\frac{\ell-1}{2} - \beta\right)/w$ and $b^* = 3$ if $\beta = 1$, else $b^* = 1$. For $0 \leq j \leq d$ define the exact achievable sum set*

$$\Sigma(d, j) = \left\{\, 3(2k - j) + \big(2l - (d - j)\big) :\ 0 \leq k \leq j,\ 0 \leq l \leq d - j \,\right\}.$$

Then the support-augmentation join of Proposition 4.14 is nonempty if and only if there are a representation $q = X^2 + Y^2$ with X even and Y odd, signs $\epsilon_A, \epsilon_B \in \{\pm 1\}$, and integers $j_A, j_B \geq 0$ with $j_A + j_B = J$, $j_A, j_B \leq d$, such that

$$\epsilon_A X \in 2\{\pm 1\} + w\,\Sigma(d, j_A), \qquad \epsilon_B Y \in b^*\{\pm 1\} + w\,\Sigma(d, j_B).$$

Proof. Every nonzero orbit has size w, so the weighted support is $N(A, B) = w(j_A + j_B) + \mathbf{1}_{|B_0|=3}$, where j_A, j_B count the magnitude-3 orbits per side. The support equation $N(A, B) = \frac{\ell-1}{2}$ forces $\mathbf{1}_{|B_0|=3} = \beta$, hence $|B_0| = b^*$ and $j_A + j_B = J$. A side row with j magnitude-3 orbits, k of them positive, and l positive among the $d - j$ magnitude-1 orbits has orbit-sum contribution exactly $3(2k - j) + (2l - (d - j))$, and every element of $\Sigma(d, j)$ arises this way; thus the achievable row sums are $A(1) \in \pm 2 + w\Sigma(d, j_A)$ and $B(1) \in \pm b^* + w\Sigma(d, j_B)$. The augmentation equation is $A(1)^2 + B(1)^2 = q$ with $A(1)$ even and $B(1)$ odd (as w is even), which is the displayed representation condition up to the signs ϵ_A, ϵ_B. □

Corollary 4.19 ($h = 3$ augmentation residue criterion). *In the setting of Proposition 4.18: if no representation $q = X^2 + Y^2$ with X even and Y odd satisfies*

$$X \equiv \pm 2 \pmod{w} \qquad \textit{and} \qquad Y \equiv \pm b^* \pmod{w},$$

then the $h = 3$ marginal at ℓ fires and no Turyn pair exists at $q = 6\ell - 1$.

Proof. Drop the support coupling in Proposition 4.18: membership in $\pm 2 + w\Sigma(d, j_A)$ forces $X \equiv \pm 2 \pmod{w}$, and likewise for Y. An empty relaxation is an empty join. □

The closed forms are machine-verified against the support-augmentation DP on every evaluable prime $h = 3$ node through $q \leq 10000$: all 71 nodes agree exactly. The criterion fires on 68 of the 71 (61 already by the residue corollary alone); the three survivors passing to the PAF layer are $q = 1445$ at 241 (closed by Proposition 4.20), and, outside the finite ledger, $q = 3845$ at 641 and $q = 3965$ at 661, whose reduced PAF joins (20 and 66 nonzero orbits) exceed present enumeration budgets.

Proposition 4.20 (Closure of $q = 1445$ at $t' = 241$). *Let $q = 1445$, so $t = 723 = 3 \cdot 241$, and take the marginal divisor $t' = 241$ with $h = 3$. The $h = 3$ support-PAF join of Corollary 4.15 is empty. Therefore no Turyn pair exists at $q = 1445$.*

Proof. Here the image $V' \leq (\mathbb{Z}/241\mathbb{Z})^\times$ has order 20, and the nonzero residues split into 12 orbits, each of size 20. The $h = 3$ energy equation forces support $120 = (241 - 1)/2$. The augmentation equation and row-sum congruences force

$$A(1) = \pm 38, \qquad B(1) = \pm 1.$$

The exact reduced join enumerates 2,148,432 admissible A-rows (1,074,216 distinct orbit-PAF side vectors) and 2,257,816 admissible B-rows. No B side vector has a complementary A vector at the required support. Thus the reduced finite system of Corollary 4.15 is empty. □

Proposition 4.21 (Component join criterion). *Assume the composite multiplier signs are all $+1$ and let*

$$\mathbb{Q}[C_{t'}]^{V'} \simeq \prod_{\kappa \in \mathcal{C}} K_\kappa$$

be the rational Wedderburn decomposition of the fixed algebra, with component maps λ_κ. Since $-1 \in V'$, inversion is trivial on the fixed algebra. A V'-invariant marginal solution exists if and only if there are component pairs

$$\lambda_\kappa(A)^2 + \lambda_\kappa(B)^2 = q \qquad (\kappa \in \mathcal{C})$$

whose inverse image in the orbit-sum basis lies in $\mathbb{Z}[C_{t'}]^{V'}$ and satisfies the fiber boxes and parities of Lemma 4.7. For every real embedding $\rho : K_\kappa \hookrightarrow \mathbb{R}$, any such component value obeys $|\rho(\lambda_\kappa(A))|, |\rho(\lambda_\kappa(B))| \leq \sqrt{q}$; hence each component search is finite in the corresponding order. If the finite component join is empty for some t', then no Turyn pair exists at q.

Proof. The semisimple algebra $\mathbb{Q}[C_{t'}]^{V'}$ is the product of the character components fixed by V'. Because $-1 \in V'$, each fixed element is also fixed by inversion, so the image of $AA^* + BB^* = q \cdot 1$ in every component is exactly $\lambda_\kappa(A)^2 + \lambda_\kappa(B)^2 = q$. Conversely, component values that invert to integral orbit coefficients satisfying the boxes/parities give an element of the marginal system. Applying any real embedding to the component equation gives a sum of two real squares equal to q, which gives the displayed bound and finiteness. The final obstruction is Proposition 4.12. □

Corollary 4.22 (Full-torus sign firing criterion)**.** *Assume the hypotheses of Proposition 4.12 and take the full torus $t' = t$. If t is prime and the multiplier signs are all $+1$, then the fiber parameter is $h = 1$ and the marginal boxes force*

$$A_0 = 0, \quad A_j \in \{\pm 1\} \ (j \neq 0), \qquad B_j \in \{\pm 1\} \quad (0 \leq j < t),$$

constant on the V-orbits. Hence full-torus existence is decided by the finite sign join

$$AA^* + BB^* = q \cdot 1, \qquad A(1)^2 + B(1)^2 = q$$

over these orbit-sign rows. If this finite join is empty, then no Turyn pair exists at q.

Proof. For $t' = t$ one has $h = 1$ in the fiber boxes of Lemma 4.7; the A-origin is therefore forced to be 0, every other A coefficient is odd of absolute value at most 1, and every B coefficient is odd of absolute value at most 1. The composite multiplier theorem makes the rows constant on V-orbits. The displayed finite join is precisely the marginal algebra equation and the trivial character equation from Lemma 4.7, so emptiness is an instance of Proposition 4.12. □

The sign join is finite but can be wide. For prime t the trivial character alone already decides the join whenever the multiplier image is large: the row sums of V-invariant sign rows are so rigid that the two-square equation overshoots. This gives the prime-t full-torus branch a zero-computation firing criterion, and, read contrapositively, a square-root bound on the multiplier image of any genuine Turyn pair of prime order.

Theorem 4.23 (Full-torus row-sum bound)**.** *Let $q = 2t - 1$ with $t \geq 3$ prime and all composite multiplier signs $+1$. Let w be the order of the image of V in $(\mathbb{Z}/t)^\times$ (w is even because $-1 \in V$), and let $d = (t-1)/w$. If*

$$(w-1)^2 \geq t, \qquad \textit{equivalently} \qquad w \geq d + 2,$$

then the full-torus sign join of Corollary 4.22 is empty, and no Turyn pair of order t exists. Contrapositively: if a Turyn pair of prime order t exists, then $w < 1 + \sqrt{t}$.

Proof. Since t is prime, V acts freely on the nonzero residues, so there are exactly d nonzero orbits, all of size w, and $wd = t-1$. The stated equivalence is $w \geq d+2 \iff w(w-1) \geq wd + w = t-1+w \iff w^2 - 2w \geq t-1 \iff (w-1)^2 \geq t$.

At $h = 1$ the boxes force sign rows: $A_0 = 0$, A equal to a sign on each nonzero orbit, B a sign everywhere. Hence

$$\lambda_1(A) = w\, s_A, \qquad \lambda_1(B) = \beta + w\, s_B,$$

where s_A and s_B are sums of d signs — so $s_A \equiv s_B \equiv d \pmod 2$ and $|s_A|, |s_B| \leq d$ — and $\beta = B_0 = \pm 1$. The trivial-character equation of Lemma 4.7 reads

$$w^2 s_A^2 + (\beta + w\, s_B)^2 = q = 2wd + 1.$$

Put $\sigma = \beta s_B$, of the same parity and range as s_B. Expanding and dividing by $2w$,

$$w\,(s_A^2 + \sigma^2) = 2(d - \sigma).$$

If d is even, then s_A, σ are even; $(s_A, \sigma) = (0, 0)$ would force $d = 0$, so $s_A^2 + \sigma^2 \geq 4$ and $4w \leq 2(d-\sigma) \leq 4d$, giving $w \leq d$. If d is odd, then s_A, σ are odd, so $s_A^2 + \sigma^2 \geq 1 + \sigma^2$, and $w \geq d+2$ would give

$$2(d-\sigma) \;=\; w\,(s_A^2 + \sigma^2) \;\geq\; (d+2)(1+\sigma^2),$$

i.e. $0 \geq d(\sigma^2 - 1) + 2(\sigma^2 + \sigma + 1)$, false because $\sigma^2 \geq 1$ and $\sigma^2 + \sigma + 1 > 0$. In both parities the equation has no solution when $w \geq d+2$, so the sign join is empty, and Proposition 4.12 applies. ☐

Remark 4.24 (Sharpness and the small-image regime). (i) The bound is sharp. For odd d and $w = d+1$ the displayed equation is solvable: $s_A = \pm 1$, $\sigma = -1$ gives $2w = 2(d+1)$, and then $q = 2wd + 1 = w^2 + (w-1)^2$. Turyn's pair at $q = 25$ realizes the boundary: $t = 13$, the image of $\langle M(25), -1\rangle$ in $(\mathbb{Z}/13)^\times$ has $w = 4$, $d = 3$, and $25 = 4^2 + 3^2$. The row-sum bound thus separates the construction regime from the obstruction regime exactly at $w = d+1$. (ii) The case $d = 1$ recovers Corollary 4.48: $w = t-1$ and $(t-2)^2 \geq t$ for $t \geq 4$. (iii) On the finite panel the theorem closes the entire prime-t full-torus branch with zero computation: all 29 branch members at $q \leq 2000$ satisfy $w \geq d+2$, including the higher-degree joins $q = 697$ ($w = 58$, $d = 6$), $q = 1521$ ($w = 76$, $d = 10$), and $q = 1765$ ($w = 126$, $d = 7$), previously closed by complete integer MITM. (iv) What remains at prime-t top nodes is the small-image regime $w \leq d+1$, i.e. $w < 1 + \sqrt{t}$. This is exactly where Turyn's construction lives ($q = p^e$ makes $\langle p, -1\rangle$ small), so no row-sum-type refinement can close it uniformly. The regime does contain multi-prime composites: the members at $q \leq 30000$ are $q = 5185 = 5\cdot 17\cdot 61$ ($t = 2593$, $w = 8$, $d = 324$), $q = 10045$ ($t = 5023$, $w = 54$), $q = 13833$ ($t = 6917$, $w = 26$), and $q = 15205$ ($t = 7603$, $w = 42$). Thus no uniform composite-q lower bound on the multiplier image can exist. Three of the four fall to the exact form of the trivial-character predicate (Corollary 4.25 below), so the regime demanding genuinely non-multiplier input is thinner than the threshold suggests; its one surviving member in range, $q = 5185$, falls to the fold-tower/stabilizer-amplification mechanism below (GRH-conditional through one class-group computation). Novelty of the row-sum mechanism relative to classical multiplier-size arguments is not audited here; the closures stand regardless.

The proof of Theorem 4.23 in fact establishes a predicate strictly stronger than the threshold: the displayed scalar equation is *necessary* for the sign join, so its exact emptiness fires below the threshold as well.

Corollary 4.25 (Exact trivial-character firing)**.** *In the setting of Theorem 4.23, if the equation*

$$w\,(s^2 + \sigma^2) = 2(d - \sigma)$$

has no integer solution with $s \equiv \sigma \equiv d \pmod 2$ and $|s|, |\sigma| \le d$, then the full-torus sign join is empty and no Turyn pair of order t exists. For $w \ge d+2$ no solution exists for any parameters (Theorem 4.23); below the threshold the exact check is a finite $O(d)$ enumeration and can still fire.

Proof. The equation was derived in the proof of Theorem 4.23 as a consequence of the trivial-character equation on V-invariant sign rows; emptiness of its admissible solution set therefore empties the sign join, and Proposition 4.12 applies. □

The exact form is decisive on the small-image census: of the four members at $q \le 30000$ listed in Remark 4.24(iv), three fire below the threshold — $q = 10045$ ($w = 54$, $d = 93$), $q = 13833$ ($w = 26$, $d = 266$), and $q = 15205$ ($w = 42$, $d = 181$) all have empty admissible solution sets (complete enumeration; each is all-plus and blind-spot, independently re-verified) — while $q = 5185$ genuinely passes, with $(s, \sigma) = (8, 4)$ realizing $5185 = 64^2 + 33^2$. The census extends unconditionally to $q \le 100000$: of 1027 blind-spot prime-t all-plus members, 1011 die by the threshold, 9 more by the exact predicate, and exactly four survive:

$$q = 5185, \quad 37825 = 5^2 \cdot 17 \cdot 89, \quad 62305 = 5 \cdot 17 \cdot 733, \quad 93721 = 17 \cdot 37 \cdot 149,$$

all, curiously, divisible by 17. Two of the four are decided by the fold-tower mechanism below (Propositions 4.28, 4.31); the other two are the sharp wall witnesses of Remark 4.32.

Small-image top nodes: stabilizer amplification. In the small-image regime the multiplier group is too small for any $M(q)$-invariance argument, but it is not static: class-group input can force partial Galois stability of the ideal generated by the character values, and stability converts — by the next lemma — into *honest new multipliers*, re-entering the regime where the row-sum bound fires. Throughout, t is prime, the multiplier signs are all $+1$, $K_V \subset \mathbb{Q}(\zeta_t)$ is the fixed field of V, and $L = K_V(i)$, a CM field with $\mu_L = \langle i \rangle$ and $\mathrm{Gal}(L/\mathbb{Q}(i)) \simeq (\mathbb{Z}/t)^\times/V$, cyclic of order $d = (t-1)/w$. A Turyn pair gives the Weil number $\gamma = \chi(\tau) \in \mathcal{O}_L$ (any character χ of order t), with $\gamma\bar{\gamma} = q$; for squarefree q the ideal (γ) selects exactly one prime from each conjugate pair above each $p \mid q$, a *pattern ideal.*

Lemma 4.26 (Stabilizer rigidity)**.** *Let $t \ge 9$ be prime with all multiplier signs $+1$, and let $\sigma = \sigma_u \in \mathrm{Gal}(L/\mathbb{Q}(i))$ satisfy $\sigma((\gamma)) = (\gamma)$. Then $\sigma(\gamma) = \gamma$, and u is an honest multiplier of the pair: $\tau^{(u)} = \tau$.*

Proof. $\sigma(\gamma)/\gamma$ is an algebraic integer all of whose archimedean absolute values equal 1 (numerator and denominator are Weil q-numbers), hence a root of unity in L (Kronecker), so $\sigma(\gamma) = i^a\gamma$ with $a \in \mathbb{Z}/4$. Applying this identity to every character conjugate and inverting the Fourier transform on C_t gives

$$\tau^{(u)} = i^a\tau + \frac{(1 - i^a)\,\chi_0(\tau)}{t} \sum_{g \in C_t} g.$$

Integrality of the correction requires $t \mid (1 - i^a)\chi_0(\tau)$ in $\mathbb{Z}[i]$. For $a \ne 0$ the norm of the left side is $N(1 - i^a) \cdot q \le 4(2t - 1) < t^2$ for $t \ge 9$, while $(1 - i^a)\chi_0(\tau) \ne 0$ because $\chi_0(\tau)\bar{\chi}_0(\tau) = q \ne 0$; so no such nonzero multiple of t exists, forcing $a = 0$ and $\tau^{(u)} = \tau$. □

Proposition 4.27 (The F_4 class-group cut at $q = 5185$)**.** *Let $q = 5185 = 5 \cdot 17 \cdot 61$, $t = 2593$, $w = 8$. In L (degree 648) the pattern ideal factors as: a pinned Gaussian 61-part ($(5 \pm 6i)\mathcal{O}_L$; one conjugate pair), a 17-pattern on $C_4 = \mathrm{Gal}(L/\mathbb{Q}(i))/D_{17}$ (four pairs), and a 5-pattern on C_{81} (81 pairs). Let $F_4 = k_4(i)$ be the degree-8 fold field, k_4 the quartic subfield of $\mathbb{Q}(\zeta_t)$. Assume the class* *group computed by* `bnfinit` *under GRH:* $\mathrm{Cl}(F_4) = C_{408} \times C_2 \times C_2$ *of order* 1632 *— consistent with the* unconditional *analytic minus class number* $h^-(F_4) = 1632\,Q$ *computed by generalized Bernoulli numbers, which pins* $Q = 1$, $h(k_4) = 1$. *Then: both primes above* 5 *and above* 61 *are principal in* F_4*; the four* 17*-side prime classes* $v_1, \dots, v_4$ *satisfy* $\sum v_j = 0$ *(exactly: their product is* $(1+4i)\mathcal{O}_{F_4}$*, verified as an ideal identity) and all four first coordinates are* $\equiv 1 \pmod 4$*; hence the folded ideal of a pattern with sign set* S *has class* $162 \sum_{j\in S} v_j \equiv 2|S| \pmod 8$ *in the* C_{408} *coordinate, nonzero for every proper nonempty* S*. Consequently every non-constant* 17*-pattern gives a non-principal folded ideal, and the* 17*-pattern of any Turyn pair at* $q = 5185$ *is constant. The order-*8 *character of* C_{408} *is the minimal certifying quotient: an order-*4 *character misses* $|S| = 2$ *and the order-*17 *character misses two pairs.*

Proof. The fold data and the fourteen non-constant folded ideals were computed exactly in PARI/gp and verified two independent ways: the class-linear arithmetic displayed, and direct `bnfisprincipal` on each exact folded ideal $(\prod_{j\in S} P_j)^{162}$ (all non-principal, classes matching). Existence of the pattern ideal downstairs forces the folded ideal principal, which by the above forces $S \in \{\emptyset, \mathrm{all}\}$: a constant pattern. □

Proposition 4.28 (Closure of $q = 5185$, conditional on $\mathrm{Cl}(F_4)$)**.** *Assume the class group of Proposition 4.27. Then no Turyn pair of order* $t = 2593$ *exists.*

Proof. By Proposition 4.27 the 17-pattern is constant, so the stabilizer of (γ) in $\mathrm{Gal}(L/\mathbb{Q}(i)) \simeq C_4 \times C_{81}$ contains $C_4 \times \{0\}$: the 61-part is always stable, the constant 17-part is stable, and $C_4 \times \{0\}$ is exactly the decomposition group D_5, which acts trivially on the 81 pairs above 5 — so the 5-pattern imposes no condition. (This coincidence, D_5 equal to the 17-pattern space, is what makes this top node decidable at one fold level.) By Lemma 4.26, every element of $C_4 \times \{0\}$ is an honest multiplier, so the pair's true multiplier image contains the unique order-32 subgroup of $(\mathbb{Z}/t)^\times$, i.e. its order w' lies in $\{32, 96, 288, 864, 2592\}$. All die: at $w' = 32$ the trivial-character equation reads $1024\, s_A^2 + \lambda_1(B)^2 = 5185$ with s_A odd, impossible ($5185 - 1024 = 4161$ is not a square, and $s_A^2 \geq 9$ overshoots); every $w' \geq 96$ satisfies $(w'-1)^2 \geq t$ and dies by Theorem 4.23. □

Remark 4.29 (Conditionality ledger and the general mechanism)*.* Every link in the $q = 5185$ chain is unconditional exact arithmetic — the pattern normal form, the pinned trivial character $5185 = 64^2 + 33^2$ (the unique representation compatible with the V-invariant row sums, and the unique one divisible by $5 + 6i$), the stabilizer lemma, the exact ideal identities, and the amplification cascade — except the single input $\mathrm{Cl}(F_4)$, which currently rests on `bnfinit` (GRH/Bach). `bnfcertify` passed for the smaller fold field F_3 but extrapolates to weeks for F_4 (discriminant $\approx 7.8 \cdot 10^{22}$); two upgrade paths are documented: the h^- pincer (unconditional $h^- \geq 1632$ plus a flag-1 quotient certification pins $\mathrm{Cl}(F_4)$ exactly) and an explicit octic class-field certificate (necessarily octic: C_8 is the minimal certifying quotient, and $\zeta_8 \notin F_4$ makes it a Kummer-descent construction). Until one lands, $q = 5185$ carries the same GRH-conditional label as the class-group layer of the prime-quotient program, in contrast to the field-free ledger of §7.2. The general lemma to extract, the *fold-tower/stabilizer-amplification dichotomy*, is: if some prime $p \mid q$ has decomposition group equal to the pattern space of another prime p', and the fold level $L^{D_{p'}}$ forces the p'-pattern constant, then the multiplier image amplifies by the full pattern space of p' and the trivial-character/row-sum cascade applies at the amplified level. Quantifying when the fold level fires (here: prime classes $\equiv 1$ mod 4 in the relevant cyclic quotient) is the next theorem obligation for this family; novelty is not yet audited.

The mechanism is now stated once, with its single class-group input isolated as a hypothesis, and it has a second instance.

Proposition 4.30 (Fold-amplification dichotomy). *Let $q = 2t - 1$ be squarefree with every prime divisor $\equiv 1 \bmod 4$, $t \geq 9$ prime, all multiplier signs $+1$, and suppose the exact predicate of Corollary 4.25 is solvable at $w = |V|$. Write $C = \mathrm{Gal}(L/\mathbb{Q}(i)) \simeq (\mathbb{Z}/t)^\times/V$, let $D_p \subseteq C$ be the decomposition group of $p \mid q$ and $X_p = C/D_p$ its pattern space. Fix a target prime $p' \mid q$, put $F = L^{D_{p'}} = k_e(i)$ with $e = |X_{p'}|$, and*

$$U(p') = \bigcap_{p'' \neq p'} D_{p''}.$$

Assume the firing condition *at F: for every non-constant p'-pattern, no choice of the other primes' fold weights makes the pushed-down pattern ideal principal in $\mathrm{Cl}(F)$. Then any Turyn pair of order t has multiplier image containing the subgroup of order $w \cdot |U(p')|$, and if the exact predicate of Corollary 4.25 is empty at every order w' with $w|U(p')| \mid w' \mid t - 1$, no Turyn pair of order t exists. Every step except the evaluation of the firing condition — which requires $\mathrm{Cl}(F)$ and carries that computation's certification tier — is unconditional exact arithmetic.*

Proof. A pair gives the Weil number $\gamma \in \mathcal{O}_L$ with $\gamma\bar{\gamma} = q$; squarefreeness makes $v_P(\gamma) + v_{\bar{P}}(\gamma) = 1$ for every prime $P \mid q$ of L, so (γ) is a pattern ideal. Pushing down by $N_{L/F}$ forces the folded pattern ideal principal in $\mathrm{Cl}(F)$; the p'-pairs fold bijectively, while each other prime contributes a free fold weight. The firing condition therefore forces the p'-pattern constant. A constant p'-pattern is Gal-stable, every $D_{p''}$ acts trivially on its own pattern space, and the 61-type pinned parts are always stable; hence $\mathrm{Stab}_C((\gamma)) \supseteq U(p')$, and Lemma 4.26 converts each stabilizing element into an honest multiplier. The enlarged image has order divisible by $w|U(p')|$, and Corollary 4.25 at the enlarged image gives the final contradiction. □

Proposition 4.31 (Closure of $q = 62305$, conditional on $\mathrm{Cl}(k_3(i))$). *Let $q = 62305 = 5 \cdot 17 \cdot 733$, $t = 31153$, $w = 44$. Take $p' = 5$ ($e = 3$, $F = k_3(i)$ of degree 6, discriminant $t^4 \cdot 4^3 \approx 6 \cdot 10^{19}$) and assume the class group computed by* `bnfinit` *under GRH: $\mathrm{Cl}(F) = C_{5799}$ — matching the* unconditional *analytic minus class number $h^-(F) = 5799$ exactly, which pins $Q = 1$ and $h(k_3) = 1$. Then the rescue-class set generated by the 17- and 733-fold weights has size one, only the two constant 5-patterns are rescued, and the firing condition of Proposition 4.30 holds. With $U = D_{17} \cap D_{733}$ of order 3, the multiplier image contains the order-132 subgroup, and all six admissible amplified orders are empty for the exact predicate. Hence no Turyn pair of order $t = 31153$ exists, conditional exactly on the one* `bnfinit` *input.*

Proof. Fold data (exact): the three conjugate pairs above 5 fold bijectively with exponent 236; 17 contributes weights in $[0, 4]$ with exponent 59; 733 weights in $[0, 59]$ with exponent 4. The rescue computation is exact linear arithmetic in C_{5799}, validated on exact ideals in PARI/gp: the identity $\prod_j P_j = (1 + 2i)\mathcal{O}_F$ holds, direct `bnfisprincipal` on the folded ideal of a non-constant pattern with 15 sampled rescue weights finds no principal instance, and the constant pattern is rescued at weight $(0, 0)$. Lemma 4.26 applies since $4q < t^2$; the six amplified levels are the orders w' with $132 \mid w' \mid 31152$, each checked empty by complete enumeration of the exact predicate. □

Remark 4.32 (The two named gaps, and the wall witnesses). (i) *The Stickelberger gap.* The firing condition cannot yet be stated as arithmetic of (q, t, w, p'): both instances reduce to "the p'-side degree-one prime classes share a nonzero image in a small cyclic quotient of $\mathrm{Cl}^-(k_e(i))$ and the rescue set degenerates," but predicting either fact currently requires the class group per instance. The missing lemma is an explicit Stickelberger/reflection description of the subgroup of $\mathrm{Cl}^-(k_e(i))$ generated by the degree-one primes above p', computable from (t, e, p') alone; until it exists, each instance costs one `bnfinit` and its certification. (ii) *Wall witnesses.* The dichotomy cannot close the other two census survivors. At $q = 37825 = 5^2 \cdot 17 \cdot 89$ every nontrivial-U fold field is infeasible (degrees 48 and 2364), $p' = 5$ has $U = 1$, and $5^2 \parallel q$ needs the valuation-vector generalization of the pattern normal form. At $q = 93721 = 17 \cdot 37 \cdot 149$ ($w = 2$, the minimal possible image) the only near-feasible fold target $p' = 37$ has amplified orders $\{10, 20, 30, 60\}$ that *survive* the exact predicate — so even a perfect class-group cut cannot close it: the dichotomy is provably insufficient there, and a refinement (partial folds at intermediate levels $k_{e'}(i)$, forcing fold weights to extremes) is the recorded direction. These two orders are the correct stress tests for any strengthened small-image criterion.

Proposition 4.33 (Prime sub-torus component criterion)**.** *Assume the composite multiplier signs are all $+1$, let $\ell \mid t$ be prime, and let H be the image of $M(q)$ in $(\mathbb{Z}/\ell\mathbb{Z})^\times$. Put $K = \mathbb{Q}(\zeta_\ell)^H$. Then*

$$\mathbb{Q}[C_\ell]^H \simeq \mathbb{Q} \times K,$$

where the first factor is the trivial component and the second is expressed in the Gaussian-period orbit basis attached to the H-orbits on $(\mathbb{Z}/\ell\mathbb{Z})^\times$. A prime sub-torus marginal solution exists if and only if there are component pairs

$$X_1^2 + Y_1^2 = q, \qquad X_K^2 + Y_K^2 = q \quad (X_K, Y_K \in \mathcal{O}_K)$$

whose inverse period transform gives integral orbit values satisfying the fiber boxes and parities. If this finite component join is empty for this prime divisor ℓ, then no Turyn pair exists at q.

Proof. For prime ℓ there is the rational decomposition $\mathbb{Q}[C_\ell] \simeq \mathbb{Q} \times \mathbb{Q}(\zeta_\ell)$. Taking H-fixed points gives $\mathbb{Q}[C_\ell]^H \simeq \mathbb{Q} \times \mathbb{Q}(\zeta_\ell)^H$. The nontrivial fixed field has degree equal to the number of nonzero H-orbits, and the corresponding period sums give the orbit-basis component transform. Proposition 4.21 then gives the two component equations, the embedding bounds, the inverse integrality condition, and the fiber boxes/parities. An empty component join therefore contradicts Proposition 4.12. □

Proposition 4.34 (Prime sub-torus period transform)**.** *In the setting of Proposition 4.33, write $m = |H|$ and $d = (\ell - 1)/m$, fix a primitive root g modulo ℓ, and let*

$$\eta_k = \sum_{r \in g^k H} \zeta_\ell^r \qquad (0 \le k < d)$$

be the Gaussian periods. Then $\eta_0, \ldots, \eta_{d-1}$ is an integral basis of $\mathcal{O}_K$, each η_k is real, the Galois group of K is cyclic with generator $\sigma\colon \eta_k \mapsto \eta_{k+1 \bmod d}$, and $\eta_0 + \cdots + \eta_{d-1} = -1$. For orbit values $X = x_0 E_{\{0\}} + \sum_k x_k E_{g^k H}$ the two components of Proposition 4.33 are

$$\lambda_1(X) = x_0 + m \sum_k x_k, \qquad \lambda_\ell(X) = \sum_k a_k \eta_k \quad \text{with } a_k = x_k - x_0.$$

Conversely, a pair $(\lambda_1, (a_k)) \in \mathbb{Z} \times \mathbb{Z}^d$ arises from integer orbit values if and only if

$$\lambda_1 \equiv m \sum_k a_k \pmod{\ell},$$

in which case $x_0 = (\lambda_1 - m \sum_k a_k)/\ell$ and $x_k = x_0 + a_k$. The fiber parities are equivalent to: on the A side every a_k is odd and $\lambda_1(A)$ is even; on the B side every a_k is even and $\lambda_1(B)$ is odd. The fiber boxes are $|x_0| \le h - 1$ and $|x_0 + a_k| \le h$.

Proof. Since ℓ is an odd prime, $\mathbb{Q}(\zeta_\ell)/\mathbb{Q}$ is tamely ramified, and the Galois conjugates η_k of the trace $\eta_0 = \mathrm{Tr}_{\mathbb{Q}(\zeta_\ell)/K}(\zeta_\ell)$ form a normal integral basis of K; they are real because $-1 \in H$ makes every coset negation-closed, and their sum is $\sum_{r\neq 0} \zeta_\ell^r = -1$. Evaluating X at the trivial character gives λ_1; evaluating at ζ_ℓ gives $x_0 + \sum_k x_k\eta_k = \sum_k (x_k - x_0)\eta_k$ after substituting $1 = -\sum_k \eta_k$. From $\lambda_1 = x_0 + m\sum_k(x_0 + a_k) = \ell x_0 + m\sum_k a_k$ the inverse formula and the congruence follow. For the parities: on the A side x_0 is even and every x_k is odd, so every a_k is odd, and $m\sum_k x_k \equiv md \equiv 0 \pmod 2$ because $md = \ell - 1$ is even, so $\lambda_1 \equiv x_0 \pmod 2$ is even; the B side is the same computation with every value odd. The boxes restate the fiber boxes of Lemma 4.7. ☐

Corollary 4.35 (Prime sub-torus firing criterion). *Assume the hypotheses of Proposition 4.34. Let $Z(q) = \{(L, M) \in \mathbb{Z}^2 : L^2 + M^2 = q\}$ and let*

$$K_\ell(q) = \Big\{(\alpha, \beta) \in \big(\textstyle\bigoplus_k \mathbb{Z}\eta_k\big)^2 : \alpha^2 + \beta^2 = q\Big\}.$$

Applying each real place to $\alpha^2 + \beta^2 = q$ gives $|\sigma^r(\alpha)|, |\sigma^r(\beta)| \le \sqrt{q}$ for $0 \le r < d$, so $K_\ell(q)$ is finite and computable inside the period coordinate box. Let $\mathcal{S}_A(q, \ell, h)$ be the set of pairs (L, α) with L a first coordinate of $Z(q)$ and $\alpha = \sum_k a_k\eta_k$ a first coordinate of $K_\ell(q)$ with every a_k odd and $L \equiv m\sum_k a_k \pmod \ell$, whose inverse transform under Proposition 4.34 satisfies the A boxes and parities; define $\mathcal{S}_B(q, \ell, h)$ likewise with every a_k even and the B boxes and parities. If $\mathcal{S}_A = \varnothing$, or $\mathcal{S}_B = \varnothing$, or no $(L, \alpha) \in \mathcal{S}_A$ has a mate $(M, \beta) \in \mathcal{S}_B$ with $(L, M) \in Z(q)$ and $(\alpha, \beta) \in K_\ell(q)$, then the prime sub-torus marginal at ℓ is impossible, and hence no Turyn pair exists at q.

Proof. By Proposition 4.34, a marginal solution is exactly a matched pair of component halves whose inverse transforms are integral, boxed, and correctly oriented; the component values of a marginal lie in the period lattice because their coordinates are the integers $x_k - x_0$. An empty join is therefore an empty marginal, and the obstruction is Proposition 4.12. ☐

For quadratic lanes the period solution set has an explicit divisor parametrization. Degree $d = 2$ forces $\ell \equiv 1 \pmod 4$, because H is then the subgroup of squares and $-1 \in H$; the component field is $K = \mathbb{Q}(\sqrt{\ell})$ with $\eta_0, \eta_1 = (-1 \pm \sqrt{\ell})/2$.

Lemma 4.36 (Quadratic two-square parametrization). *Let $\ell \equiv 1 \pmod 4$ be prime and let $q \equiv 1 \pmod 4$. Period coordinate vectors of constant parity are exactly the elements of $\mathbb{Z}[\sqrt{\ell}]$: if $a_0 \equiv a_1 \pmod 2$, then $\alpha = a_0\eta_0 + a_1\eta_1 = U + V\sqrt{\ell}$ with $U = -(a_0 + a_1)/2$ and $V = (a_0 - a_1)/2$, and conversely $a_0 = V - U$, $a_1 = -U - V$. Under this change of coordinates the A-side parity of Corollary 4.35 is $U \not\equiv V \pmod 2$, the B-side parity is $U \equiv V \pmod 2$, and the congruence $L \equiv m\sum_k a_k \pmod \ell$ becomes $L \equiv U \pmod \ell$. Moreover the pairs $(\alpha, \beta) = (U_A + V_A\sqrt{\ell},\ U_B + V_B\sqrt{\ell})$ in $\mathbb{Z}[\sqrt{\ell}]^2$ with $\alpha^2 + \beta^2 = q$ are exactly:*

- *the rational pairs $(V_A, V_B) = (0, 0)$ with $(U_A, U_B) \in Z(q)$; and*
- *for every factorization $q = ef$ in which $e = p^2 + r^2$ is a primitive two-square representation and $f = k^2 + \ell c^2$ with $c \neq 0$, the pairs*

$$(U_A, U_B) = k\,(-r, p), \qquad (V_A, V_B) = c\,(p, r),$$

where (p, r) runs over all primitive representations of e and $k \in \mathbb{Z}$, $c \in \mathbb{Z} \setminus \{0\}$ run over all representations of f.

Proof. The coordinate identities are direct computation, using $m\sum_k a_k = \frac{\ell-1}{2}(-2U) = -(\ell-1)U \equiv U \pmod{\ell}$. Expanding $\alpha^2+\beta^2=q$ in $\mathbb{Z}[\sqrt{\ell}]$ gives the pair of equations

$$U_A^2+U_B^2+\ell(V_A^2+V_B^2)=q, \qquad U_AV_A+U_BV_B=0.$$

Write $z=(U_A,U_B)$ and $w=(V_A,V_B)$ in $\mathbb{Z}^2$. If $w=0$, the equations say $z\in Z(q)$. If $w\neq 0$, write $w=cw_0$ with $w_0=(p,r)$ primitive and $c\neq 0$; the orthogonality $z\cdot w_0=0$ forces $z=k(-r,p)$ for some $k\in\mathbb{Z}$, and substituting gives $(p^2+r^2)(k^2+\ell c^2)=q$. Primitivity of w_0 makes $e=p^2+r^2$ a primitive representation, and every displayed pair conversely satisfies both equations. □

In particular a quadratic prime sub-torus marginal has a component half with $V\neq 0$ on either side only if some divisor of q is represented by $x^2+\ell y^2$ with $y\neq 0$ and cofactor a primitive sum of two squares; otherwise both side sets of Corollary 4.35 consist of rational halves (L,U) with L even and U odd on the A side, L odd and U even on the B side, and $L\equiv U \pmod{\ell}$ on each side. The firing question then reduces to a residue-matching condition modulo ℓ between the even and odd first coordinates of $Z(q)$, together with the fiber boxes. For example, at $q=1189=29\cdot 41$, $\ell=17$: no divisor of q is represented by x^2+17y^2 with $y\neq 0$, the even first coordinates $\{\pm 10,\pm 30\}$ reduce to $\{10,7,13,4\}$ modulo 17, the odd first coordinates $\{\pm 33,\pm 17\}$ reduce to $\{16,1,0\}$, no residue matches, and both side sets are empty: the $t'=17$ marginal fires, reproducing the exhaustive-MITM kill of $q=1189$ by a quantified criterion. The same criterion with the cubic period basis fires at $q=1105$, $\ell=79$. SAT controls exist throughout the lane, both with rational-only halves matched modulo ℓ (for example $q=985$, $\ell=17$) and with genuinely irrational period witnesses (for example $q=441$, $\ell=13$, where $441=(3\sqrt{13})^2+18^2$ enters the join), so the criterion is a firing predicate, not an automatic orbit-shape obstruction.

Proposition 4.37 (C_{27} full-unit component criterion)**.** *Assume $27\mid t$, put $h=t/27$ and $q=54h-1$, and suppose the image V' of the composite multiplier group on C_{27} is $(\mathbb{Z}/27\mathbb{Z})^\times$. Let the four orbits be*

$$O_0=\{0\},\quad O_1=(\mathbb{Z}/27\mathbb{Z})^\times,\quad O_2=3(\mathbb{Z}/9\mathbb{Z})^\times,\quad O_3=9(\mathbb{Z}/3\mathbb{Z})^\times.$$

For $X=x_0E_{O_0}+x_1E_{O_1}+x_2E_{O_2}+x_3E_{O_3}$ the rational components at conductors $1,3,9,27$ are

$$\begin{aligned}
\lambda_1(X)&=x_0+18x_1+6x_2+2x_3,\\
\lambda_3(X)&=x_0-9x_1+6x_2+2x_3,\\
\lambda_9(X)&=x_0-3x_2+2x_3,\\
\lambda_{27}(X)&=x_0-x_3.
\end{aligned}$$

Hence a C_{27} marginal solution exists if and only if there are four Gaussian representations

$$r_d^2+s_d^2=q \qquad (d=1,3,9,27)$$

such that applying the inverse transform

$$\begin{aligned}
x_0&=(u_1+2u_3+6u_9+18u_{27})/27,\\
x_1&=(u_1-u_3)/27,\\
x_2&=(u_1+2u_3-3u_9)/27,\\
x_3&=(u_1+2u_3+6u_9-9u_{27})/27
\end{aligned}$$

to $(u_1,u_3,u_9,u_{27})=(r_1,r_3,r_9,r_{27})$ gives the A-orbit values and applying it to (s_1,s_3,s_9,s_{27}) gives the B-orbit values, with both inverse vectors integral and satisfying the fiber boxes and parities $x_0(A)$

even with $|x_0(A)| \le h-1$, all other A-values odd of absolute value at most h, and all B-values odd of absolute value at most h. If this Gaussian component join is empty, then no Turyn pair exists at q.

Proof. For $V' = (\mathbb{Z}/27\mathbb{Z})^\times$, the fixed algebra has the four rational conductor components displayed above; the period sums are the usual Ramanujan sums for moduli $1, 3, 9, 27$. The displayed inverse matrix is the inverse of this component transform. Proposition 4.21 gives the component equations $r_d^2 + s_d^2 = q$ and says that precisely the integral inverse vectors satisfying the boxes/parities are marginal solutions. The obstruction statement is Proposition 4.12. $\square$

Proposition 4.38 (C_{27} nested congruence criterion)**.** *In the setting of Proposition 4.37, let*

$$G(q) = \{z = a + bi \in \mathbb{Z}[i] : a^2 + b^2 = q\},$$

and let $G_{eo}(q)$ (respectively $G_{oe}(q)$) denote the elements of $G(q)$ with even real part and odd imaginary part (respectively odd real part and even imaginary part). A C_{27} full-unit marginal solution exists if and only if there are

$$z_1 \in G_{eo}(q), \qquad z_3, z_9, z_{27} \in G_{oe}(q)$$

such that, in $\mathbb{Z}[i]$,

$$z_1 \equiv z_3 \pmod{27}, \qquad z_1 \equiv z_9 \pmod 9, \qquad z_1 \equiv z_{27} \pmod 3,$$

and the inverse transform

$$\begin{aligned} w_0 &= (z_1 + 2z_3 + 6z_9 + 18z_{27})/27, \\ w_1 &= (z_1 - z_3)/27, \\ w_2 &= (z_1 + 2z_3 - 3z_9)/27, \\ w_3 &= (z_1 + 2z_3 + 6z_9 - 9z_{27})/27 \end{aligned}$$

has real parts satisfying the A-boxes and imaginary parts satisfying the B-boxes. Equivalently, the C_{27} marginal fires precisely when this oriented nested congruence join is empty after the remaining box cut.

Proof. Write $z_d = r_d + is_d$. Proposition 4.37 says that the component criterion is the inverse transform applied separately to (r_1, r_3, r_9, r_{27}) and (s_1, s_3, s_9, s_{27}). Applying the inverse transform in $\mathbb{Z}[i]$ packages these two real inversions into the displayed w_j.

The inverse is integral if and only if the displayed nested congruences hold. Indeed, $w_1 \in \mathbb{Z}[i]$ is equivalent to $z_1 \equiv z_3 \pmod{27}$; then $w_2 \in \mathbb{Z}[i]$ is equivalent to $z_1 \equiv z_9 \pmod 9$; with those two congruences, $w_3 \in \mathbb{Z}[i]$ is equivalent to $z_1 \equiv z_{27} \pmod 3$. The formula for w_0 is then automatically integral. The orientation condition is exactly the fiber parity condition: $z_1 \in G_{eo}$ and $z_3, z_9, z_{27} \in G_{oe}$ make $\Re(w_0)$ even, the other real parts odd, and all imaginary parts odd; the converse follows from the same parity calculation. The remaining conditions are precisely the fiber boxes of Proposition 4.37. $\square$

Corollary 4.39 (Automatic boxes for large C_{27} marginals)**.** *In Proposition 4.38, if $h \ge 57$, then the box inequalities are automatic. Hence for $h \ge 57$ the C_{27} full-unit marginal exists if and only if the oriented nested congruence join is nonempty.*

Proof. For every $z_d \in G(q)$ we have $|z_d| = \sqrt{q} = \sqrt{54h-1}$. The displayed inverse transform gives

$$|w_0| \le \sqrt{q}, \quad |w_1| \le \frac{2}{27}\sqrt{q}, \quad |w_2| \le \frac{2}{9}\sqrt{q}, \quad |w_3| \le \frac{2}{3}\sqrt{q}.$$

When $h \ge 57$, $\sqrt{54h-1} < h$, so every coordinate of each w_j has absolute value at most h. In the non-vacuous case q is a sum of two squares, so $q \equiv 1 \pmod 4$ and hence h is odd; the even coordinate $\Re(w_0)$ is then strictly smaller than h in absolute value, so $|\Re(w_0)| \le h-1$. These are exactly the A- and B-boxes. □

Corollary 4.40 (C_{27} residue-pair firing criterion). *Let $\overline{G}_{eo}(q)$ and $\overline{G}_{oe}(q)$ be the images of $G_{eo}(q)$ and $G_{oe}(q)$ in $\mathbb{Z}[i]/27\mathbb{Z}[i]$. If*

$$\overline{G}_{eo}(q) \cap \overline{G}_{oe}(q) = \varnothing,$$

then the C_{27} full-unit marginal is impossible, hence no Turyn pair exists whenever this marginal is forced by the multiplier group. Conversely, if $h \ge 57$, then the C_{27} full-unit marginal exists if and only if this intersection is nonempty.

Proof. Any oriented nested congruence tuple in Proposition 4.38 satisfies $z_1 \equiv z_3 \pmod{27}$ with $z_1 \in G_{eo}(q)$ and $z_3 \in G_{oe}(q)$, so it gives a residue in the displayed intersection. Thus an empty intersection kills the marginal before the box cut. Conversely, if $z \in G_{eo}(q)$ and $w \in G_{oe}(q)$ have the same residue modulo 27, set $z_1 = z$ and $z_3 = z_9 = z_{27} = w$. This satisfies all nested congruences and orientations. For $h \ge 57$, Corollary 4.39 makes the box conditions automatic. □

Corollary 4.41 (C_{27} arithmetic firing alternatives). *Assume the hypotheses of Proposition 4.37. Let $\mathcal{J}_{27}(q)$ be the set of oriented nested congruence tuples of Proposition 4.38, and let $\mathcal{B}_{27}(q,h) \subset \mathcal{J}_{27}(q)$ be the subset whose inverse orbit values satisfy the fiber boxes. The C_{27} marginal is impossible, and hence no Turyn pair exists, in either of the following quantified situations:*

$$\overline{G}_{eo}(q) \cap \overline{G}_{oe}(q) = \varnothing, \qquad \text{or} \qquad h < 57 \ \text{ and } \ \mathcal{B}_{27}(q,h) = \varnothing.$$

For $h \ge 57$, the first condition is also necessary for firing within this C_{27} lane: the marginal exists if and only if the displayed residue-pair intersection is nonempty.

Proof. The empty residue-pair condition is Corollary 4.40. If $h < 57$ and $\mathcal{B}_{27}(q,h)$ is empty, then Proposition 4.38 has no boxed inverse tuple and therefore no marginal solution. The large-h equivalence is exactly Corollary 4.39 applied to Corollary 4.40. □

This is the first obstruction in the paper that sees genuinely composite quotients between the prime quotients and the full torus. It closes cases where each prime quotient passes but the data cannot coexist on a larger sub-torus; for example $q = 549$ has lifting prime quotients but is killed at the prime-power marginal $t' = 25$.

The C_{27} criteria are the $(p,e) = (3,3)$ instance of one general statement: at a prime-power node with *full-unit* multiplier image, all conductor components are rational, and the marginal is a join of $e+1$ Gaussian representations glued by nested congruences.

Proposition 4.42 (Full-unit prime-power nested criterion). *Let p be an odd prime, $e \ge 1$, $t' = p^e$, and suppose the multiplier image at t' is the full unit group $V' = (\mathbb{Z}/p^e)^\times$, with all signs $+1$ and $q = 2p^e h - 1$, h odd. The V'-orbits are $\{0\}$ and $O_j = p^j\,(\mathbb{Z}/p^{e-j})^\times$ for $0 \le j \le e-1$; write y_j for the value on O_j and y_e for the origin value. For $0 \le m \le e$, the conductor-p^m component is rational, and*

$$\lambda_{p^m} = y_e + \sum_{j \ge m} \varphi(p^{e-j})\, y_j - p^{e-m}\, y_{m-1} \qquad (\textit{the last term present only for } m \ge 1).$$

The marginal system at t' is solvable if and only if there are Gaussian integers $z_1, z_p, \ldots, z_{p^e}$ with $z_d\bar{z}_d = q$ satisfying the nested congruences

$$z_1 \equiv z_{p^m} \pmod{p^{e+1-m}} \qquad (1 \le m \le e),$$

the orientation $z_1 \in G_{eo}(q)$, $z_{p^m} \in G_{oe}(q)$, and whose inverse orbit values (real parts on the A side, imaginary parts on the B side) satisfy the fiber boxes. The inverse is automatically integral under the congruences, and the orientation is equivalent to the fiber parities.

Proof. For a character χ of conductor p^m and u ranging over $(\mathbb{Z}/p^e)^\times$, the sum of $\chi(p^j u)$-values is $\varphi(p^{e-j})/\varphi(p^{m-j}) \cdot \mu(p^{m-j})$ for $m \ge j$ and $\varphi(p^{e-j})$ for $m \le j$: nonzero only for $m - j \le 1$, giving the displayed evaluation. Taking successive differences, $\lambda_{p^m} - \lambda_{p^{m+1}} = p^{e-m}(y_m - y_{m-1})$ for $0 \le m \le e-1$ (with $y_{-1} := 0$), so integrality of all orbit values is equivalent to the adjacent congruences $z_{p^m} \equiv z_{p^{m+1}} \pmod{p^{e-m}}$, which by transitivity along the chain of moduli are equivalent to the displayed anchored form. For the parities: every $\varphi(p^{e-j})$ is even and every p^{e-m} is odd, so $\lambda_1 \equiv y_e$ and $\lambda_{p^m} \equiv y_e + y_{m-1} \pmod 2$; on the A side (y_e even, other y_j odd) this is the orientation $z_1 \in G_{eo}$, $z_{p^m} \in G_{oe}$, and on the B side (all odd) the complementary parities, exactly as in Proposition 4.38. Feasibility is then Proposition 4.21 for the rational conductor decomposition. □

Corollary 4.43 (Residue-pair collapse at p^e)**.** *If the residue sets of $G_{eo}(q)$ and $G_{oe}(q)$ modulo p^e in $\mathbb{Z}[i]$ are disjoint, the full-unit p^e marginal fires. Conversely, for $h \ge 2p^e + 3$ the boxes are automatic from $|z_d| = \sqrt{q} \le h - 1$, and a common residue gives a marginal witness by taking $z_p = \cdots = z_{p^e}$ in the matching G_{oe} class: the oriented residue join alone decides the node.*

Proof. A common residue is forced by $z_1 \equiv z_p \pmod{p^e}$, giving the firing direction. For the converse, the binding box is the A-side origin: $q = 2p^e h - 1 \le (h-1)^2$ holds iff $h^2 - (2p^e + 2)h + 2 \ge 0$, true for $h \ge 2p^e + 3$ (odd h); the remaining coordinates are bounded by $2\sqrt{q} \cdot p^{j-e} \le h$ already for small h. The construction $z_p = \cdots = z_{p^e}$ satisfies all adjacent congruences. □

Corollary 4.44 (C_{49} full-unit nested criterion)**.** *For $(p, e) = (7, 2)$: with $q = 98h - 1$ and full-unit image at $t' = 49$, the marginal is solvable iff there are z_1, z_7, z_{49} of norm q with $z_1 \equiv z_7 \pmod{49}$, $z_1 \equiv z_{49} \pmod 7$, the orientation $z_1 \in G_{eo}(q)$, $z_7, z_{49} \in G_{oe}(q)$, and boxed inverse*

$$x_0 = \tfrac{1}{49}(\lambda_1 + 6\lambda_7 + 42\lambda_{49}), \quad x_1 = \tfrac{1}{49}(\lambda_1 - \lambda_7), \quad x_2 = \tfrac{1}{49}(\lambda_1 + 6\lambda_7 - 7\lambda_{49}).$$

The residue-pair collapse (Corollary 4.43) holds with modulus 49 and automatic boxes for $h \ge 101$.

The $(7, 2)$ instance closes the two remaining prime-power secondary orders: $q = 685$ ($h = 7$) and $q = 1665$ ($h = 17$) both fire by disjoint residue pairs modulo 49, each confirmed by the exact orbit-value MITM (448×512 and $5{,}508 \times 5{,}832$ assignments); the control sweep over $q = 98k - 1$, k odd, $k \le 301$ finds 22 shape-matching orders of which 21 fire and $q = 23225$ ($h = 237$) is a genuine SAT control in the automatic-box regime. The transform and inverse were machine-derived by exact linear algebra and round-trip verified against the constructed orbit algebras at both orders. Note the full-unit shape does not collide with the $p = 7$ lift lane of Proposition 4.55: $q = 1469$ at $t' = 49$ has $|V'| = 14$, not 42.

Theorem 4.45 (Transitive two-orbit marginal obstruction)**.** *When V' is transitive on the nonzero residues of C_m for a prime $m = t'$, the marginal algebra has the two orbits $\{0\}$ and $C_m \setminus \{0\}$. Writing $A = a_0 + c\sum_{j \ne 0} y^j$ and $B = b_0 + d\sum_{j \ne 0} y^j$, the nonzero PAF constraint is*

$$2a_0c + (m-2)c^2 + 2b_0d + (m-2)d^2 = 0,$$

and the counting equation is

$$a_0^2 + (m-1)c^2 + b_0^2 + (m-1)d^2 = q.$$

If these two equations have no solution with a_0 even, $|a_0| \le h-1$, and c, b_0, d odd with absolute value at most h, then no Turyn pair of order t exists.

Proof. Transitivity gives $A = a_0 + c\sum_{j\neq 0} y^j$ and $B = b_0 + d\sum_{j\neq 0} y^j$. For any nonzero shift, the PAF of the A row has two terms involving the origin and $m-2$ off-origin/off-origin terms, hence $2a_0c + (m-2)c^2$; the B row is identical with (b_0, d). The zero-shift coefficient is the displayed sum of squares. Lemma 4.7 supplies the boxes and parities, and Proposition 4.12 gives the obstruction. □

Seven of the $q \le 2000$ T2 closures below are of this transitive type; they are certified by the two scalar equations above rather than by a general-purpose solver.

The scalar system is small enough to admit a complete arithmetic characterization: feasibility is equivalent to the existence of a pair of integer representations of q with opposite parity orientations, congruent coordinatewise modulo m. This converts the `scalar_unsat` mechanism of the branch accounting from an enumeration outcome into closed-form arithmetic in $(q, M(q))$, and it isolates exactly which two-orbit nodes can never fire.

Theorem 4.46 (Two-orbit representation criterion)**.** *Assume the hypotheses of Theorem 4.45: $m = t'$ is prime, V' is transitive on the nonzero residues, $h = t/m$, and $q = 2mh-1$. The scalar system of Theorem 4.45 has a solution satisfying the boxes and parities if and only if there are integers X, Y, X', Y' with*

$$q = X^2 + Y^2 = X'^2 + Y'^2, \qquad X,\ Y' \text{ even}, \quad Y,\ X' \text{ odd},$$

$$X \equiv X' \pmod{m}, \qquad Y \equiv Y' \pmod{m},$$

whose inverse values

$$a_0 = \frac{X + (m-1)X'}{m}, \quad c = \frac{X - X'}{m}, \quad b_0 = \frac{Y + (m-1)Y'}{m}, \quad d = \frac{Y - Y'}{m}$$

satisfy $|a_0| \le h-1$ and $|c|, |b_0|, |d| \le h$. The parities of (a_0, c, b_0, d) are automatic. Moreover, if $h \ge 2m+2$, the four box conditions are automatic, so the marginal fires if and only if no congruent, oppositely oriented pair of representations exists.

Proof. For prime m with transitive V', the fixed algebra is $\mathbb{Q}[C_m]^{V'} \simeq \mathbb{Q}\times\mathbb{Q}$, with component maps $\lambda_1(A) = a_0 + (m-1)c$ (trivial character) and $\lambda_m(A) = a_0 - c$ (any nontrivial character sends the nonzero-orbit sum to $\mu(m) = -1$), and likewise for B. By Proposition 4.21, the scalar system with its boxes and parities is equivalent to the two component equations $\lambda_\kappa(A)^2 + \lambda_\kappa(B)^2 = q$ together with integrality of the inverse transform, the parities, and the boxes. Set $(X, Y) = (\lambda_1(A), \lambda_1(B))$ and $(X', Y') = (\lambda_m(A), \lambda_m(B))$. Since a_0 is even and c, b_0, d are odd while $m-1$ is even, X and Y' are even and Y and X' are odd. The displayed inverse is the linear inversion of the component transform, and its integrality is exactly the two congruences. Conversely, given the congruences and the orientation, the parities of the inverse are automatic because m is odd: $ma_0 = X + (m-1)X'$ is even and $mc = X - X'$ is odd, so a_0 is even and c odd, and likewise b_0, d are odd.

For the automatic boxes with $h \ge 2m+2$: each component obeys $|\cdot| \le \sqrt{q}$ by the two-square equations, and a_0 is the convex combination $\frac{1}{m}X + \frac{m-1}{m}X'$, so $|a_0| \le \sqrt{q}$; the inequality $\sqrt{q} \le h-1$ is $(h-1)^2 - q = h^2 - 2(m+1)h + 2 \ge 0$, which holds for $h \ge 2m+2$. Similarly $|b_0| \le \sqrt{q} \le h$, and $|c|, |d| \le 2\sqrt{q}/m \le 2(h-1)/m \le h$ for $m \ge 3$. □

Corollary 4.47 (Parity firing for large nodes)**.** *In the setting of Theorem 4.46, if $m^2 > 4q$ — equivalently, using $q = 2mh - 1$, whenever $m \geq 8h$ — then no congruent pair exists and the two-orbit marginal fires unconditionally: any $X \equiv X' \pmod{m}$ with $|X|, |X'| \leq \sqrt{q} < m/2$ forces $X = X'$, contradicting the opposite parities. No enumeration is involved.*

Corollary 4.48 (Transitive full torus)**.** *Let $t \geq 5$ be prime with V transitive on $(\mathbb{Z}/t)^\times$, all multiplier signs $+1$. Then no Turyn pair of order t exists. Indeed at the full torus $h = 1$, so $a_0 = 0$ and $|\lambda_1(A)| = (t-1)|c| = t-1$, while $\lambda_1(A)^2 + \lambda_1(B)^2 = q$ forces $(t-1)^2 \leq 2t-1$, false for $t \geq 5$. (For $t \geq 11$ this is also the parity case $t^2 > 4q = 8t - 4$ of Corollary 4.47.)*

Corollary 4.49 (The C_3 node never fires)**.** *Let $3 \mid t$, $h = t/3 \geq 8$, and let $q = 6h - 1$ be a sum of two squares. Then the C_3 marginal is feasible. Consequently the guaranteed node $t' = 3$ of Theorem 4.66(iii) is never a certifying node for $q \geq 47$; orders with $3 \mid t$ must fire elsewhere, for instance at the $h = 3$ node $t' = t/3$, as at $q = 425$, 1445, 1625.*

Proof. Here $q \equiv 2 \pmod{3}$, so any representation $q = e^2 + o^2$ (e even, o odd) has $e^2 \equiv o^2 \equiv 1 \pmod{3}$, hence $3 \nmid e, o$. Use the same representation on both components: choose signs so that $X = \pm e \equiv X' = \pm o \pmod{3}$ and $Y = \pm o \equiv Y' = \pm e \pmod{3}$, which is possible because $e, o \equiv \pm 1 \pmod{3}$ and the four signs are independent. The boxes are automatic since $h \geq 8 = 2 \cdot 3 + 2$, so Theorem 4.46 gives feasibility. (The excluded cases $h \in \{3, 5, 7\}$ have $q \in \{17, 29, 41\}$, all prime.) $\square$

Remark 4.50 (Arithmetic form, and the audit). In $\mathbb{Z}[i]$ the criterion reads: with $D(q) = \{z : z\bar{z} = q\}$, the node (m, h) in the automatic-box regime is feasible if and only if $z_1 \equiv i\bar{z}_2 \pmod{m\mathbb{Z}[i]}$ for some $z_1, z_2 \in D(q)$ in the even-real orientation — a finite condition on the residues of the Gaussian divisors of q modulo m, computable from the factorization of q in $\mathbb{Z}[i]$. It yields closed forms per m: for $m = 3$ the condition is always satisfiable (Corollary 4.49); for $m = 5$, since $q \equiv 4 \pmod{5}$ forces exactly one of e, o to be divisible by 5 in each representation, the node is feasible if and only if representations of both types occur. The criterion is machine-verified against the exact scalar enumeration on every two-orbit node of every panel order (`-two-orbit-rep-audit`): at $q \leq 2000$, 123 nodes split into 64 feasible, 24 parity fires (Corollary 4.47; 15 of them full-torus), 32 residue fires, and 3 box cuts (65@11, 265@19, 493@19, each with $h < 2m+2$); at $q \leq 10000$, 652 nodes split 346/128/169/9 the same way, with zero disagreements. The feasible nodes are exactly the scalar local-pass rows: the complement that the structural dichotomy must route to other divisors.

Definition 4.51 (Prime-square lift marginal)**.** Let p be an odd prime and suppose $t' = p^2$. A marginal is called a *prime-square $\{\pm 1\}$-lift marginal* if the multiplier image is

$$V' = \{u \in (\mathbb{Z}/p^2)^\times : \ u \equiv \pm 1 \pmod{p}\}.$$

For $1 \leq i \leq (p-1)/2$, put

$$U_i = \{a \in \mathbb{Z}/p^2 : \ p \nmid a, \ a \equiv \pm i \pmod{p}\}, \qquad P_i = \{\pm ip\}.$$

The V'-orbits are $\{0\}$, the unit orbits U_i of size $2p$, and the p-multiple orbits P_i of size 2.

Theorem 4.52 (Prime-square lift marginal certificate)**.** *Assume $q = 2t - 1$, $p^2 \mid t$, $h = t/p^2$, and the marginal at $t' = p^2$ is a prime-square $\{\pm 1\}$-lift marginal. Let E_0, E_i^U, E_i^P be the orbit-sum basis from Definition 4.51. For an orbit-value vector x, let*

$$X = x_0 E_0 + \sum_i x_i^U E_i^U + \sum_i x_i^P E_i^P$$

and define

$$\Phi_p(x) = \big((XX^*)_O\big)_O \oplus X(1)^2,$$

where O ranges over the p orbit classes. If a Turyn pair exists, then there are orbit-value vectors a, b such that

$$\Phi_p(a) + \Phi_p(b) = (q, 0, \dots, 0, q),$$

with a_0 even and $|a_0| \leq h-1$, and all other entries odd with absolute value at most h. Consequently, if this finite quadratic system has no solution, then no Turyn pair exists at q.

Proof. Definition 4.51 gives the orbit-sum basis of $\mathcal{A}(p^2, V')$. Lemma 4.7 supplies the equation $AA^* + BB^* = q \cdot 1$, the trivial-character equation, and the same fiber boxes/parities. These are exactly the displayed finite system. □

For $p = 3$, this theorem is already a three-orbit scalar system. With $x = (x_0, x_U, x_P)$,

$$\Phi_3(x) = \begin{pmatrix} x_0^2 + 6x_U^2 + 2x_P^2 \\ 2x_0x_U + 3x_U^2 + 4x_Ux_P \\ 2x_0x_P + 6x_U^2 + x_P^2 \\ (x_0 + 6x_U + 2x_P)^2 \end{pmatrix}.$$

The orbit pattern alone is not an obstruction: some $p = 3$ lift marginals have solutions. The theorem is therefore a reusable exact certificate criterion; the next closed-form step is to find arithmetic hypotheses on q, h, p forcing the join above to be empty.

Proposition 4.53 (The $p = 3$ Gaussian component criterion)**.** *In the prime-square lift marginal with $t' = 9$, write an orbit-value vector as $x = (x_0, x_U, x_P)$, where x_U is the value on the units and x_P is the value on $\{3, 6\}$. Put*

$$\lambda_1(x) = x_0 + 6x_U + 2x_P, \quad \lambda_3(x) = x_0 - 3x_U + 2x_P, \quad \lambda_9(x) = x_0 - x_P.$$

Then a V'-invariant marginal solution exists if and only if there are three ordered representations

$$X_i^2 + Y_i^2 = q \qquad (i = 1, 3, 9)$$

such that

$$a_0 = \frac{X_1 + 2X_3 + 6X_9}{9}, \quad a_U = \frac{X_1 - X_3}{9}, \quad a_P = \frac{X_1 + 2X_3 - 3X_9}{9},$$
$$b_0 = \frac{Y_1 + 2Y_3 + 6Y_9}{9}, \quad b_U = \frac{Y_1 - Y_3}{9}, \quad b_P = \frac{Y_1 + 2Y_3 - 3Y_9}{9}$$

are integral and satisfy the fiber boxes/parities. In particular, absence of such a compatible triple of Gaussian representations is a field-free firing criterion for the $p = 3$ lift marginal.

Proof. The fixed algebra $\mathbb{Q}[C_9]^{(\mathbb{Z}/9)^\times}$ has three rational character components, represented by the trivial character, a character of conductor 3, and a primitive character of conductor 9. Evaluating $x_0E_0 + x_UE_U + x_PE_P$ at these three components gives the displayed $\lambda_1, \lambda_3, \lambda_9$. Hence $AA^* + BB^* = q \cdot 1$ is equivalent to $\lambda_i(A)^2 + \lambda_i(B)^2 = q$ for $i = 1, 3, 9$. The displayed inverse is the inverse linear transform from component values back to orbit values. The boxes and parities are those of Lemma 4.7. □

This criterion reproduces the $p = 3$ prime-square lift split in the $q \leq 2000$ blind-spot panel: it is infeasible for $q = 485, 1385, 1421, 1745$, while the known SAT controls admit compatible triples.

Proposition 4.54 (The $p = 5$ quadratic-component criterion)**.** *In the prime-square lift marginal with $t' = 25$, write an orbit-value vector as $x = (x_0, x_1, x_2, x_3, x_4)$, where x_1, x_2 are the two unit-orbit values modulo 5 and x_3, x_4 are the values on $\{\pm 5\}$ and $\{\pm 10\}$. Let u satisfy $u^2 + u - 1 = 0$. Put*

$$\begin{aligned}\lambda_1(x) &= x_0 + 10x_1 + 10x_2 + 2x_3 + 2x_4,\\ \lambda_5(x) &= (x_0 + 2x_3 + 2x_4 - 5x_2) + 5(x_1 - x_2)u,\\ \lambda_{25}(x) &= (x_0 - x_4) + (x_3 - x_4)u.\end{aligned}$$

Then a V'-invariant marginal solution exists if and only if there are representations

$$X_1^2 + Y_1^2 = q, \qquad \alpha_5^2 + \beta_5^2 = q, \qquad \alpha_{25}^2 + \beta_{25}^2 = q$$

with $\alpha_5, \beta_5, \alpha_{25}, \beta_{25} \in \mathbb{Z}[u]$, such that the inverse transform of $(X_1, \alpha_5, \alpha_{25})$ and $(Y_1, \beta_5, \beta_{25})$ gives integral orbit values satisfying the fiber boxes/parities. Explicitly, if $\lambda_5 = A + Bu$ and $\lambda_{25} = C + Du$, set

$$\Delta_U = \frac{B}{5}, \qquad \Delta_P = \frac{A - C - 2D}{5}, \qquad x_4 = \frac{\lambda_1 - C - 2D + 20\Delta_P - 10\Delta_U}{25},$$

and then

$$x_3 = D + x_4, \qquad x_0 = C + x_4, \qquad x_2 = x_4 - \Delta_P, \qquad x_1 = x_2 + \Delta_U.$$

The divisibility and box/parity conditions in this inverse transform are therefore an exact field-free firing criterion for the $p = 5$ lift marginal.

Proof. The fixed algebra decomposes into the trivial component and two quadratic components isomorphic to $\mathbb{Q}(u)$: the conductor-5 component and the conductor-25 component. Evaluating the five orbit sums in these three components gives the displayed $\lambda_1, \lambda_5, \lambda_{25}$. Since the involution is trivial on the fixed quadratic components, the marginal equation is equivalent to the three displayed two-square equations. The inverse formulas solve the displayed linear transform. Finally, Lemma 4.7 supplies the fiber boxes and parity restrictions. □

The criterion reproduces the exact MITM obstructions at $q = 549$ and $q = 949$, and it also closes the formerly unresolved $p = 5$ stress instance $q = 1649, t' = 25$. The next member, $p = 7$, is cubic rather than quadratic but has the same component shape.

Proposition 4.55 (The $p = 7$ cubic-component criterion)**.** *In the prime-square lift marginal with $t' = 49$, write an orbit-value vector as*

$$x = (x_0, x_1^U, x_2^U, x_3^U, x_1^P, x_2^P, x_3^P),$$

where x_i^U is the value on residues congruent to $\pm i \bmod 7$ and x_i^P is the value on $\{\pm 7i\}$. Let u satisfy $u^3 + u^2 - 2u - 1 = 0$. Put

$$\begin{aligned}\lambda_1(x) &= x_0 + 14(x_1^U + x_2^U + x_3^U) + 2(x_1^P + x_2^P + x_3^P),\\ \lambda_7(x) &= (x_0 + 2(x_1^P + x_2^P + x_3^P) + 7(-2x_2^U + x_3^U))\\ &\quad + 7(x_1^U - x_3^U)u + 7(x_2^U - x_3^U)u^2,\\ \lambda_{49}(x) &= (x_0 - 2x_2^P + x_3^P) + (x_1^P - x_3^P)u + (x_2^P - x_3^P)u^2.\end{aligned}$$

Then a V'-invariant marginal solution exists if and only if there are representations

$$L^2 + M^2 = q, \qquad \alpha_7^2 + \beta_7^2 = q, \qquad \alpha_{49}^2 + \beta_{49}^2 = q$$

with $\alpha_7, \beta_7, \alpha_{49}, \beta_{49} \in \mathbb{Z}[u]$, such that the inverse transform of $(L, \alpha_7, \alpha_{49})$ and (M, β_7, β_{49}) gives integral orbit values satisfying the fiber boxes/parities. Explicitly, if

$$\alpha_7 = A + Bu + Cu^2, \qquad \alpha_{49} = D + Eu + Fu^2,$$

set

$$\Delta_1 = B/7, \qquad \Delta_2 = C/7,$$
$$R = (A - D - 2E - 4F + 2C)/7, \qquad S = (L - D - 2E - 4F - 2B - 2C)/7,$$

then require these quantities and $x_3^U = (S - R)/7$ to be integral, and put

$$x_3^P = R + x_3^U, \qquad x_1^P = E + x_3^P, \quad x_2^P = F + x_3^P,$$
$$x_0 = D + 2F + x_3^P, \quad x_1^U = x_3^U + \Delta_1, \quad x_2^U = x_3^U + \Delta_2.$$

The same inverse applied to (M, β_7, β_{49}) gives the B row. If no component halves survive these inverse boxes and parities, or no surviving halves have componentwise two-square mates, then the C_{49} marginal is impossible and no Turyn pair exists.

Proof. Let $s_i = \zeta_7^i + \zeta_7^{-i}$. With $u = s_1$, one has $s_2 = u^2 - 2$ and $s_3 = 1 - u - u^2$. In the conductor-7 component, a unit orbit contributes $7s_i$ and a 7-multiple orbit contributes 2; in the conductor-49 component, the unit-orbit sums vanish and the 7-multiple orbits contribute s_i. This gives the displayed $\lambda_1, \lambda_7, \lambda_{49}$. Since the fixed algebra is real, the marginal equation is equivalent to the three displayed two-square component equations. Solving the displayed linear transform gives the inverse formulas, and Lemma 4.7 supplies the boxes and parities. □

Proposition 4.56 (Closure of $q = 1469$ at $t' = 49$)**.** *Let $q = 1469$, so $t = 735 = 3 \cdot 5 \cdot 7^2$, and take the prime-square marginal $t' = 49$ with $h = 15$. The $p = 7$ cubic-component criterion of Proposition 4.55 has no valid side vector. Hence no Turyn pair exists at $q = 1469$.*

Proof. The multiplier image is $V' = \{u \bmod 49 : \ u \equiv \pm 1 \bmod 7\}$, so the marginal is exactly the $p = 7$ prime-square lift. The component enumeration has 16 ordered integer representations of 1469 and 256 ordered representations $\alpha^2 + \beta^2 = 1469$ in $\mathbb{Z}[u]$. Applying the inverse transform of Proposition 4.55 to all component halves yields no admissible A side vector and no admissible B side vector. Thus the component join is empty, and Proposition 4.12 applies. □

Proposition 4.57 (C_{21} quadratic-component criterion)**.** *Assume $21 \mid t$, put $h = t/21$, and suppose the image of the composite multiplier group on C_{21} is*

$$H = \{1, 4, 5, 16, 17, 20\} \subset (\mathbb{Z}/21\mathbb{Z})^\times.$$

Let

$$O_0 = \{0\}, \qquad O_1 = H, \qquad O_2 = 2H = \{2, 8, 10, 11, 13, 19\},$$
$$O_3 = 3(\mathbb{Z}/7\mathbb{Z})^\times = \{3, 6, 9, 12, 15, 18\}, \quad O_4 = \{7, 14\}.$$

Write an orbit-value vector as $x = (x_0, x_1, x_2, x_3, x_4)$ in this order and set $\eta = (1 + \sqrt{21})/2$, so $\eta^2 = \eta + 5$. The conductor components are

$$\lambda_1(x) = x_0 + 6x_1 + 6x_2 + 6x_3 + 2x_4,$$
$$\lambda_3(x) = x_0 - 3x_1 - 3x_2 + 6x_3 - x_4,$$
$$\lambda_7(x) = x_0 - x_1 - x_2 - x_3 + 2x_4,$$
$$\lambda_{21}(x) = (x_0 + x_2 - x_3 - x_4) + (x_1 - x_2)\eta.$$

Consequently a C_{21} marginal solution exists if and only if there are three Gaussian representations

$$L_d^2 + M_d^2 = q \qquad (d = 1, 3, 7)$$

and one representation

$$\alpha^2 + \beta^2 = q, \qquad \alpha, \beta \in \mathbb{Z}[\eta],$$

such that, writing $\alpha = P + Q\eta$, the inverse transform

$$\begin{aligned}
x_0 &= (L_1 + 2L_3 + 6L_7 + 12P + 6Q)/21, \\
x_1 &= (L_1 - L_3 - L_7 + P + 11Q)/21, \\
x_2 &= (L_1 - L_3 - L_7 + P - 10Q)/21, \\
x_3 &= (L_1 + 2L_3 - L_7 - 2P - Q)/21, \\
x_4 &= (L_1 - L_3 + 6L_7 - 6P - 3Q)/21
\end{aligned}$$

gives integral A-orbit values satisfying the A fiber boxes/parities, and the same inverse applied to (M_1, M_3, M_7, R, S), where $\beta = R + S\eta$, gives integral B-orbit values satisfying the B boxes/parities. If this component join is empty, then no Turyn pair exists at q.

Proof. The conductor-1, 3, and 7 components are rational. The displayed $\lambda_1, \lambda_3, \lambda_7$ are the corresponding Ramanujan sums on the five H-orbits. For conductor 21, the primitive residues split into H and its complementary coset. If

$$\eta = \sum_{u \in H} \zeta_{21}^u,$$

then the complementary period is $1 - \eta$, and $\eta(1 - \eta) = -5$; hence $\mathbb{Z}[\eta] = \mathbb{Z}[(1 + \sqrt{21})/2]$ and $\eta^2 = \eta + 5$. Evaluating the five orbit sums at a primitive 21st root therefore gives the displayed λ_{21}.

Since $-1 \in H$, inversion is trivial on the fixed algebra. Thus Proposition 4.21 converts the marginal identity into the three rational two-square equations and the one $\mathbb{Z}[\eta]$ two-square equation. The displayed formulas are the inverse of the component transform. Integral inverse values satisfying the fiber boxes/parities are exactly the marginal solutions of Lemma 4.7; if no such join exists, Proposition 4.12 gives the obstruction. □

Corollary 4.58 (C_{21} arithmetic firing criterion)**.** *Assume the hypotheses of Proposition 4.57. Let $Z(q) = \{(L, M) \in \mathbb{Z}^2 : L^2 + M^2 = q\}$ and let $K_{21}(q) = \{(\alpha, \beta) \in \mathbb{Z}[\eta]^2 : \alpha^2 + \beta^2 = q\}$, with $\eta = (1 + \sqrt{21})/2$. Define $\mathcal{S}_A(q, h)$ to be the set of component halves (L_1, L_3, L_7, α) with each L_i appearing as the first coordinate of some element of $Z(q)$ and α appearing as the first coordinate of some element of $K_{21}(q)$, such that the inverse transform of Proposition 4.57 is integral and satisfies the A boxes and parities. Define $\mathcal{S}_B(q, h)$ in the same way using the B boxes. If $\mathcal{S}_A(q, h) = \varnothing$, or $\mathcal{S}_B(q, h) = \varnothing$, or no element $(L_1, L_3, L_7, \alpha) \in \mathcal{S}_A(q, h)$ has a mate $(M_1, M_3, M_7, \beta) \in \mathcal{S}_B(q, h)$ with*

$$(L_i, M_i) \in Z(q) \quad (i = 1, 3, 7), \qquad (\alpha, \beta) \in K_{21}(q),$$

then the C_{21} marginal is impossible, and hence no Turyn pair exists.

Proof. Proposition 4.57 says that every marginal solution is exactly such a matched pair of component halves whose inverse transforms satisfy the fiber boxes and parities. If one side set is empty, or if the two side sets have no componentwise two-square mate, the component join is empty. The obstruction is then Proposition 4.12. □

This criterion reproduces the compact-middle C_{21} obstructions at 545@21 and 629@21. It also shows why the signature is a criterion rather than an automatic obstruction: examples such as 2225@21 admit boxed component witnesses.

The same template fires at $t' = 57$, where it retires the last recorded exhaustive certificate of the finite panel.

Proposition 4.59 (C_{57} quadratic-component criterion)**.** *Let $q = 114h - 1$ with h odd, and suppose the multiplier image at $t' = 57$ is $V' = \ker(\chi_3\chi_{19})$, the unique index-two subgroup of $(\mathbb{Z}/57)^\times$ containing -1 that surjects onto both $(\mathbb{Z}/3)^\times$ and $(\mathbb{Z}/19)^\times$; the orbits are $\{0\}, H, 3\,(\mathbb{Z}/19)^\times, gH, \{19, 38\}$ of sizes $1, 18, 18, 18, 2$. The fixed algebra is $\mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q}(\sqrt{57})$ with conductors $1, 3, 19, 57$; the conductor-57 component takes values in $\mathbb{Z}[\eta]$, $\eta = (1 + \sqrt{57})/2$, $\eta^2 = \eta + 14$. In orbit order $(x_0, \ldots, x_4)$ the transform is*

$$\begin{aligned}
\lambda_1 &= x_0 + 18x_1 + 18x_2 + 18x_3 + 2x_4,\\
\lambda_3 &= x_0 - 9x_1 + 18x_2 - 9x_3 - x_4,\\
\lambda_{19} &= x_0 - x_1 - x_2 - x_3 + 2x_4,\\
\lambda_{57} &= (x_0 - x_2 + x_3 - x_4) + (x_1 - x_3)\,\eta,
\end{aligned}$$

with inverse of denominator 57 (writing $\lambda_{57} = a + b\eta$)

$$\begin{aligned}
x_0 &= \tfrac{1}{57}(\lambda_1 + 2\lambda_3 + 18\lambda_{19} + 36a + 18b), & x_1 &= \tfrac{1}{57}(\lambda_1 - \lambda_3 - \lambda_{19} + a + 29b),\\
x_2 &= \tfrac{1}{57}(\lambda_1 + 2\lambda_3 - \lambda_{19} - 2a - b), & x_3 &= \tfrac{1}{57}(\lambda_1 - \lambda_3 - \lambda_{19} + a - 28b),\\
x_4 &= \tfrac{1}{57}(\lambda_1 - \lambda_3 + 18\lambda_{19} - 18a - 9b).
\end{aligned}$$

The marginal at $t' = 57$ is solvable if and only if there are three Gaussian representations z_1, z_3, z_{19} of q and one representation $\alpha^2 + \beta^2 = q$ in $\mathbb{Z}[\eta]$ — equivalently, with $\alpha = a + b\eta$, $\beta = c + d\eta$: $a^2 + c^2 + 14(b^2 + d^2) = q$ and $2ab + 2cd + b^2 + d^2 = 0$, completely enumerable inside $|b|, |d| \le \sqrt{q/14}$ — whose inverse transform per side is integral and lands in the fiber boxes, with parities $(\lambda_1, \lambda_3, \lambda_{19}, a, b) \equiv (0, 1, 1, 1, 0)$ on the A side and $(1, 0, 0, 0, 0)$ on the B side modulo 2.

Proof. Identical to Proposition 4.57 with the conductor data $(1, 3, 7, 21)$, $\eta^2 = \eta + 5$ replaced by $(1, 3, 19, 57)$, $\eta^2 = \eta + 14$: the character evaluations give the displayed transform ($\eta_H + \eta_{gH} = \mu(57) = 1$, $\eta_H - \eta_{gH} = \sqrt{57}$ by the even-character Gauss sum), the parity equivalences are exact linear algebra modulo 2, and feasibility is Proposition 4.21. The transform and inverse were machine-derived and round-trip verified against the constructed orbit algebra. □

At $q = 1937$ ($h = 17$) the criterion fires by an empty component join: two valid side vectors on each side and no compatible pair, confirmed by the exact orbit-value MITM ($1{,}784{,}592 \times 1{,}889{,}568$ assignments), matching the previously recorded exhaustive kill. The control sweep over $q = 114k-1$, k odd, $k \le 301$ finds 48 shape-matching orders: 42 fire and six are SAT controls (first $q = 5585$ — which carries both shapes: its C_{49} node fires by Corollary 4.43 while its C_{57} node is solvable). With this promotion every closure in the 21-order T3-unreachable panel is a quantified firing route: no finite certificate remains.

Proposition 4.60 (The $p = 5$ prime-square lift certificate at $q = 549$)**.** *Let $q = 549$, so $t = 275$, and take the prime-square marginal $t' = 25$ with $h = 11$. The induced multiplier group V' has five orbits*

$$\begin{aligned}
O_0 &= \{0\},\\
O_1 &= \{1, 4, 6, 9, 11, 14, 16, 19, 21, 24\},\\
O_2 &= \{2, 3, 7, 8, 12, 13, 17, 18, 22, 23\},\\
O_3 &= \{5, 20\}, \qquad O_4 = \{10, 15\}.
\end{aligned}$$

Write $A = \sum_{i=0}^{4} a_i E_i$ *and* $B = \sum_{i=0}^{4} b_i E_i$. *Put*

$$\begin{aligned}
F_0(x) &= x_0^2 + 10x_1^2 + 10x_2^2 + 2x_3^2 + 2x_4^2, \\
F_1(x) &= 2x_0x_1 + 10x_1x_2 + 4x_1x_3 + 4x_1x_4 + 5x_2^2, \\
F_2(x) &= 2x_0x_2 + 5x_1^2 + 10x_1x_2 + 4x_2x_3 + 4x_2x_4, \\
F_3(x) &= 2x_0x_3 + 10x_1^2 + 10x_2^2 + 2x_3x_4 + x_4^2, \\
F_4(x) &= 2x_0x_4 + 10x_1^2 + 10x_2^2 + x_3^2 + 2x_3x_4,
\end{aligned}$$

and $L(x) = x_0 + 10x_1 + 10x_2 + 2x_3 + 2x_4$. *Then the marginal system is*

$$(F_0, \ldots, F_4, L^2)(a) + (F_0, \ldots, F_4, L^2)(b) = (549, 0, 0, 0, 0, 549),$$

with a_0 *even,* $|a_0| \leq 10$, *and* a_i, b_i *odd with absolute value at most* 11 *otherwise. This finite system has no solution. Hence no Turyn pair exists at* $q = 549$.

Proof. The displayed forms are the coefficient equations of Lemma 4.7 in the five orbit-sum basis. The stated boxes are the $h = 11$ fiber boxes. The exact certificate is a side-separated meet-in-the-middle over the integer orbit values: enumerate all $11 \cdot 12^4 = 228096$ possible a-vectors and all $12^5 = 248832$ possible b-vectors, storing the exact vector $(F_0, \ldots, F_4, L^2)$. No a-side vector and b-side vector sum to $(549, 0, 0, 0, 0, 549)$. By Proposition 4.12, this infeasible marginal rules out a Turyn pair at $q = 549$. □

This five-orbit certificate is the first prime-power marginal model beyond the two-orbit theorem and is the $p = 5$ instance of Theorem 4.52. It shows the shape of the next structural task: prime quotients can lift separately, while a prime-power marginal may compress the problem to a finite algebraic join whose infeasibility must then be proved by additional arithmetic, modular, energy, or box/parity input.

The same component strategy extends beyond prime-power sub-tori. The last open case of the $q \leq 2000$ harvest, $q = 1325$, is decided by the analogous criterion at the composite marginal $t' = 51$, where the two nonrational components live in the real quartic subfield of $\mathbb{Q}(\zeta_{17})$.

Proposition 4.61 (The quartic-component criterion at $t' = 51$). *Let* $q = 2t - 1$ *with* $51 \mid t$, $h = t/51$, *and suppose the multiplier image in* $(\mathbb{Z}/51)^\times$ *is the order-*8 *product group*

$$V' = \{\pm 1 \bmod 3\} \times \{\pm 1, \pm 4 \bmod 17\} = \{1, 4, 13, 16, 35, 38, 47, 50\}.$$

The V'*-orbits on* $\mathbb{Z}/51$ *are* $\{0\}$; *four unit orbits* $U_0, \ldots, U_3$ *of size* 8; *four* 3*-multiple orbits* $T_0, \ldots, T_3$ *of size* 4; *and the* 17*-multiple orbit* $\{17, 34\}$ *of size* 2. *Here* k *indexes the cosets* $3^k\langle 4\rangle$ *of* $\langle 4\rangle = \{1, 4, 13, 16\}$ *in* $(\mathbb{Z}/17)^\times$: *the mod-*17 *image of* U_k *is the coset* $3^k\langle 4\rangle$ *with each residue hit twice, and that of* T_k *is the same coset with each residue hit once. Let* $\eta_k = \sum_{r \in 3^k\langle 4\rangle} \zeta_{17}^r$ *be the quartic Gaussian periods and* $F \subset \mathbb{Q}(\zeta_{17})$ *the real quartic subfield, so that* $\mathcal{O}_F = \mathbb{Z}\eta_0 \oplus \cdots \oplus \mathbb{Z}\eta_3$ *and* $\mathrm{Gal}(F/\mathbb{Q})$ *shifts the index* k *cyclically. For an orbit-value vector* $x = (x_0; u_0, \ldots, u_3; p_0, \ldots, p_3; x_{17})$ *put (indices modulo* 4*)*

$$\begin{aligned}
\lambda_1(x) &= x_0 + 2x_{17} + 8\sum\nolimits_k u_k + 4\sum\nolimits_k p_k, \\
\lambda_3(x) &= x_0 - x_{17} - 4\sum\nolimits_k u_k + 4\sum\nolimits_k p_k, \\
\lambda_{17}(x) &= (x_0 + 2x_{17}) + \sum\nolimits_k (2u_k + p_k)\,\eta_k, \\
\lambda_{51}(x) &= (x_0 - x_{17}) + \sum\nolimits_k (p_k - u_k)\,\eta_{k+3}.
\end{aligned}$$

Then the marginal equation is equivalent to the four two-square equations

$$\lambda_1(a)^2+\lambda_1(b)^2=q, \qquad \lambda_3(a)^2+\lambda_3(b)^2=q \qquad \textit{in } \mathbb{Z},$$

$$\lambda_{17}(a)^2+\lambda_{17}(b)^2=q, \qquad \lambda_{51}(a)^2+\lambda_{51}(b)^2=q \qquad \textit{in } \mathcal{O}_F.$$

Every element occurring in the $\mathcal{O}_F$ equations has all four real embeddings bounded by $\sqrt{q}$, so both $\mathcal{O}_F$ equations have finitely many, exactly enumerable solutions. Writing $\lambda_{17}=\sum_k a_k\eta_k$ and $\lambda_{51}=\sum_k b_k\eta_k$ in the period basis (via $1=-\sum_k \eta_k$), the inverse transform is

$$P=\frac{\lambda_1-4\sum_k a_k}{17}, \qquad C=\frac{\lambda_3-4\sum_k b_k}{17}, \qquad x_{17}=\frac{P-C}{3}, \qquad x_0=x_{17}+C,$$

$$u_k=\frac{(a_k+P)-(b_{k+3}+C)}{3}, \qquad p_k=u_k+b_{k+3}+C.$$

A V'-invariant marginal solution exists if and only if some choice of the four representations makes both inverse transforms integral and satisfy the fiber boxes/parities of Lemma 4.7.

Proof. Since V' is the full product of $\{\pm1\}\subset(\mathbb{Z}/3)^\times$ and $\langle 4\rangle\subset(\mathbb{Z}/17)^\times$, the fixed algebra $\mathbb{Q}[C_{51}]^{V'}$ decomposes by conductor as $\mathbb{Q}\times\mathbb{Q}\times F\times F$: the conductor-3 component is fixed by all of $(\mathbb{Z}/3)^\times$, hence rational; the conductor-17 and conductor-51 components are fixed by $\langle 4\rangle$, hence land in F (for conductor 51, the ζ_3-part of each character sums to $\zeta_3+\zeta_3^2=-1$ over the $\{\pm1 \bmod 3\}$ factor). The dimensions $1+1+4+4$ match the ten orbits, and the periods are an integral basis of $\mathcal{O}_F$ permuted cyclically by $\mathrm{Gal}(F/\mathbb{Q})$.

Evaluating orbit sums: under the conductor-17 character $j\mapsto \zeta_{17}^{j \bmod 17}$, the orbit U_k evaluates to $2\eta_k$, T_k to η_k, and $\{17,34\}$ to 2; under the conductor-51 character $j\mapsto\zeta_{51}^j=\zeta_3^{2j}\zeta_{17}^{6j}$, the ζ_3-factor sums to -1 on U_k and on $\{17,34\}$ and to $+1$ on T_k (whose elements are $0 \bmod 3$), while multiplication by $6\in 3^3\langle 4\rangle$ shifts the coset index by 3; hence $U_k\mapsto-\eta_{k+3}$, $T_k\mapsto\eta_{k+3}$, $\{17,34\}\mapsto -1$. This gives the displayed evaluations. Since $-1\in V'$, every component value is fixed by the involution, so the marginal equation $AA^*+BB^*=q\cdot 1$ is equivalent to the componentwise two-square equations; the embedding bound follows because $\sigma(\lambda(A))^2\le\sigma(\lambda(A))^2+\sigma(\lambda(B))^2=q$ at each real place. For the inverse: with $P=x_0+2x_{17}$ and $C=x_0-x_{17}$, the period coordinates are $a_k=(2u_k+p_k)-P$ and $b_{k+3}=(p_k-u_k)-C$, so $\sum_k a_k=2\sum u_k+\sum p_k-4P$ and $\lambda_1-4\sum_k a_k=17P$; similarly $\lambda_3-4\sum_k b_k=17C$; the per-index 2×2 blocks $(a_k+P,\ b_{k+3}+C)=(2u_k+p_k,\ p_k-u_k)$ invert as displayed. The boxes and parities are those of Lemma 4.7. ☐

Proposition 4.62 (Closure of $q=1325$ at the $t'=51$ marginal)**.** *Let $q=1325=5^2\cdot 53$, $t=663=3\cdot 13\cdot 17$, and take the marginal $t'=51$, $h=13$, whose multiplier image is exactly the V' of Proposition 4.61. Then:* (i) *the ordered integer representations of 1325 as a sum of two squares are the 24 pairs arising from $35^2+10^2=34^2+13^2=29^2+22^2=1325$;* (ii) *the equation $\alpha^2+\beta^2=1325$ has exactly 48 ordered solutions in $\mathcal{O}_F$, all lying in the quadratic subfield $\mathbb{Q}(\sqrt{17})\subset F$: the 24 rational pairs together with the orbit of*

$$1325=30^2+\big(5\sqrt{17}\big)^2=\big(24-3\sqrt{17}\big)^2+\big(18+4\sqrt{17}\big)^2$$

under conjugation, negation, and swap; (iii) *exactly two A-side and two B-side orbit-value vectors pass the inverse transform with the $h=13$ boxes and parities, namely*

$$\pm a^\dagger:\quad x_0=0,\ u_k=p_k=-1,\ x_{17}=13; \qquad \pm b^\dagger:\quad x_0=13,\ u_k=p_k=-1,\ x_{17}=3,$$

with components $(\lambda_1,\lambda_3,\lambda_{17},\lambda_{51})(a^\dagger) = (-22,-13,29,-13)$ *and* $(\lambda_1,\lambda_3,\lambda_{17},\lambda_{51})(b^\dagger) = (-29,10,22,10)$*;*
(iv) *no combination survives: for every sign choice,*

$$\lambda_3(\pm a^\dagger)^2 + \lambda_3(\pm b^\dagger)^2 = 13^2 + 10^2 = 269 \neq 1325.$$

Hence the $t' = 51$ *marginal system is infeasible, and by Proposition 4.12 no Turyn pair exists at* $q = 1325$*.*

Proof. (i) is elementary. For (ii), every solution has all four real embeddings in $[-\sqrt{q},\sqrt{q}]$, hence lies in a finite lattice box in the Minkowski embedding of $\mathcal{O}_F$ (400,753 integer vectors); an exact meet-in-the-middle on the period coordinates of α^2 over this box produces exactly the 48 stated pairs. (The enumeration was reproduced by a structurally independent square-root–recovery method, and the solution set is closed under the Galois action, negation, and swap, as it must be.) For (iii), run the complete join of Proposition 4.61: over the $24 \times 24 \times 48 \times 48$ component choices, the divisibility conditions ($17 \mid \lambda_1 - 4\sum a_k$, $17 \mid \lambda_3 - 4\sum b_k$, the mod-3 block conditions) and the $h = 13$ boxes/parities leave exactly the displayed rows; two independent implementations of the reconstruction agree. Item (iv) is a 2×2 hand check on the λ_3 component. Everything is integer arithmetic in the fixed order $\mathbb{Z}\eta_0 \oplus \cdots \oplus \mathbb{Z}\eta_3$; no number field beyond the explicit quartic periods, no class-group input, and no solver is involved. □

The same criterion applied out of panel at $q = 6425 = 5^2 \cdot 257$ (the next $q \equiv 1 \bmod 4$ with $51 \mid t$, the same V', and a two-square representation) is again infeasible at $t' = 51$, with zero surviving side vectors. As a positive control, relaxing the $q = 1325$ fiber boxes to $|\cdot| \leq 29$ produces eight exact joins, each verified as an identity in the orbit algebra; the $q = 1325$ obstruction is therefore genuinely a fiber-box obstruction, not an artifact of the component compatibility search.

4.5 The divisor lattice and existential selection

The criteria above are stated one divisor at a time. This subsection organizes them over the whole divisor lattice of t and proves the structural layer of the existential divisor-selection problem: marginal feasibility is monotone along the lattice, the orbit model at every divisor is computed by $M(q)$ alone, criterion-equipped divisors always exist, and the top node is not a relaxation but the exact multiplier-reduced decision. The consequence is that, for composite all-plus q, the converse statement "no Turyn pair exists at q" is *equivalent* to the existential statement "some divisor's marginal system is empty." The selection quantifier itself is therefore no longer an obligation; what remains open for the full converse is exactly the per-family firing content, made precise in Remark 4.68 below.

Throughout this subsection $q = 2t - 1$ is composite with t odd, the composite multiplier signs are all $+1$, and for $t' \mid t$ we write $S(t')$ for the finite (possibly empty) solution set of the marginal system of Lemma 4.7 at t': pairs of V'-invariant vectors in $\mathcal{A}(t', V')$ satisfying the orbit-algebra equations, the trivial-character equation, and the fiber boxes and parities for $h = t/t'$.

Lemma 4.63 (Lattice functoriality)**.** *Let* $t'' \mid t' \mid t$ *and let* $\pi : \mathbb{Z}[C_{t'}] \to \mathbb{Z}[C_{t''}]$ *be the natural surjection. Then* $\pi(S(t')) \subseteq S(t'')$*. Consequently*

$$U(q) = \{\, t' \mid t : \; S(t') = \emptyset \,\}$$

is an up-set in the divisor lattice of t*: if* $t'' \in U(q)$ *and* $t'' \mid t' \mid t$*, then* $t' \in U(q)$*.*

Proof. Write $h = t/t'$ and $h' = t'/t''$, so the fiber parameter at t'' is $h'h = t/t''$; both h and h' are odd because t is odd. The map π is a ring homomorphism, so it preserves the equation $AA^* + BB^* = q \cdot 1$, and evaluation at the trivial character is unchanged, so the trivial-character equation is preserved. The image of V' under $(\mathbb{Z}/t')^\times \to (\mathbb{Z}/t'')^\times$ is V'' (both are images of V), so π carries V'-invariance to V''-invariance. For the boxes and parities: an off-origin cell of $\pi(A)$ is a sum of h' cells of A, each odd with absolute value at most h; a sum of h' odd integers is odd, and the bound is $h'h$. The origin cell of $\pi(A)$ is the origin cell of A (even, absolute value at most $h-1$) plus $h'-1$ odd cells, hence even with absolute value at most $(h-1)+(h'-1)h = h'h-1$. The B side is the same computation without the origin exception. These are exactly the boxes and parities of Lemma 4.7 at t'' with fiber parameter $h'h$. The up-set property is the contrapositive: if $S(t') \neq \emptyset$ then $S(t'') \supseteq \pi(S(t')) \neq \emptyset$. ☐

Lemma 4.64 (Orbit census)**.** *For $f \mid t$ let V_f denote the image of V in $(\mathbb{Z}/f)^\times$. Then for every divisor $t' \mid t$:* (i) *the V'-orbit of an element $j \in \mathbb{Z}/t'$ with* $\gcd(j, t') = t'/f$ *has size $|V_f|$, the number of such orbits is $\varphi(f)/|V_f|$, and*

$$\#\Omega(t', V') = \sum_{f \mid t'} \frac{\varphi(f)}{|V_f|};$$

(ii) *the rational Wedderburn decomposition of Proposition 4.21 is indexed by conductors,*

$$\mathbb{Q}[C_{t'}]^{V'} \;\simeq\; \prod_{f \mid t'} \mathbb{Q}(\zeta_f)^{V_f},$$

and the conductor-f component is a totally real field of degree $\varphi(f)/|V_f|$. In particular the fiber parameter, orbit sizes, orbit count, and component degree profile at every divisor of t are determined by t and $M(q)$ alone, before any enumeration.

Proof. (i) The elements of $\mathbb{Z}/t'$ with $\gcd(j, t') = t'/f$ are the $(t'/f)j_0$ with $j_0 \in (\mathbb{Z}/f)^\times$, and $u \in V'$ acts by $j_0 \mapsto uj_0 \bmod f$, i.e., through V_f. The multiplication action of V_f on $(\mathbb{Z}/f)^\times$ is free, so every orbit on this stratum has size $|V_f|$ and there are $\varphi(f)/|V_f|$ of them; summing over $f \mid t'$ exhausts $\mathbb{Z}/t'$. (ii) The characters of $C_{t'}$ of conductor f span a block of $\mathbb{Q}[C_{t'}]$ isomorphic to $\mathbb{Q}(\zeta_f)$, on which V' acts through the Galois action of V_f; the fixed part is $\mathbb{Q}(\zeta_f)^{V_f}$, of degree $\varphi(f)/|V_f|$, totally real because $-1 \in V_f$. The degrees sum to the orbit count of (i), as both equal $\dim_{\mathbb{Q}} \mathbb{Q}[C_{t'}]^{V'}$. ☐

Lemma 4.65 (Exactness at the top node)**.** *$S(t) \neq \emptyset$ if and only if a Turyn pair of order t exists.*

Proof. At $t' = t$ the fiber parameter is $h = 1$: the boxes force the A-origin to be 0 and every other cell of A and every cell of B to be a sign. If a Turyn pair exists, it is itself V-invariant by Theorem 3.7 in the all-plus case (no translate is needed) and symmetric, hence defines an element of $S(t)$. Conversely, an element of $S(t)$ is a pair (A, B) of V-invariant sign rows with zero A-origin; since $-1 \in V$, both rows are symmetric, so $A^* = A$ and $B^* = B$, and the marginal equation reads $A^2 + B^2 = q \cdot 1$: a Turyn pair. Compare Proposition 4.5, of which this is the solution-set form. ☐

Theorem 4.66 (Existential divisor selection)**.** *Let $q = 2t - 1$ be composite with t odd and all composite multiplier signs $+1$. Then:* (i) *no Turyn pair exists at q if and only if $S(t') = \emptyset$ for some divisor $t' \mid t$, if and only if $t \in U(q)$, if and only if $U(q) \neq \emptyset$;* (ii) *$U(q)$ is an up-set, so its minimal elements form an antichain, and certifying $S(t') = \emptyset$ at any single node closes q;* (iii) *the divisors carrying a written finite criterion are nonempty for every t and are located by Lemma 4.64: every prime $\ell \mid t$ carries the period component criterion (Proposition 4.33, Corollary 4.35) of degree $(\ell - 1)/|V_\ell|$; if $3 \mid t$, the node $t' = 3$ is transitive two-orbit (Theorem 4.45, since $V_3 = (\mathbb{Z}/3)^\times$) and* *the node $t' = t/3$ carries the $h = 3$ support hierarchy (Proposition 4.14, Corollary 4.15); and the top node $t' = t$ carries the exact sign decision (Lemma 4.65; for prime t, Corollary 4.22);* (iv) *if t is composite and every proper divisor node is feasible, then $U(q) \subseteq \{t\}$, and nonexistence at q is equivalent to emptiness of the top node alone.*

Proof. (i) If $S(t') = \emptyset$ for some t', then no Turyn pair exists by Proposition 4.12. If no Turyn pair exists, then $S(t) = \emptyset$ by Lemma 4.65, so $t \in U(q)$ and $U(q) \neq \emptyset$. Finally $U(q) \neq \emptyset$ implies, by the up-set property of Lemma 4.63, that $t \in U(q)$, hence $S(t) = \emptyset$ and no pair exists. (ii) is Lemma 4.63. (iii) $V_3 \supseteq \{\pm 1\} = (\mathbb{Z}/3)^\times$, so the $t' = 3$ orbit census is one zero orbit plus one transitive nonzero orbit; the remaining memberships are the stated hypotheses of the cited criteria, all decided by Lemma 4.64. (iv) is (i) together with the up-set property. □

Corollary 4.67 (Projection-image criterion). *Let $t''_1, \ldots, t''_k$ and t' be divisors of t with $t''_i \mid t'$ for each i, and let $R(t') \supseteq S(t')$ be any finite relaxation defined by a subset of the marginal constraints at t' (for instance the boxes, parities, and V'-invariance alone). If*

$$\left\{\, x \in R(t') :\ \pi_i(x) \in S(t''_i) \text{ for } i = 1, \ldots, k \,\right\} = \emptyset,$$

then $S(t') = \emptyset$ and no Turyn pair exists at q.

Proof. $S(t') \subseteq R(t')$, and $\pi_i(S(t')) \subseteq S(t''_i)$ by Lemma 4.63; so $S(t')$ is contained in the displayed set. Emptiness then gives $S(t') = \emptyset$, and Proposition 4.12 applies. □

The $q = 441$ obstruction of §7.2 is the instance $t' = 221$, $\{t''_1, t''_2\} = \{13, 17\}$, with R the per-side boxed V-invariant rows: the C_{13}-projection of any boxed cover row is $\equiv \pm 1 \pmod 8$ on nonzero orbits, while every element of $S(13)$ has nonzero side values ± 3, so already the A-side compatible set is empty. Corollary 4.67 is the general form of that obstruction, available at every node of the lattice against any family of exactly solved quotient nodes.

Remark 4.68 (What the selection theorem does and does not provide). Theorem 4.66 closes the quantification layer of the divisor-selection problem: the correct quantifier is existential over all divisors of t; it is complete (part (i)); the search space it ranges over is monotone (part (ii)), so a nonexistence proof may stop at any certified node; family membership at every node is computed from $M(q)$ by Lemma 4.64 before any enumeration; and criterion-equipped nodes always exist (part (iii)). It does *not* prove that a written criterion fires at some node for every composite q: by part (i), that statement is equivalent to Conjecture 1.3 itself for the all-plus branch. The open content of the full converse is therefore isolated in per-family firing theorems — when is the period join of Corollary 4.35 empty, when does the $h = 3$ hierarchy fire, and their analogues — together with the top-node analysis of part (iv). The finite-panel branch accounting of §7.2 is the instantiation of the theorem at $q \leq 2000$: each closure records one certified minimal element of $U(q)$. Two panel facts are worth recording in lattice terms. First, only 7 of the 21 recent panel closures occur at the divisor preferred by the shape-ranking heuristic: minimal elements of $U(q)$ are not located by orbit shape alone, which is why the quantifier must remain existential. Second, every composite-t case in the panel has a certified *proper* minimal node — the top node was needed only for prime t, where no proper node exists. Thus the family of part (iv), composite t with all proper nodes feasible, has no known member with $q \leq 2000$; whether it is empty is precisely the gluing question, now stated as a property of the lattice rather than of one example.

An explicit lane where the converse holds. The selection theorem and the $h = 3$ closed forms combine into the program's first infinite-family-shaped partial converse: on the lane $q = 6\ell - 1$ with $\ell \equiv 3 \pmod 4$ prime, the firing conditions hold *deterministically* for every composite q with two prime factors — no density or equidistribution input is needed. Three elementary facts collapse the lane.

Lemma 4.69 (Anchor: no ± 1 factors at composite shifted primes)**.** *Let $\ell \geq 3$ be odd and let $q = 6\ell - 1$ be composite. Then no prime divisor of q is congruent to $\pm 1 \pmod \ell$. If moreover ℓ is prime, every prime divisor p of q has $\mathrm{ord}_\ell(p) \geq 3$; and if $\ell \equiv 3 \pmod 4$, the odd part of $\mathrm{ord}_\ell(p)$ is at least 3.*

Proof. An odd $p > 1$ with $p \equiv \pm 1 \pmod \ell$ satisfies $p \geq 2\ell - 1$, because $\ell \pm 1$ are even. Then $c = q/p < (6\ell - 1)/(2\ell - 1) < 7/2$, and c is odd, so $c \in \{1, 3\}$; $c = 3$ contradicts $q \equiv -1 \pmod 3$, and $c = 1$ contradicts compositeness. The order statements follow since $\mathrm{ord}_\ell(p) \leq 2$ means $p \equiv \pm 1 \pmod \ell$, and $v_2(\mathrm{ord}_\ell(p)) \leq v_2(\ell - 1) = 1$ when $\ell \equiv 3 \pmod 4$. □

Lemma 4.70 (Semiprime multiplier image)**.** *Let $\ell \equiv 3 \pmod 4$ be prime and $q = 6\ell - 1 = p_1 p_2$ with $p_1 \neq p_2$ prime, both $\equiv 1 \pmod 4$. Let $x = p_1 \bmod \ell$, n_0 the odd part of $\mathrm{ord}_\ell(x)$. Then $\{p_1, p_2\} \equiv \{1, 5\} \pmod{12}$, the multiplier image satisfies $M(q)|_\ell = \langle x^2 \rangle$, cyclic of odd order $n_0 \geq 3$ (independent of which factor is used), all multiplier signs are $+1$, and*

$$w = \left|V_\ell\right| = 2n_0 \geq 6, \qquad d = \frac{\ell - 1}{w} \text{ odd.}$$

Proof. $q \equiv 5 \pmod{12}$ forces $\{p_1, p_2\} \bmod 12 \in \{\{1, 5\}, \{7, 11\}\}$, and $p_i \equiv 1 \pmod 4$ selects $\{1, 5\}$; all powers of the p_i are $\equiv 1 \pmod 4$, so every multiplier sign is $+1$. Write elements of $(\mathbb{Z}/12\ell)^\times \cong (\mathbb{Z}/4)^\times \times (\mathbb{Z}/3)^\times \times (\mathbb{Z}/\ell)^\times$ as triples and let $y = p_2 \bmod \ell$. With $p_1 \equiv 1$, $p_2 \equiv 5 \pmod{12}$ (the mirror case is symmetric), $\langle p_1 \rangle = \{(1, 1, x^a)\}$ and $\langle p_2 \rangle = \{(1, 2^b, y^b)\}$, so intersecting forces b even and $M(q)|_\ell = \langle x \rangle \cap \langle y^2 \rangle$. From $q \equiv -1 \pmod \ell$ we get $y = -x^{-1}$, hence $\langle y^2 \rangle = \langle x^2 \rangle \subseteq \langle x \rangle$ and $M(q)|_\ell = \langle x^2 \rangle$, of order the odd part n_0 of $\mathrm{ord}_\ell(x)$ (as $v_2(\mathrm{ord}_\ell(x)) \leq 1$); $n_0 \geq 3$ by Lemma 4.69, and $y = -x^{-1}$ gives the same odd part on the other side. Finally $-1 \notin \langle x^2 \rangle$ (odd order), so $w = |\langle x^2, -1 \rangle| = 2n_0$, and $v_2(w) = 1 = v_2(\ell - 1)$ makes d odd. □

Theorem 4.71 (Semiprime partial converse on the lane $q \equiv 5 \pmod 6$)**.** *Let $\ell \equiv 3 \pmod 4$ be prime and let $q = 6\ell - 1$ be composite with $\Omega(q) = 2$. Then no Turyn pair of order $t = 3\ell$ exists.*

Proof. q is not a square: $q \equiv -1 \pmod \ell$ and -1 is not a quadratic residue for $\ell \equiv 3 \pmod 4$. So $q = p_1 p_2$ with distinct primes. If some $p_i \equiv 3 \pmod 4$, both are (mod-12 classification), each to the first power, and Proposition 4.1 applies. Otherwise Lemma 4.70 gives all-plus signs, $w = 2n_0 \geq 6$ and d odd at the divisor $t' = \ell$ ($h = 3$, the node of Theorem 4.66(iii)); then $(\ell - 1)/2 = (w/2)d \equiv w/2 \pmod w$ and $w/2 \geq 3 \notin \{0, 1\}$, so the $h = 3$ support stage fires by Corollary 4.16, and Proposition 4.12 concludes. □

Theorem 4.72 (Two-prime-support extension)**.** *Let $\ell \equiv 3 \pmod 4$ be prime and $q = 6\ell - 1 = p_1^{e_1} p_2^{e_2}$ composite, $p_1 \neq p_2$ prime. Let n_i be the odd part of $\mathrm{ord}_\ell(p_i)$ and $m_i = n_i / \gcd(n_i, \mathrm{oddpart}(e_i))$. If some $p_i \equiv 3 \pmod 4$ has e_i odd, no Turyn pair exists by two squares. Otherwise, if $\max(m_1, m_2) \geq 3$ — in particular whenever some e_i is a power of two — no Turyn pair of order $t = 3\ell$ exists.*

Proof. The signed branch would force q to be a square (Lemma 4.2), excluded as above; so the signs are all $+1$. Squaring $x^{e_1} y^{e_2} \equiv -1 \pmod \ell$ gives $\langle x^{2e_1} \rangle = \langle y^{2e_2} \rangle =: H$; even powers of p_1 and p_2 are trivial mod 12, so $H \subseteq M(q)|_\ell$. The order of H has odd part m_1 (equivalently m_2), so $w \geq 2\max(m_1, m_2) \geq 6$ with $v_2(w) = 1$, and the firing argument of Theorem 4.71 repeats. □

Theorem 4.73 (Conditional infinitude)**.** *Assume Dickson's conjecture [35] for the admissible pair* $\ell(c) = 20c + 11$*,* $v(c) = 24c + 13$*. Then there are infinitely many* c *with both values prime, and for each,* $q = 5v(c) = 6\ell(c) - 1$ *satisfies the hypotheses of Theorem 4.71. Hence, conditionally, Turyn pairs fail to exist for infinitely many composite* q*; the first members are* $q = 65, 185, 785$*.*

Theorem 4.74 (Dichotomy under a congruence-restricted Chen input)**.** *Assume the following analytic hypothesis* (C)*: there are infinitely many primes* $\ell \equiv 3 \pmod 4$ *with* $\Omega(6\ell - 1) \leq 2$*. For every such* ℓ*, the converse Conjecture 1.3 holds at* $q = 6\ell - 1$*: if* q *is prime a Turyn pair exists and the conjecture is vacuous there; if* q *is composite, Theorem 4.71 denies a Turyn pair. Consequently, under* (C)*, at least one of the following is true: (i)* $6\ell - 1$ *is prime for infinitely many primes* $\ell \equiv 3 \pmod 4$*; (ii) there are infinitely many composite* q *admitting no Turyn pair.*

Remark 4.75 (Status of hypothesis (C))*.* The 2026-07-06 literature audit found no retrievable statement covering exactly (C). What is in print: Chen's P_2 bound [34] for the degree-one case $p + 2$; Richert's weighted-sieve theorem giving $f(p) = P_{2 \deg f+1}$ for irreducible f, whose degree-one specialization covers $6p - 1 = P_2$ over *unrestricted* primes p (attribution as summarized by Irving [36]); and the abundance of Chen primes in fixed arithmetic progressions for the form $p+2$ (Lewulis [37]). The missing step is only the residue restriction $\ell \equiv 3 \pmod 4$ on the sieving variable, which the standard weighted-sieve machinery with fixed modulus is expected to absorb routinely; until a derivation appendix or a primary citation is in place, the dichotomy is stated conditionally on (C) rather than attributed to Chen.

Remark 4.76 (The parity wall, and the lane's calibration)*.* Unconditional infinitude of the composite members is a parity-problem instance: "$\Omega(6\ell - 1) = 2$ for infinitely many primes ℓ" is of the same class as "$p + 2$ is a semiprime infinitely often," open despite Chen's theorem, and the standard E_2/GPY technology does not apply because ℓ itself must be prime. For $\omega(q) \geq 3$ the deterministic collapse fails structurally: the product relation $\prod p_i^{e_i} \equiv -1 \pmod \ell$ constrains the join of the cyclic groups $\langle p_i \rangle$, never their intersection. Calibration at $q \leq 10^7$ (62,929 lane primes ℓ): 18.7% of lane members are Theorem 4.71 kills, 18.5% die classically, 17.9% have q prime, and 41.4% sit behind the parity wall — of which 99.98% pass the gcd-of-orders proxy for $w \geq 4$, so the obstruction to closing them is proof technology, not truth. The first Theorem-4.71 members in the blind-spot $\{1, 5\}$ mod 12 class (the two-squares members, e.g. $q = 473, 497$, fall to the classical obstruction) are $q = 65$, 185, 785, 905, 1073, all verified against the exact multiplier computation, and the framework provably cannot collide with Turyn's construction: prime q has its own factor $\equiv -1 \pmod \ell$ ($w = 2$, every criterion silent), and odd prime powers reach a prime ℓ only through the $p = 5$ tower, which lands $\ell \equiv 1 \pmod 4$.

4.6 The lane at general ω: an averaged-equidistribution reduction

Remark 4.76 localizes the failure of Theorem 4.71 at $\omega(q) \geq 3$ and prices unconditional infinitude at a parity problem. This subsection removes the first limitation and reprices the second. At general ω the multiplier image on the lane is still computed by a gcd of orders (Lemma 4.77), the criteria still fire deterministically whenever that image exceeds $\{\pm 1\}$ (Proposition 4.78), and a character-pinning identity (Lemma 4.80) determines the residue status of the one possible prime factor of q above $\sqrt{q}$ from the small factors. The open analytic content is then isolated in an *averaged equidistribution* input of Bombieri–Vinogradov type for cubic Kummer splitting (Hypothesis EQ_3 below) together with a structural control of the factors of $6\ell - 1$ above the sieve level; under the hypothesis alone we prove the small-factor obstruction count (Proposition 4.82). Everything before the hypothesis is unconditional.

Lemma 4.77 (Multiplier image on the lane at general ω)**.** *Let $\ell \equiv 3 \pmod 4$ be prime and let $q = 6\ell - 1 = p_1^{e_1} \cdots p_k^{e_k}$ be composite. Put $n_i = \mathrm{ord}_\ell(p_i)$, $g = \gcd(n_1, \ldots, n_k)$, and let g_0 be the odd part of g. Then the image $M(q)|_\ell$ of $M(q)$ in $(\mathbb{Z}/\ell)^\times$ is cyclic of order g_0 or $2g_0$, and*

$$V_\ell = \big\langle M(q)|_\ell, -1\big\rangle, \qquad w = |V_\ell| = 2g_0, \qquad v_2(w) = 1, \qquad d = \frac{\ell - 1}{w} \text{ odd,}$$

independent of the exponents e_i. Moreover every sign branch reduces to the all-plus branch: $q \equiv 5 \pmod{12}$ is never a square, so Lemma 4.2 excludes the sign-alternating case.

Proof. All prime factors of q are coprime to 12, and $(\mathbb{Z}/12\ell)^\times \cong (\mathbb{Z}/12)^\times \times (\mathbb{Z}/\ell)^\times$; write $x_i = p_i \bmod \ell$. Since $v_2(\ell - 1) = 1$, every n_i has $v_2(n_i) \le 1$, and -1 is the unique involution of the cyclic group $(\mathbb{Z}/\ell)^\times$, contained in every even-order subgroup.

Lower bound. Every odd square is 1 mod 4 and every square prime to 3 is 1 mod 3, so $p_i^2 \equiv 1 \pmod{12}$ and $\langle p_i^2\rangle = \{1\} \times \langle x_i^2\rangle$. Hence $K := \bigcap_i \langle p_i^2\rangle \subseteq M(q)$ projects isomorphically to $\bigcap_i \langle x_i^2\rangle$. The order of x_i^2 is the odd part of n_i (as $v_2(n_i) \le 1$); in a cyclic group an intersection of subgroups has order the gcd of their orders, and odd part commutes with gcd, so $|K| = \gcd_i \mathrm{oddpart}(n_i) = g_0$. As g_0 is odd, $-1 \notin K|_\ell$, and $w \ge 2g_0$.

Upper bound. Projecting, $M(q)|_\ell \subseteq \bigcap_i \langle x_i\rangle = C_g$, the unique subgroup of order g. If g is odd then $|\langle C_g, -1\rangle| = 2g = 2g_0$; if g is even then $v_2(g) = 1$, so C_g already contains the involution and $\langle C_g, -1\rangle = C_g$ has order $g = 2g_0$. Either way $w \le 2g_0$; the 2-part of $M(q)|_\ell$ itself never needs to be resolved.

Exponent-independence is definitional ($M(q) = \bigcap_{p|q}\langle p\rangle$ sees only the distinct primes), and $w = 2g_0 \mid \ell - 1$ with g_0 odd makes $d = (\ell - 1)/(2g_0)$ odd. ☐

For $k = 2$ the formula specializes to Lemma 4.70 ($g_0 = n_0$). Notably, the proof never uses the product relation $\prod_i p_i^{e_i} \equiv -1 \pmod \ell$ that drove the semiprime argument: both bounds are relation-free. The relation returns as the central actor in Lemma 4.80.

Proposition 4.78 (Deterministic kill at general ω)**.** *Let $\ell \equiv 3 \pmod 4$ be prime and $q = 6\ell - 1$ composite. If $g_0 = \mathrm{oddpart}\big(\gcd_{p|q} \mathrm{ord}_\ell(p)\big) \ge 3$, then no Turyn pair exists at q. Equivalently: a Turyn pair at composite $q = 6\ell - 1$ forces $w = 2$, i.e. the multiplier image in $(\mathbb{Z}/\ell)^\times$ collapses to $\{\pm 1\}$.*

Proof. Lemma 4.77 gives $w = 2g_0 \ge 6$ and d odd, so $(\ell - 1)/2 = (w/2)\, d \equiv w/2 \pmod w$ with $w/2 \ge 3$, hence $(\ell - 1)/2 \bmod w \notin \{0, 1\}$ and the prime support-sum closed form (Corollary 4.16) fires at the node $t' = \ell$ of $t = 3\ell$; Proposition 4.12 concludes. ☐

Lemma 4.79 (Residue reformulation of the bad set)**.** *With ℓ, q as above and, for an odd prime $r \mid \ell - 1$, write $S_r \le (\mathbb{Z}/\ell)^\times$ for the subgroup of rth powers.*

1. *If some odd prime $r \mid \ell - 1$ has $p \bmod \ell \notin S_r$ for every $p \mid q$, then $g_0 \ge r \ge 3$ and Proposition 4.78 kills q.*

2. *If $g_0 = 1$, then for every odd prime $r \mid \ell - 1$ some $p \mid q$ has $p \bmod \ell \in S_r$.*

Proof. S_r is the unique subgroup of order $(\ell - 1)/r$, so $x \in S_r \iff \mathrm{ord}(x) \mid (\ell - 1)/r$. If $x \notin S_r$ then $r \mid \mathrm{ord}(x)$, giving (i); if $r \nmid \mathrm{ord}(x)$ then $\mathrm{ord}(x)$ divides $(\ell - 1)/r^{v_r(\ell - 1)}$, which divides $(\ell - 1)/r$, so $x \in S_r$, giving (ii). ☐

Lemma 4.80 (Character pinning). *Let $\ell \equiv 3 \pmod 4$ be prime, $q = 6\ell - 1$ composite, $r \mid \ell - 1$ an odd prime, and χ a character of $(\mathbb{Z}/\ell)^\times$ of order r. Then $\chi(-1) = 1$, so $q \equiv -1 \pmod \ell$ gives $\prod_{p^e \| q} \chi(p)^e = 1$. Consequently, writing $q = mP^\epsilon$ with P the (at most one, automatically first-power) prime factor exceeding $\sqrt{q}$ and m the $\sqrt{q}$-smooth cofactor,*

$$\epsilon = 1 \implies \chi(P) = \prod_{p^e \| m} \chi(p)^{-e} :$$

the residue status of P at every odd prime $r \mid \ell - 1$ is determined by the factors of q below $\sqrt{q}$.

Proof. $\chi(-1)^2 = 1$ and $\chi(-1) \in \mu_r$ with r odd force $\chi(-1) = 1$; apply χ to $q \equiv -1 \pmod \ell$. Two prime factors $> \sqrt{q}$ would multiply past q. Finally $P \notin S_r \iff \chi(P) \neq 1$, since $S_r = \ker \chi$. □

The lane's ℓ-dependent prime factors are the reason no fixed Chebotarev statement applies directly: the splitting field of "p is an rth power residue mod ℓ" varies with $p \mid 6\ell - 1$. Lemma 4.80 removes the one factor that cannot be averaged over (the large one); for the small factors, the right input is equidistribution *on average over the base*. We restrict to the sub-lane $\ell \equiv 7 \pmod{12}$ (equivalently $3 \mid \ell - 1$), where $r = 3$ is always available.

Definition 4.81 (Cubic Kummer average). For squarefree $d = p_1 \cdots p_r$ with $(d, 6) = 1$ let

$$M_d = \mathbb{Q}\big(\zeta_3,\ p_1^{1/3}, \ldots, p_r^{1/3},\ \zeta_{12d}\big), \qquad G_d = \mathrm{Gal}(M_d/\mathbb{Q}),$$

and let $C_d \subseteq G_d$ be the union of conjugacy classes cut out by the conditions: $\ell \equiv 7 \pmod{12}$, and for each i, $\ell \equiv 6^{-1} \pmod{p_i}$ and p_i a cubic residue mod ℓ (for $\ell \equiv 1$ mod 3 the latter says ℓ splits completely in $\mathbb{Q}(\zeta_3, p_i^{1/3})$). Its density is an *exact* product,

$$\delta_d := \frac{|C_d|}{|G_d|} = \frac{1}{4} \prod_{i=1}^{r} \frac{1}{3\,\varphi(p_i)},$$

because the Kummer layers are independent ($\mathrm{Gal}(\mathbb{Q}(\zeta_3, p_1^{1/3}, \ldots, p_r^{1/3})/\mathbb{Q}(\zeta_3)) \cong (\mathbb{Z}/3)^r$: no product $\prod p_i^{a_i}$, $a_i \in \{0, 1, 2\}$ not all zero, is a cube) and no cubic Kummer subfield is abelian over $\mathbb{Q}$, so the Kummer part is linearly disjoint from $\mathbb{Q}(\zeta_{12d})$ over $\mathbb{Q}(\zeta_3)$. For fixed $\theta \in (0, \frac{1}{2})$, *Hypothesis* $\mathrm{EQ}_3(\theta)$ asserts: for every $A > 0$,

$$\sum_{\substack{d \le x^{2\theta} \\ d \text{ squarefree},\ (d,6)=1}} \Big| \pi_{C_d}(x; M_d) - \delta_d \,\mathrm{Li}(x) \Big| \ll_A \frac{x}{(\log x)^A}.$$

(The variant with prescribed cubic-symbol values at each p_i, needed for character-twisted sums, is defined analogously; the two nontrivial symbol values are exchanged by conjugation in each S_3 layer, so only their real combinations are class functions, which is all such sums use.)

Proposition 4.82 (Small-factor obstruction count). *Fix $\theta \in (0, \frac{1}{2})$ and assume $\mathrm{EQ}_3(\theta)$. There is $c(\theta) > 0$ such that, for all large x, the number of primes $\ell \le x$ with $\ell \equiv 7 \pmod{12}$, $q = 6\ell - 1$ composite, and every prime $p \le x^\theta$ dividing q a cubic non-residue mod ℓ, is at least $c(\theta)\, x/(\log x)^{4/3}$.*

Proof. Let $z = x^\theta$, $D = z^2 = x^{2\theta}$, and let $\mathcal{A}$ be the set of primes $\ell \le x$ with $\ell \equiv 7 \pmod{12}$, of size $X = (\frac{1}{4} + o(1))\, x/\log x$ (prime number theorem in the fixed progression). For a prime $p \le z$,

$(p,6)=1$, put $E_p=\{\ell\in\mathcal{A}:\ p\mid 6\ell-1 \text{ and } p \text{ a cubic residue mod } \ell\}$; note $2,3\nmid q$ automatically. For squarefree d composed of such primes,

$$A_d := \Big|\bigcap_{p\mid d} E_p\Big| = \pi_{C_d}(x;M_d) + O\big(\omega(d)\big) = g(d)\,X + r_d, \qquad g(d)=\prod_{p\mid d}\frac{1+o(1)}{3\varphi(p)},$$

with the product density exact (Definition 4.81) and $\sum_{d\le D}|r_d| \ll_A x(\log x)^{-A}$ by $\mathrm{EQ}_3(\theta)$ (the $O(\omega(d))$ terms contribute $O(D\log D)$, absorbed). The density has sieve dimension $\kappa=\frac{1}{3}$, since $g(p)=\frac{1}{3}\cdot\frac{1+o(1)}{p-1}$ on the sieving primes. The beta-sieve lower bound of dimension $\frac{1}{3}$ at $s=\log D/\log z=2$ applies — the sieving limit satisfies $\beta(\frac{1}{3})\le\beta(\frac{1}{2})=1<2$, the half-dimensional value (see [30, Ch. 11], [29]) — and gives

$$\#\{\ell\in\mathcal{A}:\ \ell\notin E_p\ \forall p\le z\} \ \ge\ \big(f_{1/3}(2)-o(1)\big)\,X\prod_{p\le z}\big(1-g(p)\big)\ -\ \sum_{d\le D}|r_d|,$$

with $f_{1/3}(2)>0$. By the Mertens-type theorem for sieve densities of dimension κ ([29]), $\prod_{p\le z}(1-g(p))\asymp_\theta(\log x)^{-1/3}$, so the main term is $\gg_\theta x/(\log x)^{4/3}$ and the remainder is negligible. A survivor ℓ has every prime factor $p\le z$ of q a cubic non-residue; survivors with q prime number $O(x/\log^2 x)=o\big(x/(\log x)^{4/3}\big)$ (upper-bound sieve for the pair $(\ell,6\ell-1)$, [29]) and are removed. □

Remark 4.83 (From the small factors to a family: status). Three calibrations. *(i) What is not yet proved.* Members of the set in Proposition 4.82 are not certified nonexistence orders: q may have prime factors in $(x^\theta,\sqrt{q}\,]$ about which the proposition says nothing, and only the at-most-one factor above $\sqrt{q}$ is pinned by Lemma 4.80. Converting the count into an infinite family of Proposition 4.78 kills requires exactly (a) a structure count — sub-lane primes with $6\ell-1=(x^\theta\text{-smooth})\cdot(\text{prime})$ — which the linear sieve at Bombieri–Vinogradov level does not reach (sifting the middle range to $\sqrt{6x}$ at level $x^{1/2-\varepsilon}$ has $s=1$, below the linear sieving limit; per fixed smooth cofactor the count is a two-linear-forms prime pair, a parity-type problem), and (b) a joint (vector-sieve) average running the Kummer conditions relative to the structured set, which EQ_3 as stated does not supply. A 2026-07-06 bookkeeping attempt at (a)+(b) was retracted on verification, gapped at both points; the almost-all version of the statement (nonexistence at all but $o(x/\log x)$ of the composite lane members, via Lemma 4.79(ii) at moderately large $r\mid\ell-1$) reduces to rth-power analogues of EQ_3 uniformly for odd $r\le(\log x)^{1+\epsilon}$ *plus the same two open points. (ii) GRH tier.* Under GRH, $\mathrm{EQ}_3(\theta)$ holds for every $\theta<\frac{1}{8}$: the effective Chebotarev theorem of Lagarias–Odlyzko [31] bounds each term by $x^{1/2}\big(\log|d_{M_d}|+n_{M_d}\log x\big)\ll x^{1/2}d^{1+o(1)}\log x$ (as $n_{M_d}\le 6^{\omega(d)}\varphi(12d)=d^{1+o(1)}$), and summing over $d\le x^{2\theta}$ gives $x^{1/2+4\theta+o(1)}$; with the class-proportional form of the error term the same computation gives $\theta<\frac{1}{4}$. Note what GRH delivers is the hypothesis, not the theorem it feeds: points (a) and (b) of (i) remain open under GRH — the per-modulus $x^{1/2}$ barrier is the same — so the lane density program is *not* GRH-complete as of this draft. *(iii) Placement.* EQ_3 is a level-of-distribution statement — Bombieri–Vinogradov for cubic Kummer splitting, averaged over the base — with no parity obstruction; the nearest technology is the average-Artin theorem of Stephens [32] and the cubic Kummer-sum equidistribution (Kummer-on-average) results of Heath-Brown–Patterson [33], neither directly sufficient. Calibration through $q\le 2\cdot 10^5$: the lane has exactly two members with $w=2$ ($q=9185$ and $q=117185$), and the ω-stratified frequencies of the cubic-non-residue conditions match the exact Chebotarev densities of Definition 4.81. Chapter-and-verse verification of the classical sieve inputs cited above (beta-sieve limits, dimension-κ Mertens) is pending, per the 2026-07-06 audit discipline.

4.7 Quotient descent and the rigidity criterion

Fix a prime $m \mid t$ and write $t = mh$. The ring surjection $\mathbb{Z}[C_t] \to \mathbb{Z}[C_m]$ (induced by $x \mapsto g$, $C_m = \langle g \rangle$) sends a Turyn pair (r, s) to symmetric $A, B \in \mathbb{Z}[C_m]$ with

$$A^2 + B^2 = q \cdot 1 \qquad \text{in } \mathbb{Z}[C_m]. \tag{5}$$

Each coefficient of A, B is a sum of h (or $h-1$, at the identity of A) original entries over a residue class; so all coefficients are bounded by h with fixed parities. A zero solution count for (5) proves nonexistence upstairs; a positive count carries no information.

Write $n = (m-1)/2$, $u_j = g^j + g^{-j}$, and $K = \mathbb{Q}(\zeta_m + \zeta_m^{-1})$, the real cyclotomic field of degree n, with $\mathbb{Z}$-basis $1, u_1, \ldots, u_n$ of $\mathbb{Z}[C_m]^+$ and $A = A_0 + \sum_j A_j u_j$. Rationally $\mathbb{Q}[C_m]^+ \cong \mathbb{Q} \times K$ (trivial character and $g \mapsto \zeta_m$), and (5) becomes an ordinary two-squares equation in $\mathbb{Z}$ together with

$$\alpha^2 + \beta^2 = q, \qquad \alpha = A_0 + \sum_j A_j \big(\zeta_m^j + \zeta_m^{-j}\big) \in \mathcal{O}_K, \tag{6}$$

and likewise β for B. Since $1, u_1, \ldots, u_{n-1}$ is an integral basis of $\mathcal{O}_K$ and $\sum_j u_j = -1$ at $g \mapsto \zeta_m$, we have $\alpha = (A_0 - A_n) + \sum_{j<n}(A_j - A_n)u_j(\zeta_m)$, so $\alpha \in \mathbb{Q}$ iff $A_1 = \cdots = A_n$, i.e. iff A is constant off the identity. Call solutions with A, B both constant off the identity the *constant branch*: writing $A = a_0 + s\sum_j u_j$, $B = b_0 + c\sum_j u_j$, it is equivalent to a pair of Gaussian representations $X^2 + Y^2 = q$, $U^2 + V^2 = q$ with $X = a_0 + (m-1)s$, $U = a_0 - s$, $Y = b_0 + (m-1)c$, $V = b_0 - c$, subject to $X \equiv U$, $Y \equiv V \pmod m$ and the parity/box conditions on a_0, s, b_0, c. Checking for a constant-branch witness is thus a finite loop over the representations of q as a sum of two squares.

Definition 4.84. Let $p \nmid 2m$ and let $f = f_K(p)$ be the residue degree of p in K (the order of p in $(\mathbb{Z}/m)^\times/\{\pm 1\}$). Call p *rigid for* m if either *(i)* $p^f \equiv 3 \pmod 4$ (equivalently $p \equiv 3 \bmod 4$ and f odd), or *(ii)* $f = n$ and $p \equiv 1 \pmod 4$.

Theorem 4.85 (Rigidity)**.** *Let m be an odd prime and $\gcd(q, 2m) = 1$. If every prime $p \mid q$ is rigid for m, then every solution of (6) has $\alpha, \beta \in \mathbb{Z}$; equivalently every symmetric solution of (5) is constant off the identity. No class-number or unit hypothesis is used.*

Proof. Let $L = K(i)$ (abelian of degree $2n$, conductor $\mid 4m$), c the nontrivial element of $\mathrm{Gal}(L/K)$ (complex conjugation in every embedding), and $\gamma = \alpha + i\beta \in \mathcal{O}_L$, so $\gamma\, c(\gamma) = q$ and $|\tau(\gamma)|^2 = q$ at every archimedean τ.

Step 1 ($\mu_L = \{\pm 1, \pm i\}$). If $\zeta_w \in L$ with $m \mid w$ then $\mathbb{Q}(\zeta_m) \subseteq L$; both have degree $2n$, so they are equal and $i \in \mathbb{Q}(\zeta_m)$, impossible for odd conductor m. Hence $w \mid 4$, and $w = 4$.

Step 2 (primes above q). Each $p \mid q$ is unramified in L; with $k_p = v_p(q)$, every $P \mid p$ satisfies $v_P(\gamma) + v_{c(P)}(\gamma) = k_p$.

Step 3 (rigid (i)). If $p^f \equiv 3 \pmod 4$, the residue field of K at p has order $\equiv 3 \bmod 4$, so $x^2 + 1$ is irreducible there and every $P \mid p$ is inert in L/K ($c(P) = P$); then $2v_P(\gamma) = k_p$, k_p is even, and the p-part of (γ) is $\big(\prod_{P|p} P\big)^{k_p/2} = (p^{k_p/2})$, generated by a rational integer. (If k_p is odd there is no solution and the theorem holds vacuously.)

Step 4 (rigid (ii)). If $f = n$ and $p \equiv 1 \pmod 4$, write $p = \pi\overline{\pi}$ in $\mathbb{Z}[i]$. As K is totally real, $\mathrm{Gal}(L/\mathbb{Q}) \cong \mathrm{Gal}(K/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{Q}(i)/\mathbb{Q})$ and the Frobenius of p is $(\sigma_p, 1)$ of order $f = n$, so there are exactly 2 primes of L above p, namely $\pi\mathcal{O}_L, \overline{\pi}\mathcal{O}_L$ (comaximal, swapped by c, product $p\mathcal{O}_L$). The p-part of (γ) is $(\pi^a \overline{\pi}^{k_p - a})$, generated by a Gaussian integer.

Step 5 (Kronecker). Let $w \in \mathbb{Z}[i]$ be the product of the generators from Steps 3–4; then $(\gamma) = (w)$ and $w\overline{w} = q$, so $u = \gamma/w \in \mathcal{O}_L^\times$ has $|\tau(u)| = 1$ everywhere, hence $u \in \mu_L = \{\pm 1, \pm i\}$ (Step 1) and

$\gamma \in \mathbb{Z}[i]$. Writing $\gamma = x + iy$ ($x, y \in \mathbb{Z}$): if $\beta \neq y$ then $i = (\alpha - x)/(y - \beta) \in K$, contradicting K real; so $\alpha = x, \beta = y \in \mathbb{Z}$. The group-ring statement follows by Section 4.7's dictionary (rational K-components $\Leftrightarrow$ constant off the identity). □

The content is case (ii), where $p \equiv 1 \pmod 4$ and every self-conjugacy tool is silent.

Remark 4.86. The two rigid cases are precisely those in which the valuation vector of (γ) above p is *forced* to be that of a Gaussian integer. If $p \equiv 1 \bmod 4$ has $f < n$ there are $2n/f \geq 4$ primes above p and non-uniform exponent vectors of non-Gaussian shape appear; if $p \equiv 3 \bmod 4$ has f even (possible only for $m \equiv 1 \bmod 4$) the primes of K split in L and the same freedom appears. Whether such a non-Gaussian ideal solution has a generator of absolute value $\sqrt{q}$ everywhere is a class-group and unit question—the subject of §§4.8–4.9.

Two corollaries supply the obstruction.

Corollary 4.87 (All-rigid obstruction)**.** *If every $p \mid q$ is rigid for some prime $m \mid t$ and the constant branch of (5) has no witness, then no Turyn pair of order t exists. The witness check is the finite loop over Gaussian representations of q; no row enumeration is involved.*

Corollary 4.88 (Prime order)**.** *Let $t = m > 3$ be prime ($h = 1$). The folding boxes force $A_0 = 0$ and all other coefficients in $\{\pm 1\}$, so a constant solution has $A = d(\sum_j u_j)$, $d = \pm 1$, and $B = b_0 + c\sum_j u_j$ with $b_0, c \in \{\pm 1\}$, whence $\alpha = -d$, $\beta = b_0 - c \in \{0, \pm 2\}$ and $\alpha^2 + \beta^2 \leq 5 < q$. Hence if every $p \mid q$ is rigid for t, no Turyn pair of prime order t exists. At $t = 3$, $q = 5 = 1^2 + 2^2$ shows the bound is sharp.*

Theorem 4.89. *No Turyn pair of order* 43 *or* 73 *exists. Equivalently, there is no negacyclic $W(86, 85)$ or $W(146, 145)$ of Turyn-pair type, and Turyn's construction has no analogue at $q = 85$ or $q = 145$.*

Proof. For $t = 43$ ($n = 21$): modulo 43, $5^{21} \equiv -1$ and $\operatorname{ord}(17) = 21$, so both divisors of $q = 85 = 5 \cdot 17$ have $f = 21 = n$ and are $\equiv 1 \bmod 4$ (rigid (ii)); apply Corollary 4.88. For $t = 73$ ($n = 36$): $5^{36} \equiv 29^{36} \equiv -1 \pmod{73}$, so both divisors of $q = 145 = 5 \cdot 29$ have $f = 36 = n$; apply Corollary 4.88. □

These are the first prime-order cases in the blind spot of Remark 4.4. The case $t = 43$ was independently confirmed by an exhaustive meet-in-the-middle sweep ($2^{21} \times 2^{22}$ symmetric rows, zero solutions); $t = 73$ (2^{36} rows per side) rests on the theorem. A scan (Section 5) applies Corollary 4.87 over every prime $m \mid t$ and closes 62 of the 111 blind-spot cases.

Remark 4.90 (Consistency). For prime-power $q = p^e$ the hypothesis of Corollary 4.88 can never hold—rigid (i) contradicts $q \equiv 1 \bmod 4$, and rigid (ii) forces p to generate $(\mathbb{Z}/t)^\times/\{\pm 1\}$ with $e \geq n$, giving $q \geq 3^n > 2t - 1$ for $n \geq 3$—exactly as required, since Turyn's construction realizes all prime powers.

Remark 4.91 (Sharpness). When some $p \mid q$ is nonrigid, nonconstant solutions of (6) occur and lift to nonconstant symmetric solutions of (5) (coordinates $(A_0, \ldots, A_n)$):

- $q = 1105 = 5 \cdot 13 \cdot 17$, $m = 7$ ($13 \equiv -1 \bmod 7$ nonrigid, $f = 1 < 3$): $A = (-6, 3, -15, -1)$, $B = (-7, -5, -9, 13)$ satisfy $A^2 + B^2 = (1105, 0, 0, 0)$ (96 nonconstant K-level pairs exist here).
- $q = 1989 = 3^2 \cdot 13 \cdot 17$, $m = 5$ ($3 \equiv 3 \bmod 4$ has $f = 2$ even, the split escape): $A = (-14, 9, 19)$, $B = (5, -19, 9)$ satisfy $A^2 + B^2 = (1989, 0, 0)$.

Conversely, on cases closed by Corollary 4.87 the enumeration finds no nonconstant K-level pair (checked at $q = 153, 265, 405, 545, 657, 685$), while the control $q = 601$ (prime) shows 24. These certificates settle only the quotient's constancy behaviour; they neither construct nor exclude Turyn pairs upstairs. That lift question is the next subject.

4.8 The nonrigid case: three exact conditions

When some $p \mid q$ is nonrigid for m, we give an exact criterion for a symmetric quotient solution, replacing the earlier ad hoc layer terminology of the working notes by the standard trichotomy for a norm-form realization problem: *existence* of a solution to the relevant norm equation, the *congruences* it must satisfy, and the *size* (coefficient bound) it must respect.

Keep $L = K_m(i)$, c complex conjugation, and p_m the unique prime of $\mathcal{O}_K$ above m ($\mathcal{O}_K/p_m = \mathbb{F}_m$, $u_j \mapsto 2$). Set the congruence modulus

$$M_B = 2\, p_m \mathcal{O}_L.$$

For $\theta = c_0 + c_1u_1 + \cdots + c_{n-1}u_{n-1} \in \mathcal{O}_K$ and $X \in \mathbb{Z}$ define $\mathrm{Recon}_{m,h}(X, \theta; \epsilon)$, $\epsilon \in \{R, S\}$, by

$$b(\theta) = c_0 + 2(c_1 + \cdots + c_{n-1}), \quad w = \tfrac{X-b(\theta)}{m}, \quad \mathrm{Recon} = (c_0 + w, c_1 + w, \ldots, c_{n-1} + w, w),$$

defined only when $w \in \mathbb{Z}$ and the row satisfies the Turyn boxes (off-identity entries odd, $|\cdot| \le h$; the R identity entry with zero-diagonal parity and $|a_0| \le h-1$; the S identity entry odd, $|a_0| \le h$). Write $\mu_4 = \{\pm 1, \pm i\}$, $\Gamma(q) = \{\gamma \in \mathcal{O}_L : \gamma\, c(\gamma) = q\}/\mu_4$, and Sq for the compressed square signature in $\mathbb{Z}[C_m]^+$. Say an ideal I is *unit-admissible* if $I\, c(I) = (q)$ and $I = (\delta)$ has a generator with $\delta\, c(\delta) = q$ (equivalently $\delta c(\delta)/q \in \mathcal{O}_K^\times$ is a relative unit norm from $\mathcal{O}_L^\times$). Define the source/target ray sets

$$S_B(m, q) = \{\mu\, \sigma(\gamma) \bmod M_B : \gamma \in \Gamma(q), \mu \in \mu_4, \sigma \in \{1, c\}\},$$

$$T_B(m, q) = \{r \bmod M_B :\ r \equiv 1 \pmod{2\mathcal{O}_L},\ r \equiv x + iy \pmod{p_m},\ x^2 + y^2 = q,\ x \text{ even},\ y \text{ odd}\}.$$

Lemma 4.92 (Reconstruction). *Under $\mathbb{Q}[C_m]^+ \cong \mathbb{Q} \times K$ a symmetric row maps to (X, θ); it is integral iff $X \equiv \theta \pmod{p_m}$, and is then the unique row $\mathrm{Recon}_{m,h}(X, \theta; \epsilon)$, with identity coefficient $a_0 = (X + 2\,\mathrm{Tr}_{K/\mathbb{Q}}(\theta))/m$.*

Proof. Inverse DFT gives the displayed row; integrality is $m \mid X - b(\theta)$, exactly $X \equiv \theta \pmod{p_m}$ since $u_j \mapsto 2$. Using $\mathrm{Tr}(1) = n$, $\mathrm{Tr}(u_j) = -1$, $m = 2n+1$: $X + 2\,\mathrm{Tr}(\theta) = X + 2(nc_0 - c_1 - \cdots - c_{n-1}) = m(c_0 + w)$. □

Lemma 4.93 (Ideal/unit layer). *Solutions of $\alpha^2 + \beta^2 = q$ in $\mathcal{O}_K$ are exactly $\gamma = \alpha + i\beta \in \mathcal{O}_L$ with $\gamma\, c(\gamma) = q$; their ideals satisfy $I\, c(I) = (q)$, and such an ideal contributes a norm-q generator iff it is unit-admissible. Two such generators differ by an element of μ_4.*

Proof. $(\alpha + i\beta)c(\alpha + i\beta) = \alpha^2 + \beta^2$ gives the first claim. If $I = (\delta)$ with $Ic(I) = (q)$ then $\delta c(\delta) = \varepsilon q$, $\varepsilon \in \mathcal{O}_K^\times$; $u\delta$ has norm q iff $uc(u) = \varepsilon^{-1}$, i.e. unit-admissibility. Two norm-q generators differ by a relative norm-one unit; such a unit has all absolute values 1, so by Kronecker is a root of unity, and $\mu_L = \mu_4$ (Step 1 of Theorem 4.85). □

Lemma 4.94 (Parity and ray gluing). *For a Turyn-compatible pair the components satisfy $\theta_R \equiv 1$, $\theta_S \equiv 0 \pmod{2\mathcal{O}_K}$; the trivial values have X even, Y odd; and $\theta_R \equiv X$, $\theta_S \equiv Y \pmod{p_m}$. Hence $\gamma = \theta_R + i\theta_S \in T_B(m, q) \bmod M_B$, and conversely membership in T_B yields exactly these conditions.*

Proof. The parities are the compressed-coordinate form of the Turyn boxes (for R the zero-diagonal makes the constant coordinate odd and the rest even; for S all basis coordinates even); summing gives the parity of X, Y. Lemma 4.92 gives the p_m gluing. As $2\mathcal{O}_L$ and $p_m\mathcal{O}_L$ are coprime, CRT combines them into the single modulus M_B. □

We may now state the three conditions and the criterion.

(N) Norm equation. There is a Weil generator $\gamma \in \Gamma(q)$ (equivalently, a unit-admissible ideal).

(C) Congruence. $S_B(m,q) \cap T_B(m,q) \neq \varnothing$: some oriented conjugate of γ meets the target ray set modulo M_B.

(R) Realization. The rows $\mathrm{Recon}_{m,h}(x,\alpha;R)$, $\mathrm{Recon}_{m,h}(y,\beta;S)$ are defined inside their coefficient boxes.

Theorem 4.95 (Uniform realization criterion)**.** *Fix a congruence-admissible generator $\gamma = \alpha + i\beta$ and a signed two-square pair (x,y), so that $\alpha^2+\beta^2 = q$ in $\mathcal{O}_K$ and $x^2+y^2 = q$ in $\mathbb{Z}$. Then condition* (R) *holds if and only if the two rows*

$$R = \mathrm{Recon}_{m,h}(x,\alpha;R), \qquad S = \mathrm{Recon}_{m,h}(y,\beta;S)$$

are defined. When they are defined,

$$\mathrm{Sq}(R) + \mathrm{Sq}(S) = (q,0,\dots,0)$$

automatically.

More explicitly, for $\theta = c_0 + \cdots + c_{n-1}u_{n-1}$ and $X \in \mathbb{Z}$, put $w = (X - b(\theta))/m$ and $a_0 = c_0 + w$. The row $\mathrm{Recon}_{m,h}(X,\theta;\epsilon)$ *is defined exactly when $w \in \mathbb{Z}$, all tail entries*

$$c_1 + w, \dots, c_{n-1} + w, w$$

are odd and have absolute value at most h, and the head entry satisfies the corresponding identity box:

$$\epsilon = R: \ |a_0| \le h-1, \ a_0 \equiv h-1 \pmod 4; \qquad \epsilon = S: \ |a_0| \le h, \ a_0 \equiv 1 \pmod 2.$$

Proof. The explicit head and tail tests are just the definition of $\mathrm{Recon}_{m,h}$, with Lemma 4.92 identifying the head coefficient as $a_0 = (X + 2\,\mathrm{Tr}_{K/\mathbb{Q}}(\theta))/m$. Thus the two reconstructed rows are unique whenever they exist.

Let $\Phi : \mathbb{Q}[C_m]^+ \to \mathbb{Q} \times K$ be the usual product map, sending a row to its trivial value and its K-component. This map is injective. If the displayed rows are defined, then

$$\Phi(R^2+S^2) = (x^2+y^2, \alpha^2+\beta^2) = (q,q) = \Phi(q),$$

so $R^2 + S^2 = q$ in $\mathbb{Q}[C_m]^+$, hence in $\mathbb{Z}[C_m]^+$ because the rows are integral. The compressed square-signature equation is this equality in coordinates. Conversely, any Turyn-compatible rows with the fixed data must be the two reconstructions by Lemma 4.92; hence realization for that data is precisely their definedness. □

Theorem 4.96 (Prime-quotient descent)**.** *Turyn-compatible symmetric rows $R, S \in \mathbb{Z}[C_m]^+$ with $R^2 + S^2 = q$ exist if and only if conditions* (N), (C), (R) *hold simultaneously for some $\gamma = \alpha + i\beta \in \Gamma(q)$, μ_4-orientation, optional conjugation, and signed two-square pair (x,y). If* (C) *fails the quotient is closed with no coordinate enumeration; if* (C) *holds the question reduces to the finite check* (R).

Proof. Given rows R, S with trivial values X, Y and components θ_R, θ_S, set $\gamma = \theta_R + i\theta_S$; Lemma 4.93 gives $\gamma \in \Gamma(q)$, Lemma 4.94 puts its residue in T_B, and Lemma 4.92 identifies the rows as the stated reconstructions with $\mathrm{Sq}(R) + \mathrm{Sq}(S) = (q, 0, \ldots, 0)$; the conditions are necessary. Conversely, given the data, the ray condition supplies the parity and gluing (Lemma 4.94), and condition (R) says the two unique reconstructions are inside the boxes. Theorem 4.95 then gives the exact square signature, so these are Turyn-compatible rows with $R^2 + S^2 = q$. $\square$

Two examples calibrate the conditions.

- $q = 1377 = 3^4 \cdot 17$, $m = 13$: condition (C) fails—no unit-admissible generator meets $T_B(13, 1377)$, so the C_{13} quotient is closed. This closure is unconditional (bnfcertify certifies the class data of $K_{13}(i)$) and independent of coefficient enumeration.
- $q = 185 = 5 \cdot 37$, $m = 31$, $h = 3$: condition (C) *passes* but (R) fails. All 24 unit-admissible generators reconstruct the R-row inside the box, but the S-row identity coefficient is forced to $|a_0| = 5 > h = 3$ (the surviving two-square eigenvalue is $y \in \{\pm 13\}$, giving $a_0(S) = (y + 2\,\mathrm{Tr}(\theta_S))/31 = \pm 5$). Thus (R) is a genuinely separate obstruction, a thin-fiber ($h = 3$) phenomenon at the trivial character. This quotient-lane certificate depends on the $m = 31$ class list; the unconditional closure of $q = 185$ used in Section 5 comes instead from the field-free reduced decision of Section 4.3.

On the field-tractable subpanel where (C) passes, the uniform realization criterion gives the following compact certificate. Rows marked "yes" have bnfcertify-verified class data; rows marked "cond." are not used as certified closures. The rows are emitted by `condition_r_realization.py` (a diagnostic-table generator retained in the companion source repository, not among the shipped `code/` scripts — it pulls in the PARI/GP-driven layer-B subtree) and are provided as the machine-readable cache `code/data/condition_r_realization_qmax2000_degree40.jsonl`.

m	q	h	(R)	lifts	$\min \lvert a_0(R)\rvert$	$\min \lvert a_0(S)\rvert$	cert.	block
31	185	3	WIPEOUT	0	2	5	cond.	S-head
5	369	37	PASS	64	4	3	yes	lift
13	441	17	PASS	8	0	15	yes	lift
17	441	13	PASS	8	0	5	yes	lift
5	549	55	PASS	64	2	3	yes	lift
5	1089	109	PASS	8	0	11	yes	lift
7	1469	105	PASS	384	4	1	yes	lift
13	1585	61	PASS	48	16	5	yes	lift

Remark 4.97 (The identity-coefficient bound). The (R) failure at $q = 185$ is forced, not an artifact. By Lemma 4.92 and $|\sigma(\theta)| \le \sqrt{q}$ at each embedding, $|a_0| \le (\sqrt{q} + 2n\sqrt{q})/m = \sqrt{q}$; on the S-side a_0 is a nonzero odd integer $\le \sqrt{q}$ in absolute value. Moreover a general constant-off-identity solution is excluded whenever $2 \cdot (\text{largest two-square component of } q) < m$ and $m > h$: then $X \equiv U \pmod{m}$ with $|X - U| < m$ forces $s = 0$ (likewise $t = 0$), collapsing the row to a scalar $a_0^2 + b_0^2 = q$ that cannot fit $|a_0|, |b_0| \le h$ since $q = 2mh - 1 > 2h^2$. For $q = 185$, $m = 31$: $2 \cdot 13 < 31$, so the constant branch is empty and only the nonconstant $|a_0(S)| = 5$ obstruction remains.

4.9 Where the remaining obstruction lives

Condition (C) is the only step whose evaluation, as stated, requires constructing $K_m(i)$: the target T_B is local (two-square residues and parities), but the source residues $\gamma \bmod M_B$ are those of the Weil generators. Whether they can be read from local splitting data is governed by the class group.

Let $C_{\mathrm{loc}}(m,q)$ be the finite set of ideals I with $Ic(I)=(q)$ built from local prime data: for each $P\mid p$ $(p\mid q)$, if $P=c(P)$ the valuation is forced to $e_p/2$ (existing only for e_p even), and if $P\neq c(P)$ choose exponents a and e_p-a on the pair. Let $\mathrm{Cl}_B(L)$ be the ray class group modulo M_B.

Definition 4.98 (Ray defect)**.** $D_B(m,q)=\{\mathrm{cl}_B(I): I\in C_{\mathrm{loc}}(m,q),\ I \text{ unit-admissible}\}\subseteq \mathrm{Cl}_B(L)$.

Lemma 4.99 (Local factor lemma)**.** *The ideals I with $Ic(I)=(q)$ and $(I,M_B)=1$ are exactly $C_{\mathrm{loc}}(m,q)$.*

Proof. Since $(q,2m)=1$, every prime dividing I lies above a rational $p\mid q$ and is prime to M_B; $v_P(I)+v_{c(P)}(I)=e_p$ forces $2v_P(I)=e_p$ when $P=c(P)$ and allows independent choices on distinct conjugate pairs otherwise. □

Proposition 4.100 (Ray criterion up to the defect)**.** *Condition (C) holds for (m,q) iff there are $I\in C_{\mathrm{loc}}(m,q)$, $r\in T_B(m,q)$, $\mu\in\mu_4$, $\sigma\in\{1,c\}$ with I unit-admissible and $\mathrm{cl}_B(\sigma I)=[\mu^{-1}r]_B$ in $\mathrm{Cl}_B(L)$. Equivalently, (C) is decided by local split-prime choices together with the single global image $D_B(m,q)$.*

Proof. By Lemma 4.99 the ideal of any norm-q generator lies in $C_{\mathrm{loc}}(m,q)$; Theorem 4.96 identifies (C) with $\mu\sigma(\gamma)\equiv r \pmod{M_B}$, which says σI has a generator $\equiv \mu^{-1}r$, i.e. the displayed ray-class equality. □

Corollary 4.101 (Trivial-defect product formula)**.** *If the relevant unit-admissible local factors are principal with known residues modulo M_B, then $S_B(m,q)=\mu_4\{1,c\}\cdot(\text{forced Gaussian residue})\cdot\prod_{\text{split pairs}}(\text{local prime-pair residues})$, and (C) holds iff this product meets $T_B(m,q)$.*

The Corollary's hypothesis is exactly principality of the primes above q, and it fails often.

Proposition 4.102. *Let $\mathrm{Im}_q=\langle[P]:P\mid q\rangle\subseteq \mathrm{Cl}(L)$. The source residues in* (C) *are a product of explicit local (principal-prime-factor) residues —computable without building $K_m(i)$—if and only if $\mathrm{Im}_q=0$. If $\mathrm{Im}_q\neq 0$, a Weil generator is produced only through the global class group, and $D_B(m,q)$ is a genuine, nonzero global datum.*

Theorem 4.103 (Absorbed ray criterion)**.** *Let $G_B=(\mathcal{O}_L/M_B)^\times$. Let $\Lambda_B\subseteq G_B$ be the subgroup generated by the following field-constant ambiguities: the residues of μ_4; the local Hilbert–90 norm-one residues $z/c(z)$ with $z\in G_B$; and the residues of the global relative-unit corrections $u/c(u)$ with $u\in\mathcal{O}_L^\times$. For each unit-admissible $I\in C_{\mathrm{loc}}(m,q)$ choose one generator γ_I with $\gamma_I c(\gamma_I)=q$, and put*

$$\overline{S}_B(m,q)=\{\sigma(\gamma_I)\Lambda_B:\ I\in C_{\mathrm{loc}}(m,q)\ \text{unit-admissible},\ \sigma\in\{1,c\}\}\subseteq G_B/\Lambda_B.$$

Let

$$\overline{T}_B(m,q)=\{r\Lambda_B:\ r\in T_B(m,q)\}.$$

Then condition (C) *holds if and only if*

$$\overline{S}_B(m,q)\cap\overline{T}_B(m,q)\neq\varnothing.$$

Equivalently, the only global input needed by the congruence condition is the finite absorbed image $\overline{D}_B(m,q):=\overline{S}_B(m,q)$ of the unit-admissible ray defect.

Proof. Theorem 4.96 and Lemma 4.94 identify (C) with the existence of an oriented conjugate of a norm-q generator whose residue lies in $T_B(m,q)$. By Lemma 4.99, the ideal of such a generator is an element of $C_{\mathrm{loc}}(m,q)$, and by Proposition 4.100 the only obstruction to reading its residue as a local product is the ray class of that ideal. Changing the principal local Hilbert–90 choices, the relative-unit correction in the norm equation, or the fourth-root orientation multiplies the residue by an element of Λ_B; conversely, the generators of Λ_B are precisely these absorbed choices. Thus a unit-admissible local ideal contributes one well-defined coset $\sigma(\gamma_I)\Lambda_B$ in G_B/Λ_B, independent of the auxiliary generator choices. The displayed intersection is exactly the projected source condition $S_B(m,q) \cap T_B(m,q) \neq \varnothing$. □

On the field-tractable panel of nonrigid survivors ($q \leq 2000$, field degree ≤ 40), $\mathrm{Im}_q \neq 0$ for half the cases, including the frontier $q = 185$, where the offending classes have odd order 41 in $\mathrm{Cl}(K_{31}(i))$ ($h_L = 5084$). Moreover quadratic genus theory does not decide it: the 2-part obstructions lie in $\mathrm{Cl}(L)^2$, below the genus quotient $\mathrm{Cl}(L)/2\,\mathrm{Cl}(L)$ (at $m = 17$ the class is $4 = 2 \cdot (\mathbb{Z}/8)$; at $m = 41$ it has order 4), and are invisible to genus characters; individual primes above q can even be genus-visible on the sum-of-two-squares domain (for $m = 17$ the first case is $q = 13837 = 101 \cdot 137$), but in a unit-admissible (principal) ideal the genus images cancel. Hence (C) is not, in general, a local test; the residual global input is the odd part of Im_q (no reciprocity handle) together with a 2-part accessible at best to Rédei/higher reciprocity. This is a negative structural result: it locates the obstruction precisely rather than removing it. The following table, from the field-tractable panel, exhibits the absorbed congruence quotient. Here $|\overline{D}_B|$ is the number of residual source cosets in the target-relevant quotient, "hit" is the number meeting $\overline{T}_B$, and "cert." records whether the class data for K_m and $K_m(i)$ have been certified by bnfcertify; rows marked "cond." are not used as certified closures. The compact rows are emitted by `condition_c_absorption.py` (a diagnostic-table generator retained in the companion source repository, not among the shipped `code/` scripts) and are provided as the machine-readable cache `code/data/condition_c_absorption_qmax2000_degree40.jsonl`.

m	q	h	(C)	$\lvert\overline{D}_B\rvert$	hit	cert.	defect
31	185	3	PASS	3	1	cond.	essential
41	245	3	WIPEOUT	3	0	cond.	essential
5	369	37	PASS	2	1	yes	trivial
37	369	5	WIPEOUT	2	0	cond.	trivial
13	441	17	PASS	2	1	yes	trivial
17	441	13	PASS	3	1	yes	essential
5	549	55	PASS	2	1	yes	trivial
29	985	17	WIPEOUT	3	0	cond.	essential
5	1089	109	PASS	2	1	yes	trivial
13	1377	53	WIPEOUT	2	0	yes	trivial
7	1469	105	PASS	2	1	yes	trivial
29	1565	27	WIPEOUT	3	0	cond.	essential
13	1585	61	PASS	3	1	yes	essential
41	1885	23	WIPEOUT	4	0	cond.	essential

5 Closed orders

We summarize the composite $q \leq 2000$ closed by the hierarchy. Counts in the self-conjugacy-blind population are not additive: the rigidity and T3 reduced decisions overlap.

Layer	Mechanism	Effect for $q \leq 2000$
Two squares (§4.1)	$\ell \equiv 3 \bmod 4$ to odd power	211 closed
Self-conjugacy (§4.2)	inert tower, Prop. 4.3	13 further closed
Rigidity scan (§4.7)	prime-quotient rigidity	62 of the 111 blind-spot cases
T3 reduced decision (§4.3)	composite multipliers + MITM	85 of the 111 blind-spot cases
Rigidity $\cup$ T3	combined unconditional coverage	95 of the 111 blind-spot cases
Marginal decision (§4.4)	composite/full sub-tori	closes all 16 remaining cases
Full hierarchy	all field-free layers above	every blind-spot multi-prime case $q \leq 2000$ closed
Nonrigid descent (§4.8)	(N)/(C)/(R), field-tractable panel	structural diagnostics and conditional local certificates

The rigidity scan reproduces without enumeration the five cases previously closed by exact quotient computation ($q = 65, 153, 265, 405, 657$) and adds 57 (many at prime t via Corollary 4.88), among them

$$q = 85,\ 145,\ 205,\ 445,\ 481,\ 565,\ 745,\ 793,\ 865,\ 925,\ 1261,\ 1345,\ 1537,\ 1717,\ 1813,\ \ldots$$

The T3 reduced decision closes 85 blind-spot cases. It overlaps the rigidity scan in 52 cases and adds the following 33 unconditional closures:

$$\begin{gathered}185,\ 225,\ 245,\ 325,\ 365,\ 369,\ 441,\ 485,\ 585,\ 605,\ 685,\ 697,\ 725,\\ 801,\ 985,\ 1017,\ 1089,\ 1145,\ 1157,\ 1233,\ 1285,\ 1305,\ 1377,\ 1417,\\ 1465,\ 1513,\ 1521,\ 1525,\ 1665,\ 1685,\ 1765,\ 1885,\ 1989.\end{gathered}$$

This list includes $q = 185$, formerly the first frontier for the nonrigid quotient lane; $q = 441$, where both prime quotients pass and the obstruction is invisible to the quotient criteria; and the all-rigid-with-witness cases $q = 485, 685, 725$, where quotient descent is permanently silent.

After combining the unconditional rigidity scan with the original T3 harvest, there were 16 residual self-conjugacy-blind cases through 2000. The marginal decision closes all of them:

$$\begin{gathered}377,\ 425,\ 545,\ 549,\ 629,\ 1025,\ 1105,\ 1205,\ 1325,\ 1445,\ 1469,\\ 1565,\ 1585,\ 1625,\ 1649,\ 1937.\end{gathered}$$

Here 1205 and 1565 are full-torus T3 decisions; 1585 is killed already at the $t' = 61$ marginal (and also at the full torus by the CP-SAT formulation); 1325, the final residual, is killed by the quartic-component criterion at $t' = 51$ (Proposition 4.62); the others are killed at proper composite sub-tori. Thus the current field-free hierarchy closes every self-conjugacy-blind multi-prime composite $q \leq 2000$. The field-tractable (N)/(C)/(R) tables above remain useful for locating quotient phenomena, but the T3/T2 field-free layers supersede several old conditional closure claims in the headline harvest.

q	killing t'	$h = t/t'$	certificate
377	27	7	C_{27} boxed congruence criterion (Cor. 4.41)
425	71	3	$h = 3$ support-augmentation obstruction (Prop. 4.14)
545	21	13	C_{21} component join criterion (Cor. 4.58)
549	25	11	$p = 5$ component criterion (Prop. 4.54)
629	21	15	C_{21} component join criterion (Cor. 4.58)
1025	27	19	C_{27} residue-pair criterion (Cor. 4.40)
1105	79	7	prime sub-torus period criterion (Cor. 4.35); also HiGHS MILP
1205	603	1	T3 MITM
1325	51	13	quartic component criterion (Prop. 4.62)
1445	241	3	$h = 3$ support-PAF obstruction (Prop. 4.20)
1469	49	15	$p = 7$ cubic component criterion (Prop. 4.56)
1565	783	1	T3 MITM
1585	61	13	prime sub-torus period criterion (Cor. 4.35); also exhaustive MITM and HiGHS MILP
1625	271	3	$h = 3$ support-sum obstruction (Prop. 4.14)
1649	25	33	$p = 5$ component criterion; also exhaustive MITM at $t' = 55$
1937	57	17	C_{57} quadratic-component criterion (Prop. 4.59); also exhaustive MITM and MILP at $t' = 323$

Remark 5.1 (What "closed" asserts, restated). Each entry above is a proof that no Turyn *pair* of the corresponding order t exists—equivalently, no negacyclic $W(2t, 2t-1)$ of Turyn-pair type, no binary almost-perfect sequence of period $4t$ of the corresponding kind. It is *not* a proof that no Hadamard matrix of order $4t$ exists. For q within the negacyclic $v < 1000$ classification the nonexistence is a known data point and the contribution is the proof; the criterion also closes cases (e.g. larger prime t via Corollary 4.88) at orders beyond exhaustive reach.

6 Related work

The Turyn lineage. Turyn [1]; Whiteman's cyclotomic re-proof [6]; Yamamoto–Yamada's relative Gauss sums [7]; Lang–Schneider's uniqueness up to equivalence through order 99 [8]; the recent Williamson-type searches of Kharaghani–Mohammadian–Tayfeh-Rezaie [9].

Equivalent objects. Negacyclic conference matrices and the $v < 1000$ classification [11, 12]; almost-perfect autocorrelation sequences and their cyclic-divisible-difference-set description [13, 14]; the two-circulant-core construction [15]; the classical $\mathrm{BGW}(q+1, q, q-1)$ from Bose's affine difference set and Jungnickel's equivalence [16, 17, 18]; almost-perfect polyphase sequences [19, 20, 21].

Method. Rigid case (i) is Turyn's self-conjugacy mechanism [2] in relative form; no novelty is claimed for it. Rigid case (ii) is the composite-conductor analogue of the order-driven field descent of Leung and Schmidt—the anti-field-descent method [27, Lemma 4.6, Thm 4.5] and its nonsquare refinement [28, Thm 22]—which are proved for odd prime-power conductor p^a; the field of relevance here, $L = K_m(i)$, has even composite conductor $4m$, and Schmidt's composite-conductor descent [26] cannot reach the conclusion (its $F(M, N)$ is a multiple of $\mathrm{rad}(M)$, so it never strips m; concretely $F(172, 85) = 172$, no descent, whereas Theorem 4.85 lands solutions in $\mathbb{Z}[i]$). The contribution is the exact criterion (Definition 4.84), the corollaries it feeds, and the sharpness certificates that bound it, not a new descent method. The multiplier theorems are likewise normal-form refinements of classical multiplier theory for affine difference sets [18]. The composite group $M(q) = \cap_{p|q}\langle p\rangle$ is the natural common-decomposition subgroup; before any novelty claim it should be checked against the divisible-difference-set multiplier literature. The marginal obstruction is a finite quotient consequence of the same multiplier theorem. No such audit is attempted here. The unconditional closures from Sections 4.3–4.4 do not depend on that novelty posture.

Honesty on the closed facts. As facts, the nonexistence of Turyn pairs of orders $43, 73$ (periods $172, 292$), and also $q = 185, 441$, lie within the DGS/Eades computational range for negacyclic conference matrices. The almost-perfect-sequence papers of Wolfmann [13] and Bradley–Pott [14] treat these periods, and a divisible-difference-set multiplier argument there could conceivably cover some of the same closures. Their full texts were not obtained; even so, for the all-1 mod 4 cases any such argument would rest on multipliers and counting, not self-conjugacy, and would differ from the quotient-descent mechanism here.

Multiplier and RDS shadows (2026-07-06 audit). The composite multiplier theorem should be read as a Turyn-normal-form specialization of classical multiplier theory. In the group-developed relative-difference-set setting, multiplier lifting and fixed-translate theorems are standard (the Pott/Lam and Arasu–Jungnickel–Ma–Pott lineage, as summarized in Gordon's cyclic-RDS survey), and circulant weighing-matrix searches use the same multiplier-orbit and intersection-number constraints (Arasu–Gordon–Zhang). What is used here is the common-decomposition subgroup $M(q)$ acting on an arbitrary Turyn pair, where the zero diagonal removes the translate and the $R+iS$ normal form pins the signs. Likewise, the quotient maps and Gaussian-period component algebras throughout are classical; the novelty claimed for Theorem 4.66 is the marginal-complete organization for Turyn pairs — the exact top node, monotone emptiness on the divisor lattice, and finite per-family firing criteria attached to specified nodes — not the quotienting of group rings or the periods themselves. The fold layer of Proposition 4.30 uses standard ideal-class obstructions in CM fields, in the same broad family as self-conjugacy and Leung–Schmidt field descent; its contribution is the stabilizer amplification interface, and any closure using it retains the certification tier of the class-group computation.

Scope of the "first determination" claims. The closures at $q = 1469$, 1937, 1325 (and, conditionally, 5185, 62305) are apparently first determinations in the negacyclic/Turyn-pair literature searched in the 2026-07-06 audit: they lie beyond the DGS/Eades–Seberry $v < 1000$ negacyclic classification and the Lang–Schneider range, and exact-order searches (including the associated Hadamard/TCC orders and APS periods) found no hit in accessible sources. This is not a statement about arbitrary Hadamard matrices, and it remains subject to the closed-table caveat above. The semiprime-lane theorem is new only in the same audited sense: its proof combines classical shadows — common-decomposition multipliers and finite quotient marginals — with the lane-specific anchor and support congruence, and no prior statement excluding all semiprime $q = 6\ell - 1$, $\ell \equiv 3 \pmod 4$, was found.

7 What remains

Conjecture 1.3 is open. The hierarchy above closes every self-conjugacy-blind multi-prime composite $q \leq 2000$, but the proof status is not simply "the finite search got bigger." The present state separates into three parts: the prime-quotient conceptual layers that are now exact criteria, the finite-panel branch accounting for the field-free multiplier/marginal layers, and the uniform theorem obligations still needed for a full converse.

Conjecture 7.1 (Marginal-complete obstruction program). *Let $q = 2t - 1$ be composite. Then at least one holds:* (1) *a classical two-squares or self-conjugacy obstruction fires;* (2) *the full T3 multiplier-reduced system is infeasible;* (3) *for some proper divisor $t' \mid t$, the V'-invariant marginal system of Proposition 4.12 is infeasible;* (4) *all such finite marginals pass, but their data do not glue to one element of $\mathbb{Z}[C_t]$.*

7.1 Closed conceptual layers

The prime-quotient descent has two layers that are no longer informal obstacles. Condition (C), the absorbed ray criterion, is an exact global criterion: Proposition 4.102 shows that $\mathrm{Im}_q \subseteq \mathrm{Cl}(K_m(i))$ is essential when it is nonzero, and the remaining class-group input is isolated in the finite residual image $\overline{D}_B(m, q) \subseteq G_B/\Lambda_B$. Condition (R), the realization criterion, is also exact: after (N) and (C), a candidate realizes precisely when the two reconstruction rows lie in their coefficient boxes. The $q = 185$ quotient certificate remains the model local wipeout—(C) passes, but every survivor overflows the S-head box by Remark 4.97—although the unconditional closure of $q = 185$ used in the headline harvest is the field-free T3 decision.

The field-free multiplier layer is likewise no longer an all-plus-only theorem framework. The T3 solver still refuses sign-alternating inputs as an implementation safety rule, but Proposition 4.9 gives the signed marginal system whenever an $\epsilon = -1$ branch occurs. Lemma 4.2 reduces any post-two-squares signed case to the square branch $q = n^2$ with all primes of n congruent to 3 mod 4. The first multi-prime classical survivors in the signed audit, through $q \leq 50000$, are 29241, 40401, and 43681; all are closed by Corollaries 4.10 and 4.11. Thus the sign guard is not a foundational gap. A full converse must still route any future signed square branch into a signed firing criterion, just as the all-plus branch must be routed into an all-plus firing criterion.

7.2 Finite-panel branch accounting

The field-free harvest should be read at the level of branches, not merely as a list of closed examples. After the two-squares and self-conjugacy filters, there are 111 self-conjugacy-blind multi-prime composites $q \leq 2000$. The rigidity scan closes 62 of them, T3 closes 85, their union closes 95, and the marginal layer closes the remaining 16:

$$377,\ 425,\ 545,\ 549,\ 629,\ 1025,\ 1105,\ 1205,\ 1325,\ 1445,\ 1469,$$
$$1565,\ 1585,\ 1625,\ 1649,\ 1937.$$

The first-determination cases $q = 1469$, $q = 1937$, and $q = 1325$ all have $2t \geq 1000$, lie outside the negacyclic classification range, and pass every prime-quotient test. The last of them, $q = 1325$ with $t = 663 = 3 \cdot 13 \cdot 17$, has solvable prime marginals $t' = 3, 13, 17$; it is closed only at the $t' = 51$ marginal by the quartic-component criterion (Propositions 4.61–4.62).

The executable scaffold for the existential divisor theorem is

```
orbit_signature_scan.py  -existential-divisor-report  -summary-only.
```

Unlike the older lane census, it evaluates actual firing predicates on every proper divisor. By Theorem 4.66(i), each closure in this accounting records one certified minimal element of the empty-node up-set $U(q)$, and the branch label records the certifying criterion; the census formula of Lemma 4.64 is machine-verified against the constructed orbit algebras by `-divisor-lattice-audit`. With the two-orbit, $h = 3$ support-augmentation, $p = 3$, $p = 5$, $p = 7$, C_{27}, C_{21}, C_{51}, prime sub-torus, full-torus sign, and secondary exact-marginal predicates in place, the $q \leq 2000$ branch decomposition is:

81	proper-divisor quantified fires,
29	exact full-torus sign fires,
1	projection-image gluing obstruction ($q = 441$).

Thus neither a finite-certificate branch nor a secondary-certificate branch remains in this panel: the last four secondary orders were promoted to written criteria — 685 and 1665 by the full-unit

C_{49} instance of Proposition 4.42, 1937 by the C_{57} criterion (Proposition 4.59), and 1885 by the degree-4 period join of Corollary 4.35 under the measured box gate — and every closure in the 21-order T3-unreachable panel is a quantified firing route. The old proper-divisor component branch is consumed by the prime sub-torus join and the $q = 441$ gluing check. The old prime-order full-torus component branch is consumed by the row-sum bound (Theorem 4.23): all 29 full-torus branch members, including the higher-degree cases $q = 697$, 1521, and 1765 formerly closed by complete integer MITM, satisfy $w \geq d + 2$ and fire with zero computation. Every `scalar_unsat` row is an instance of the two-orbit representation criterion (Theorem 4.46) and its corollaries, certified by closed-form arithmetic rather than enumeration; Remark 4.50 records the node-level audit.

Cross-prime gluing. Passing (N),(C),(R) at every prime quotient does *not* produce a global element of $\mathbb{Z}[C_t]^+$; the quotient data must be compatible under the common projections and the integral boxes. At $q = 441 = 21^2$, $t = 221 = 13 \cdot 17$, both prime quotients admit valid nonrigid symmetric rows (C_{13} and C_{17} each lift; see the table in §4.9), so the prime-quotient criteria give no obstruction. The exact side-set gluing check does: the C_{13} and C_{17} quotient side-solution sets each contain two A-rows and two B-rows, but no boxed V-orbit row on C_{221} projects to the corresponding side sets. Already in the C_{13} projection, the nonzero quotient coordinates of a cover row have the form $4(x_1 + x_3 + x_6 + x_7) + x_9$ and $4(x_2 + x_4 + x_5 + x_{10}) + x_{12}$, hence are $8k \pm 1$; the local quotient solutions require ± 3. Thus $q = 441$ is a projection-image gluing obstruction before the full T3 PAF join. At $q = 549$, both prime quotients lift, but the prime-power marginal $t' = 25$ is infeasible. These examples show that composite marginals and projection gluing are strictly stronger than all prime-quotient tests taken separately.

7.3 Plan toward the full converse

The next theorem must replace the finite T2/T3 harvest by a structural statement about orbit algebras. Orbit shape gives *compression*; it does not by itself give an obstruction. The full converse program therefore has two layers:

compression families	$\mathcal{A}(t', V') = \mathbb{Z}[C_{t'}]^{V'}$ has a small or rigid orbit model,
firing criteria	arithmetic, congruence, energy, box/parity, or gluing arguments prove that the compressed finite system has no global witness.

The marginal algebra lemma, the transitive two-orbit theorem (Theorem 4.45), and the prime-square lift certificate (Theorem 4.52) are now compression frameworks. They become obstructions only when their associated finite joins are shown infeasible. This caution is essential: the two-orbit marginal at $q = 1325, t' = 3$ is solvable, and prime-square lift marginals through $q \leq 2000$ split between SAT and UNSAT examples.

The remaining work is therefore the following.

1. Keep the composite multiplier foundation explicit. Theorem 3.7 and Lemma 3.6 are the source of all $M(q)$-invariance used by T3 and by the marginal system.

2. Unify the finite component criteria rather than multiplying special cases. Proposition 4.21 is the general template: the fixed algebra decomposes into conductor components, each component gives a two-square equation over a fixed subfield, the embedding bound $|\sigma(\lambda)| \leq \sqrt{q}$ makes the component search finite, and the inverse transform imposes the integral boxes and parities. The C_{27} Gaussian congruence criteria (Propositions 4.37–4.38), the $p = 3$ Gaussian

criterion (Proposition 4.53), the $p = 5$ quadratic criterion (Proposition 4.54), the C_{21} quadratic criterion (Proposition 4.57), the $q = 1325, t' = 51$ quartic criterion (Propositions 4.61–4.62), and the prime sub-torus period criterion (Proposition 4.34 with Corollary 4.35) are instances of this framework. The $h = 3$ support-augmentation obstruction (Proposition 4.14) is the complementary small-fiber criterion.

3. The selection layer of the existential divisor-selection problem is now a theorem. The group

$$M(q) = \bigcap_{p|q} \langle p \rangle \subset (\mathbb{Z}/4t)^\times$$

controls every V'-orbit structure in the precise sense of Lemma 4.64, and Theorem 4.66 shows that the existential quantifier over all divisors $t' \mid t$ is complete and monotone: nonexistence at q is *equivalent* to emptiness of some marginal node, the empty nodes form an up-set, and criterion-equipped nodes always exist. This absorbs the earlier obligation to "quantify over all divisors rather than a shape-ranked divisor." What it deliberately does not provide (Remark 4.68) is a proof that a written criterion *fires* at some node for every composite q — by Theorem 4.66(i) that statement is the all-plus converse itself. The remaining structural work is therefore the per-family firing theorems: arithmetic hypotheses on $(q, M(q))$ under which the period join, the $h = 3$ hierarchy, the component joins, or the top-node sign join is provably empty. The first of these is now in place: the two-orbit family is completely characterized by Theorem 4.46, with the zero-computation parity case (Corollary 4.47), the transitive full torus (Corollary 4.48), and the never-firing C_3 node (Corollary 4.49); the prime $h = 3$ support stage likewise has the closed form of Corollary 4.16, and the prime-t top node has the row-sum bound (Theorem 4.23), which closes the whole full-torus branch whenever $(w-1)^2 \geq t$ and bounds the multiplier image of any genuine prime-order pair by $w < 1 + \sqrt{t}$. The $h = 3$ hierarchy is now closed-form through its augmentation stage at prime nodes (Proposition 4.18, Corollary 4.19); this is the routing theorem for the dominant local-pass class, because Corollary 4.49 shows the C_3 node never fires while Theorem 4.66(iii) guarantees the $h = 3$ node $t' = t/3$ for exactly those orders. The small-image prime-t top nodes ($w \leq d + 1$) now have a stated mechanism with two decided instances: the fold-amplification dichotomy (Proposition 4.30, with Lemma 4.26) closes $q = 5185$ and $q = 62305$, each GRH-conditional through exactly one class-group input, cascading into the exact trivial-character predicate; the census to $q \leq 100000$ leaves four survivors of which these are two, and the family's remaining obligations are the Stickelberger lemma for the firing condition, unconditionality of the two instances, and a refinement for the two wall witnesses (Remark 4.32). The secondary branch is now empty: the full-unit prime-power criterion (Proposition 4.42), the C_{57} criterion (Proposition 4.59), and the measured box gate for degree-4/5 period joins promote its last four orders. The families still needing firing characterizations are the very wide period joins (the lattice-reduced enumeration now evaluates every node up to $\sim 2 \cdot 10^7$ true lattice points; the surviving refusals 8865, 8905, 9265 sit at 10^9–10^{42} points and need an argument, not enumeration), the $h = 3$ PAF stage at wide nodes, the composite component joins beyond the instances above, and composite-t top nodes.

4. The remaining finite-certificate branch has been replaced in the finite panel. The case $q = 1469$ is now closed at the $p = 7$ prime-square marginal $t' = 49$ by the cubic component criterion (Proposition 4.56), so the current $q \leq 2000$ accounting has no HiGHS/MILP-only branch. The remaining obligation is to unify these component criteria and the $h = 3$ hierarchy into broader firing theorems rather than isolated low-degree instances.

5. Keep the signed branch as a theorem-facing obstruction branch, not an implementation guard. Lemma 4.2 reduces any post-two-squares signed case to $q = n^2$ with every prime divisor of n congruent to 3 mod 4. Proposition 4.9 and Corollary 4.10 give the signed finite system, and Corollary 4.11 closes the full-unit prime-torus subcase analytically. The signed audit through $q \leq 50000$ finds three multi-prime classical survivors in this square branch: 29241, 40401, and 43681; all three are closed by the signed full-torus criteria. Thus $\epsilon = -1$ is no longer a separate implementation-side blocker. The structural divisor theorem must still route any future signed square branch into a signed firing criterion, just as it must route all-plus branches into their firing criteria.

6. The $q = 441$ projection-image gluing obstruction is generalized: Corollary 4.67 states it for an arbitrary node against any family of exactly solved quotient nodes, as a consequence of lattice functoriality. In lattice terms the remaining gluing question is sharp (Remark 4.68): is the family of Theorem 4.66(iv) — composite t with every proper node feasible — empty? No member with $q \leq 2000$ is known; a member would be a q whose converse is decided only at the exact full-torus node.

7. Use wider sweeps only as evidence. Under the current branch split, the same scaffold was refreshed through $q \leq 10000$ with dynamic secondary exact checks, dynamic full-torus sign joins, and dynamic $h = 3$ support joins disabled. With the lattice-reduced period enumeration in place (an LLL-reduced embedding-cube slicer with exact sup-norm filtering, gated by the true point-count predictor $(2\sqrt{q})^d/\sqrt{\ell^{d-1}}$ rather than the coordinate-box size), it reports 415 proper-divisor quantified fires, 135 full-torus fires (82 by the zero-computation row-sum bound of Theorem 4.23), 1 remaining full-torus sign lane — the small-image case $q = 5185$, decided GRH-conditionally by Proposition 4.28 but kept in the field-free accounting pending certification — 56 local-pass quantified lanes, 3 recorded secondary exact marginal certificates, no recorded finite certificates, 1 projection-gluing obstruction, and 3 proper component-only lanes: 8865, 8905, 9265, refused honestly by the predictor at $\approx 2.0 \cdot 10^9$, $2.4 \cdot 10^{42}$, and $1.2 \cdot 10^{11}$ lattice points. The formerly refused nodes 7045@271 (1,815,037 points) and 4705@181 (15,123,779 points) are now exact `no_side_vector` fires, each cross-checked against the coordinate-box enumerator where both run. The run is deliberately evidence-only: the disabled dynamic branches mark where the structural theorem must look, not a substitute for that theorem.

References

[1] R. J. Turyn, An infinite class of Williamson matrices, *J. Combin. Theory Ser. A* **12** (1972), 319–321.

[2] R. J. Turyn, Character sums and difference sets, *Pacific J. Math.* **15** (1965), 319–346.

[3] R. J. Turyn, Sequences with small correlation, in: *Error Correcting Codes* (H. B. Mann, ed.), Wiley, New York, 1968.

[4] J. Williamson, Hadamard's determinant theorem and the sum of four squares, *Duke Math. J.* **11** (1944), 65–81.

[5] M. Hall, Jr., *Combinatorial Theory*, Blaisdell, Waltham, Mass., 1967.

[6] A. L. Whiteman, An infinite family of Hadamard matrices of Williamson type, *J. Combin. Theory Ser. A* **14** (1973), 334–340.

[7] K. Yamamoto and M. Yamada, Williamson Hadamard matrices and Gauss sums, *J. Math. Soc. Japan* **37** (1985), 703–717.

[8] W. Lang and E. Schneider, Turyn type Williamson matrices up to order 99, *Des. Codes Cryptogr.* **62** (2012), 79–84.

[9] H. Kharaghani, A. Mohammadian, and B. Tayfeh-Rezaie, A search for Hadamard matrices of Williamson type, arXiv:2605.08661 (2026).

[10] J. M. Goethals and J. J. Seidel, Orthogonal matrices with zero diagonal, *Canad. J. Math.* **19** (1967), 1001–1010.

[11] P. Delsarte, J. M. Goethals, and J. J. Seidel, Orthogonal matrices with zero diagonal. II, *Canad. J. Math.* **23** (1971), 816–832.

[12] J. Seberry, Remarks on negacyclic matrices, preprint, 2016.

[13] J. Wolfmann, Almost perfect autocorrelation sequences, *IEEE Trans. Inform. Theory* **38** (1992), 1412–1418.

[14] A. Pott and S. P. Bradley, Existence and nonexistence of almost-perfect autocorrelation sequences, *IEEE Trans. Inform. Theory* **41** (1995), 301–304.

[15] S. Barrera Acevedo, P. Ó Catháin, and H. Dietrich, Cocyclic two-circulant core Hadamard matrices, *J. Algebraic Combin.* **55** (2022), 201–215.

[16] R. C. Bose, An affine analogue of Singer's theorem, *J. Indian Math. Soc.* **6** (1942), 1–15.

[17] D. Jungnickel, On automorphism groups of divisible designs, *Canad. J. Math.* **34** (1982), 257–297.

[18] D. Jungnickel and H. Kharaghani, Balanced generalized weighing matrices and their applications, *Matematiche (Catania)* **59** (2004), 225–261.

[19] H. D. Lüke, Almost-perfect polyphase sequences with small phase alphabet, *IEEE Trans. Inform. Theory* **43** (1997), 361–362.

[20] H. D. Lüke, Almost-perfect quadriphase sequences, *IEEE Trans. Inform. Theory* **47** (2001), 2607–2608.

[21] X. Zeng, L. Hu, and Q. Liu, A novel method for constructing almost perfect polyphase sequences, in: *Coding and Cryptography — WCC 2005*, LNCS **3969**, Springer, 2006, pp. 346–353.

[22] J. A. Armario and D. L. Flannery, Almost supplementary difference sets and quaternary sequences with optimal autocorrelation, arXiv:1911.08828 (2019).

[23] N. A. Balonin and D. Ž. Đoković, Negaperiodic Golay pairs and Hadamard matrices, arXiv:1508.00640 (2015).

[24] J. Jedwab and T. Pender, Two constructions of quaternary Legendre pairs of even length, arXiv:2408.08472 (2024).

[25] I. S. Kotsireas, C. Koutschan, and A. Winterhof, Quaternary Legendre pairs II, arXiv:2408.16318 (2024).

[26] B. Schmidt, Cyclotomic integers and finite geometry, *J. Amer. Math. Soc.* **12** (1999), 929–952.

[27] K. H. Leung and B. Schmidt, The anti-field-descent method, *J. Combin. Theory Ser. A* **139** (2016), 87–131.

[28] K. H. Leung and B. Schmidt, Nonexistence results on generalized bent functions, *J. Combin. Theory Ser. A* **163** (2019), 1–33.

[29] H. Halberstam and H.-E. Richert, *Sieve Methods*, Academic Press, London, 1974.

[30] J. Friedlander and H. Iwaniec, *Opera de Cribro*, Amer. Math. Soc. Colloq. Publ. **57**, Providence, 2010.

[31] J. C. Lagarias and A. M. Odlyzko, Effective versions of the Chebotarev density theorem, in: *Algebraic Number Fields* (A. Fröhlich, ed.), Academic Press, London, 1977, 409–464.

[32] P. J. Stephens, An average result for Artin's conjecture, *Mathematika* **16** (1969), 178–188.

[33] D. R. Heath-Brown and S. J. Patterson, The distribution of Kummer sums at prime arguments, *J. Reine Angew. Math.* **310** (1979), 111–130.

[34] J. R. Chen, On the representation of a large even integer as the sum of a prime and the product of at most two primes, *Sci. Sinica* **16** (1973), 157–176.

[35] L. E. Dickson, A new extension of Dirichlet's theorem on prime numbers, *Messenger of Math.* **33** (1904), 155–161.

[36] A. J. Irving, Almost-prime values of polynomials at prime arguments, arXiv:1410.3333 (2014).

[37] P. Lewulis, Chen primes in arithmetic progressions, arXiv:1601.02873 (2016).

A Character liftings and a fillability dichotomy

The same pair object arises from the construction side, uniformly in the order m of the developing character. This section is self-contained and independent of §§4–4.9; its *existence* content is classical (the balanced generalized weighing matrix $\mathrm{BGW}(q+1, q, q-1)$ from Bose's affine difference set [16, 17, 18]), and what is new is the explicit two-coset pair normal form, the fillability dichotomy, and the arithmetic specializations.

Remark A.1 (Notation for this section). Throughout Section A, m denotes the order of a multiplicative character (a divisor of $q-1$), a role *distinct* from the prime $m \mid t$ of §§4.7–4.9; we write $E = \mathbb{F}_{q^2}$ for the quadratic extension (called K in the source note), reserving K for the real cyclotomic field above; and $B(x, y)$ denotes an alternating form (not a Williamson block).

A.1 Notation

Let $q \equiv 1 \pmod 4$ be an odd prime power, $t = (q+1)/2$ odd, $\mathbb{F} = \mathbb{F}_q$, $E = \mathbb{F}_{q^2}$, with $x \mapsto x^q$ generating $\mathrm{Gal}(E/\mathbb{F})$. Fix a primitive $\omega \in E^\times$; $g = \omega^{q+1}$ (a primitive element of $\mathbb{F}^\times$); $\zeta = \omega^{q-1}$ of order $q + 1 = 2t$ with $\zeta^t = -1$; $\theta = \omega^t$ (so $\theta^q = -\theta$, spanning the trace-zero $\mathbb{F}$-line of E); the nondegenerate alternating $\mathbb{F}$-form $B(x, y) = (xy^q - x^q y)/\theta$; a character χ of $\mathbb{F}^\times$ of exact order

$m \mid q-1$ (extended by $\chi(0)=0$); and $\xi=\chi(g)$, a primitive m-th root of unity, so $\chi(g^m)=1$ and $\chi(-1)=\xi^{(q-1)/2}\in\{\pm1\}$. Following Turyn's ordering with 4 replaced by $2m$, set the integer exponents

$$e_j = 2mj \ (0\le j\le t-1), \qquad e_j = t+2m(j-t)\ (t\le j\le 2t-1), \tag{7}$$

points $x_j=\omega^{e_j}$, and the $(q+1)\times(q+1)$ matrix $M_{jk}=\chi(B(x_j,x_k))$, written in $t\times t$ blocks with $R=M_{11}$, $S=M_{12}$, first rows $c_k=R_{0k}$, $s_k=S_{0k}$. For a length-t sequence u, $\mathrm{PAF}_u(k)=\sum_j u_j\overline{u_{j+k}}$. Two identities recur: the scaling rules $B(\lambda x,y)=B(x,\lambda y)=\lambda B(x,y)$, $B(\omega^k x,\omega^k y)=g^k B(x,y)$, $B(y,x)=-B(x,y)$; and the entry formula $B(\omega^a,\omega^b)=\omega^{a+b}(\zeta^b-\zeta^a)/\theta$.

A.2 The normal form

Lemma A.2 (Coverage)**.** *$(x_j)_{0\le j\le 2t-1}$ represents $\mathrm{PG}(1,q)=E^\times/\mathbb{F}^\times$.*

Proof. $\omega^a\equiv\omega^b \bmod \mathbb{F}^\times$ iff $(q+1)\mid a-b$. Any common divisor of $q-1$ and t divides $2t-(q-1)=2$; as t is odd, $\gcd(q-1,t)=1$, hence $\gcd(m,t)=1$. So $2mj \bmod 2t$ $(0\le j\le t-1)$ hits each even residue once and $t+2m(j-t)$ hits each odd residue once. □

Lemma A.3 (Orthogonality)**.** *$MM^*=qI_{q+1}$, for any representatives.*

Proof. $B(x_j,x_k)=0$ iff $k=j$, so each row has q unimodular entries and $(MM^*)_{jj}=q$. For $j\ne k$, $(MM^*)_{jk}=\sum_{l\ne j,k}\chi(B(x_j,x_l)/B(x_k,x_l))$; the map $y\mapsto(B(x_j,y),B(x_k,y))$ is an $\mathbb{F}$-linear bijection onto $\mathbb{F}^2$, so the ratio takes each value of $\mathbb{F}^\times$ once as $[y]$ ranges over the other points, and $\sum_{u\in\mathbb{F}^\times}\chi(u)=0$. □

Theorem A.4 (Normal form)**.** *All four blocks of M are circulant, and*

$$M=\begin{bmatrix} R & S\\ \chi(-1)\,S & \chi(-g)\,R\end{bmatrix},$$

where: (1) *R has zero diagonal, $c_k\ne0$ for $k\ne0$, and $c_{t-k}=\chi(-1)c_k$;* (2) *S is symmetric ($s_{t-k}=s_k$) with all entries nonzero;* (3) *all nonzero entries are m-th roots of unity and $M^T=\chi(-1)M$;* (4) *$RR^*+SS^*=qI_t$, equivalently $\mathrm{PAF}_r(k)+\mathrm{PAF}_s(k)=0$ for $k\not\equiv0 \pmod t$.*

Proof. Circulant blocks. Incrementing both indices within a block pair raises both exponents in (7) by $2m$; by the scaling rules the argument of χ multiplies by g^{2m} and $\chi(g^{2m})=\xi^{2m}=1$. A wrap replaces $e+2m$ by $e+2m-2mt$; since $\omega^{2mt}=g^m\in\mathbb{F}^\times$ and $\chi(g^{\mp m})=1$, every entry depends only on $(k-j) \bmod t$ within its block. $M_{22}=\chi(-g)R$. By (7), $\zeta^t=-1$, $\omega^{2t}=g$: $B(\omega^t,\omega^{t+2mk})=g\,\omega^{2mk}(1-\zeta^{2mk})/\theta=-g\,B(1,\omega^{2mk})$; apply χ. *S symmetric.* With $\omega^{2mt}=g^m$, $\chi(g^m)=1$: $B(1,\omega^{t-2mk})/B(1,\omega^{t+2mk})=g^{-2mk}$ (using $\zeta^{t\pm2mk}=-\zeta^{\pm2mk}$), and $\chi(g^{-2mk})=1$; the common factor $1+\zeta^{2mk}\ne0$. $M_{21}=\chi(-1)S$, $M^T=\chi(-1)M$. B alternating gives $M_{kj}=\chi(-1)M_{jk}$; with $S=S^T$ this yields the claim. *Diagonal and zeros.* $c_0=\chi(B(1,1))=0$; $c_k=0$ iff $t\mid k$ (as $\gcd(m,t)=1$); $s_k\ne0$ by parity; and $c_{t-k}=\chi(-1)c_k$ from circulancy and the alternating rule. *Pair identity.* The top-left block of $MM^*=qI$ (Lemma A.3) is RR^*+SS^*. □

Corollary A.5 (Turyn 1972, the case $m=2$)**.** *For $m=2$: $\chi(-1)=1$, $\chi(-g)=-1$, so $M=\left[\begin{smallmatrix} R & S\\ S & -R\end{smallmatrix}\right]$ is the symmetric conference matrix with R,S symmetric circulants and $R^2+S^2=qI$; $M+iI$ is a Butson–Hadamard $\mathrm{BH}(q+1,4)$, and $A=I+R$, $B=I-R$, $C=D=S$ is a Williamson quadruple of order $4t=2(q+1)$.*

A.3 The obstruction

Theorem A.6 (Fillability dichotomy). *A unimodular δ with $(M+\delta I)(M+\delta I)^* = (q+1)I$ exists iff $m \le 2$.*

Proof. By Lemma A.3 the condition is $\bar{\delta}M + \delta M^* = 0$; since $M^* = \chi(-1)\overline{M}$, entrywise off the diagonal $M_{jk}^2 = -\delta\chi(-1)/\bar{\delta}$ is constant, i.e. all off-diagonal entries lie in $\{\pm c\}$. But Theorem A.4 exhibits c_k (in M_{11}) and $\chi(-g)c_k = \pm\xi c_k$ (in M_{22}); both in $\{\pm c\}$ forces $\xi \in \{\pm 1\}$, i.e. $m \le 2$. Conversely $m = 2$ admits $\delta = \pm i$ (Corollary A.5). $\square$

Theorem A.6 explains why Turyn's Hadamard–Williamson endpoint was never extended past the quadratic character: for $m \ge 3$ the invariant content is the circulant pair (R, S) itself, an *almost-perfect complementary m-ary pair* of odd length t with a single zero entry, existing for every prime power $2t-1 \equiv 1 \pmod 4$ and every $m \mid 2t-2$. The closest relatives are the quaternary Legendre pairs of [24, 25], which require fully unimodular entries and PAF-sum $\equiv -2$ and live at length $(q-1)/2$ (the affine side) rather than $(q+1)/2$ (the projective side).

Remark A.7 (Quaternionic completion, and the exact boundary). The pair can be re-assembled: $N = \left[\begin{smallmatrix} R & S \\ -S^* & R^* \end{smallmatrix}\right]$ satisfies $NN^* = qI_{q+1}$ for every m (the circulant blocks commute), has zero diagonal, and is unimodular off it. A unimodular diagonal $D = \mathrm{diag}(d_j)$ completes it, $(N+D)(N+D)^* = (q+1)I$, iff $N_{jk}N_{kj} = -d_j d_k$ off the diagonal. In the cross blocks the left side is $-|S_{jk}|^2 = -1$, deleting every constraint on S; on a diagonal block it is $\chi(-1)c_{k-j}^2$, forcing d constant there and the condition c_k^2 constant. If χ^2 is nontrivial, the Hadamard square $M \circ M$ is the character matrix of χ^2, so $(R\circ R)(R\circ R)^* + (S\circ S)(S \circ S)^* = qI_t$; with $R \circ R = c^2(J-I)$ this forces $(S\circ S)(S\circ S)^* = (q-1)I - (t-2)J$, and evaluating on the all-ones vector gives $|\sum_k s_k^2|^2 = -t^2 + 4t - 2 < 0$ for $t \ge 5$. Hence the quaternionic completion exists iff $m \le 2$ or $t = 3$; the only new case is $(q, m) = (5, 4)$, a BH$(6,4)$ already reached at $q = 5$. The obstruction is intrinsic to the phase spread of c_k^2, not to the 2×2 arrangement.

A.4 Arithmetic specializations at the trivial character

Evaluating $RR^* + SS^* = qI$ on the all-ones vector gives, with $\rho = \sum_k c_k$ and $\sigma = \sum_k s_k$ in $\mathbb{Z}[\zeta_m]$,

$$|\rho|^2 + |\sigma|^2 = q, \tag{8}$$

a decomposition of q into two $\mathbb{Z}[\zeta_m]$-norms produced by character sums.

Proposition A.8. *For $m = 4$ and $q \equiv 5 \pmod 8$: $\rho = 0$ and $\sigma \in \mathbb{Z}[i]$ has $|\sigma|^2 = q$; the construction witnesses $q = a^2 + b^2$.*

Proof. $\chi(-1) = \xi^{(q-1)/2} = \xi^2 = -1$ (as $(q-1)/2 \equiv 2 \bmod 4$, $\xi = \pm i$). By Theorem A.4(1), $c_{t-k} = -c_k$, and the pairs $(k, t-k)$ partition the nonzero indices (t odd), so $\rho = 0$; then (8) gives $|\sigma|^2 = q$ with $\sigma \in \mathbb{Z}[i]$. $\square$

For $q \equiv 1 \pmod 8$ one has $\chi(-1) = 1$, so $\rho \in 2\mathbb{Z}[i]$; numerically more holds, and the basic 2-adic divisibility is elementary.

Proposition A.9 (The quartic 2-adic refinement). *For $m = 4$ and $q \equiv 1 \pmod 8$, $\rho \equiv 0 \pmod{(1+i)^3}$ in $\mathbb{Z}[i]$. Hence 8 divides $|\rho|^2$, and (8) decomposes q as a sum of two Gaussian norms with the first norm divisible by 8.*

Proof. Here $\chi(-1) = 1$, so Theorem A.4(1) gives $c_{t-k} = c_k$. Since $c_0 = 0$ and t is odd,

$$\rho = 2 \sum_{k=1}^{(t-1)/2} c_k.$$

Each c_k in the sum is a Gaussian unit. In the quotient $\mathbb{Z}[i]/(1+i)$ all four Gaussian units are congruent to 1, while $(t-1)/2 = (q-1)/4$ is even because $q \equiv 1 \pmod 8$. Therefore the half-row sum is divisible by $(1+i)$. Since $2 = -i(1+i)^2$, the displayed formula for ρ gives $\rho \in (1+i)^3\mathbb{Z}[i]$. $\square$

Remark A.10. In the computed prime cases $q \leq 457$, $q \equiv 1 \pmod 8$, the stronger diagonal pattern $\rho = a(1 \pm i)$ with a even also holds (and σ lies on a coordinate axis), recovering the classical-looking form $q = 8x^2 + b^2$ inside this normalization. That extra phase statement is not used here.

Only $|\rho|^2$, $|\sigma|^2$, and $\rho = 0$ are invariant under the global scalar $\chi(c)$; statements about the phase of σ alone are not, and are avoided.

A.5 Dictionary and related objects

Proposition A.11 (Twisted-circulant form)**.** *Set $w_e = \chi(B(1, \omega^e))$, so $w_{e+2t} = \xi w_e$. Let C be the $2t \times 2t$ ξ^{-1}-circulant $C_{jk} = w_{(k-j) \bmod 2t}\, \xi^{-[k<j]}$. Then M is monomially equivalent to C; in particular C has zero diagonal, m-th-root entries off it, and $CC^* = qI$. For $m = 2$, C is the negacyclic conference matrix of [10, 11].*

Before applying χ, the matrix $[B(x_i, x_j)]$ organized by the Singer cycle encodes Bose's affine difference set [16]: a $(q+1, q-1, q, 1)$ relative difference set in $E^\times \cong \mathbb{Z}_{q^2-1}$ relative to $\mathbb{F}^\times$. By Jungnickel's theorem [17] (quoted as [18, Thm 3.6]), cyclic relative difference sets are equivalent to ω-circulant balanced generalized weighing matrices; the family $\mathrm{BGW}(q+1, q, q-1)$ relevant here is the $d = 1$ case, appearing in [18] as the generalized-conference example $\mathrm{BGW}(5, 4, 3)$. Our C (hence M) is the character image of such a representative; Proposition A.11 makes the untwisting into the two-coset pair form explicit, extending Turyn's remark that his construction "is equivalent to the one in [4]" [1] from $m = 2$ to all m.

Remark A.12 (Single-sequence unrolling). Extending $u_e = \chi(B(1, \omega^e))$ over $e \in \mathbb{Z}$ gives $u_{e+2t} = \xi u_e$, a period-$m(q+1)$ sequence with m zeros per period; its full-period autocorrelation is $mq\,\xi^{-r}$ at shifts $2tr$ and 0 otherwise—an almost-perfect m-phase sequence. Almost-perfect polyphase sequences with small phase alphabet go back to Lüke [19, 20], and Zeng–Hu–Liu construct almost-perfect quadriphase families of exactly these lengths $m(p^J + 1)$ [21]; we verify only that the parameters coincide. For $m = 2$ the pair (r, s) is the input to the two-circulant-core construction [15]; the underlying almost-perfect autocorrelation sequences were introduced by Wolfmann [13] and classified via cyclic divisible difference sets by Bradley–Pott [14]. Turyn himself derived, on p. 321 of [1], a perfect-autocorrelation quaternary circulant from the pair; our (R, S) for $m \geq 3$ is the character-theoretic extension of that page.

Adjacent modern objects: the odd-length single quaternary sequences of Armario–Flannery [22] are fully unimodular with $\mathrm{PAF} = \pm 1$ (no quartic characters, no pair); negaperiodic Golay pairs [23] are binary of even length; the quaternary Legendre pairs [24, 25] live on the affine side. Every clause of Theorems A.4–A.6, Corollary A.5, and Propositions A.8 and A.9 have been verified entry by entry, in exact arithmetic, for 59 pairs (q, m) with q prime up to 113 and $m \mid q-1$, $m \leq 12$ (zero failures), and Remark A.7 likewise.

Remark A.13 (Further directions on the construction side). Two questions are left open. *(1)* For $q \equiv 3 \pmod 4$ (t even) the two-coset split in (7) breaks; an offset-1 ordering with m odd appears to be the correct skew analogue. *(2)* The fillability dichotomy suggests a general question: among character images of ω-circulant BGW matrices, which admit *monomial* completions to Butson–Hadamard matrices? Theorem A.6 settles diagonal fillings; row/column rescalings before filling are not addressed here.

B Verification status and provenance

Tiers. Tier 0: unconditional, complete proof in this document. Tier 1: conditional on GRH. Tier 2: conditional on a named explicit hypothesis (EQ_3, hypothesis (C), Dickson). Tier 3: numerical evidence only, no theorem claimed.

Principal claims. Selection theorem, T0. Two-orbit criterion, T0. $h = 3$ hierarchy, T0. Row-sum bound and exact form, T0. Rigidity criterion, T0. $q \leq 2000$ closure and first determinations $1469, 1937, 1325$, T0 with a closed-table novelty caveat. $q = 5185$, $q = 62305$, T1. Semiprime lane theorem, T0. Lane collapse at general ω (Proposition 4.78), T0. Dickson family and the hypothesis-(C) dichotomy, T2. Small-factor sieve count (Proposition 4.82), T2 (EQ_3); GRH implies $\mathrm{EQ}_3(\theta)$ for $\theta < 1/8$ (T1) but does not close the lane's remaining structure count, so the lane density program is not GRH-complete. Census calibrations at $q \leq 10^4$–10^7, T3.

Citations (source-verified 2026-07-07). The bibliography entries for Halberstam–Richert, Friedlander–Iwaniec, Lagarias–Odlyzko, Stephens, and Heath-Brown–Patterson were checked against public records and are bibliographically accurate (e.g. Lagarias–Odlyzko pp. 409–464; Stephens *Mathematika* **16** (1969) 178–188; Heath-Brown–Patterson *J. Reine Angew. Math.* **310** (1979) 111–130). One in-text precision point remains: the chapter pointer for Halberstam–Richert could not be independently confirmed, so those citations are now given without a chapter number.

Provenance. This document was written by large language models (Claude) under human direction, with machine-verified audits for the finite claims and independent-session cross-verification for the theoretical ones; one conditional theorem announced during development (a conditional infinite family on the lane) was retracted after verification and does not appear in this document. The working record, including the retraction, is preserved in the companion notes.